High-Fidelity CFD Simulation of Single-Phase Natural Circulation Loop Experiments: Verification and Validation

by

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Abstract

Natural circulation loops are integral to passive safety systems in advanced nuclear reactor designs, yet their computational modeling remains challenging due to the strong coupling between buoyancy-driven flow, conjugate heat transfer, and temperature-dependent fluid properties. This thesis works to demonstrate a comprehensive Verification and Validation study of a three-dimensional compuational fluid dynamics simulation of the single-phase natural convection experimental facility operated by Texas A&M University and validate this model against the experimental data of Reis, Seo, and Hassan (2024). The facility used two fluids, water and HITEC molten salt, a eutectic nitrate salt mix chosen for its low melting point and similar behavior to molten salts intended for use in molten salt reactors. For both working fluids, four experiments were conducted each, varying the loop's primary heater power between each case and studying the effects that change had on the velocity profiles within the system. Simulations of the system were performed in OpenFOAM v13 using a structured, fully-conformal, purely-hexahedral mesh generated with the `classy_blocks` library python library. A segmented first-order model was first built to reduce energy accounting uncertainties associated with the experiment, and better account for passive heat losses throughout the system.

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Nomenclature

Roman Symbols

A	Area	m^2
c_p	Specific heat capacity at constant pressure	$\mathrm{J/(kg\,K)}$
D	Pipe inner diameter	m
f	Friction factor	—
F_s	Safety factor (GCI)	—
g	Gravitational acceleration	$\mathrm{m/s^2}$
h	Heat transfer coefficient	$\mathrm{W/(m^2\,K)}$
H	Loop height	m
k	Thermal conductivity	$\mathrm{W/(m\,K)}$
L	Length; loop perimeter	m
$\dot{m}$	Mass flow rate	kg/s
p	Observed order of convergence	—
p_{rgh}	Pressure minus hydrostatic head	Pa
Q	Heat transfer rate	W
q	Heat flux	$\mathrm{W/m^2}$
r	Radius; mesh refinement ratio	m; —
R	Thermal resistance	$\mathrm{m^2\,K/W}$
T	Temperature	K
U	Velocity magnitude	m/s

Greek Symbols

β	Volumetric thermal expansion coefficient	$1/\mathrm{K}$
Δ	Difference	—
ε	Emissivity; relative error	—
μ	Dynamic viscosity	$\mathrm{Pa\,s}$
ν	Kinematic viscosity	$\mathrm{m^2/s}$
ρ	Density	$\mathrm{kg/m^3}$
σ	Stefan–Boltzmann constant	$\mathrm{W/(m^2\,K^4)}$

Dimensionless Numbers

$\mathrm{Gr} = g\beta\Delta T L^3/\nu^2$	Grashof number
$\mathrm{Nu} = hD/k$	Nusselt number
$\mathrm{Pr} = \nu/\alpha = \mu c_p/k$	Prandtl number
$\mathrm{Ra} = \mathrm{Gr} \cdot \mathrm{Pr}$	Rayleigh number
$\mathrm{Re} = \rho U D/\mu$	Reynolds number
$\mathrm{Ri} = \mathrm{Gr}/\mathrm{Re}^2$	Richardson number

Abbreviations

AMI	Arbitrary Mesh Interface
BC	Boundary Condition
CFD	Computational Fluid Dynamics
CHT	Conjugate Heat Transfer
GCI	Grid Convergence Index
IAPWS	International Association for the Properties of Water and Steam
MSR	Molten Salt Reactor
NCC	Non-Conformal Coupling
NCL	Natural Circulation Loop
PIV	Particle Image Velocimetry
QoI	Quantity of Interest
RANS	Reynolds-Averaged Navier–Stokes
V&V	Verification and Validation

Chapter 1

Introduction

1.1 Motivation

The global demand for electricity is large and ever-increasing. As Earth's population continues to grow, countries develop, and living standards rise; technology advances as well, compounding on the expanding demand for power with electric cars and the explosive growth of energy-intensive data centers for greater and greater artificial intelligence workloads. To sustain this growth, it is imperative that we prioritize the capacity of our power infrastructure [1]. That increased demand cannot be met by simply continuing to expand fossil-fuel facilities; climate change demands that new capacity, and eventually the existing fossil-fuel fleet, be filled by low-carbon sources [2].

Nuclear power is one of few technologies that can satisfy both the capacity and climate demands facing the near future. Nuclear's ability to supply large amounts of reliable, carbon-free energy independent of factors that may inhibit other carbon-free technologies such as weather and geography. Much of the renewed interest in nuclear power is centered on new, Generation IV reactor concepts. Among theses concepts are Molten Salt Reactors (MSRs), that promise higher outlet temperature, leading to greater thermal efficiency, while being operable a low pressures and having passive safety characteristics that their light-water

predecessors cannot match [3].

One of the characteristics that contribute to the passive safety of some MSR designs is natural circulation. Natural circulation is the transport of fluids resultant from the buoyant forces that arise from the thermal expansion and contraction of a fluid from heating and cooling without any external pumping required. Since the driving mechanism of this transport is from underlying physics, natural circulation can continue to remove heat from a reactor core even in total absence of electrical power or operator action, making it an attractive choice for emergency (and sometime normal) heat transport in advanced reactor designs [4]. Designing around these physics make the ability to make accurate predictions of flow rates, temperature distributions, and stability behavior an essential part of licensing and safe operation [5]. This requirement for accurate predictions necessitates accurate data, which is sparsely available for designs relying on working fluids other than water, such as MSRs.

Acquiring that predictive ability experimentally is difficult. Performing natural circulation experiments in reactor-like conditions are expensive and tricky. Molten salts are often corrosive to materials that are not specialized to deal with them. This difficulty compounds with the high temperatures needed to melt reactor salts like FLiBe, further complicating instrumentation for acquiring experimental data. Further still, the thermophysical property data necessary to interpret such experiments often carries appreciable uncertainty or is not agreed upon in the associated literature [6, 7]. Consequently, high-quality validation data for molten salt natural circulation is scarce, leading to a reliance on modeling and simulation for reactor design and licensing. Models and simulations themselves can only be relied upon once they are systematically validated against existing experimental data. Beyond that, development of these models is a challenge in and of itself; complications such as the tight coupling of buoyancy-driven flow with its own temperature field and the fluid's own temperature dependent properties, conjugate heat transfer through solid domains such as pipe walls and external insulation, as well as even subtle geometric details of the facility's

assembly [8].

The Texas A&M University (TAMU) natural circulation loop, operated by Reis, Seo, and Hassan [9], represents the state of the art for such validation data. The experiment used particle image velocimetry (PIV) to measure two-dimensional velocity fields midway up the riser leg of the loop and recorded steady-state thermocouple readings around the loop, for two working fluids: water as a baseline, and HITEC, a eutectic molten salt mixture. These experiments were conducted in identical loop geometries at four constant heater powers each until thermal equilibrium was achieved. This two-fluid approach allows us to isolate the effects of the fluid properties from those of the loop geometry, making the TAMU dataset an attractive validation target, especially for demonstrating the unique difficulties and different approaches needed for properly modeling the natural circulation of molten salt.

The need for validated models increases further when the development of new, digital-twin technology for advanced nuclear systems is considered; particularly relevant for this thesis are those in development at the University of Texas at Austin. Digital twins are anticipated to improve both the efficiency and safety of operating nuclear systems [10], but the reduced-order models and neural networks used for these applications must, at their core, be trained and evaluated against data with known quality. A rigorously validated CFD model can serve as both a reference solution and synthetic training data generator for quantities and locations that physical instrumentation cannot reach or did not measure. That intention increases the pressure for present work, as models developed downstream from this work inherit the quality of the data produced here, whose credibility must be quantified.

This thesis attempts to help fill the need for modeling and simulation of molten salt natural circulation loops (NCLs). To do so, a complete, automated computational fluid dynamics (CFD) verification and validation (V&V) pipeline for NCLs has been developed using OpenFOAM V13 [11] and tested against all eight published test cases from TAMU's experimental facility [9]. In doing so, this thesis documents challenges particular to an NCL

and how they were tackled, namely closing the energy balance of the experiment, contending with long entry lengths prohibiting classic fully-developed flow from being achieved, and flow features including secondary recirculation that undermine one-dimensional, cross-sectionally averaged models commonly used for loop analysis.

1.2 Objectives

The objectives of this thesis are:

1. To develop a complete, automated CFD modeling pipeline for the TAMU natural circulation loop using OpenFOAM V13, that accepts geometry, physics, and study configuration files for a given experiment or suite of cases and automatically generates a fully conformal, fully hexahedral mesh, applies proper initial and boundary conditions from those configuration files, pre-process to ensure desired data collection is properly mapped to the mesh and captured appropriately, execute the simulation to convergence, and post-process the data to be rendered in an overview PDF to easily determine how well a given configuration aligns with the experimental results.

2. To create a tunable first-order model to aid in closing the large energy balance uncertainties associated with the experimental data, as, more than anything else, the energy balance across the experiment's domain determines how close we can replicate the experiment within the simulation. Once created and tuned, this first order model can be wired in directly to the aforementioned automated simulation pipeline to dynamically set the boundary conditions for the CFD to adhere to.

3. To develop a custom boundary condition model (`rcExternalTemperature`) that models cylindrical conduction through multiple solid layer beyond the simulated fluid mesh, factoring in conductive, convective, and radiative heat transfer while replacing the slab-geometry assumption of OpenFOAM v13's stock `externalTemperature` boundary condition.

4. To perform systematic code verification using multiple mesh refinement levels to conduct grid convergence studies, using Richardson extrapolation and the Grid Convergence Index (GCI), following the procedure of Celik ́ et al. [12] to ensure not only that the results presented here are accurate and reliable, but to also provide high-quality data for the continued Digital-Twin efforts for advanced nuclear systems developed at the University of Texas at Austin.

5. Lastly, the ultimate goal of this work is to fully validate the simulation pipeline against all eight experimental test cases from Reis et al. [9], comparing simulated temperatures, time-averaged velocity profiles, and mass flow rates against experimental measurements, with the goal of replicating those results within 1σ of uncertainty or otherwise determining the source of any remaining discrepancy in the results.

1.3 Thesis Outline

The remainder of this thesis is organized as follows:

- **Chapter 2** reviews the historical and modern literature regarding natural circulation loop experiments and simulation, CFD simulation methods, especially pertaining to the simulation of closed-loop natural circulation, molten salt thermal hydraulics, material properties of molten HITEC salt and water, and V&V methodology.

- **Chapter 3** presents the methodology used from beginning to end of the V&V pipeline. In this chapter the structure of the pipeline is displayed along with the procedure for using the pipeline. This procedure includes a summary of the configuration file structure, integration and discussion of how the first order model works and is implemented to the simulation pipeline from the experimental data, along with the implementation of Richardson extrapolation, the Grid Convergence Index, and the validation error assessment procedure.

- **Chapter 4** describes the experimental facility that was used in the foundational work for this thesis. In this chapter, the relevant procedures, instrumentation, working fluids, experimental test matrix, and results will be discussed. Within this chapter, caveats, along with any other relevant disclaimers discovered during the conducting of the work within this thesis will be addressed.

- **Chapter 5** details the computational modeling approach at each step throughout the pipeline. This section proceeds stepwise through the pipeline to explain the approach that was taken to create the first order model and attempt to close the energy balance of the system. From here, the details of the mesh and its generation are expounded upon, followed by the numerical schemes and solution settings used within the OpenFOAM simulation for each case. Next, the actual conduction of the simulation and utilization of high-performance computing (HPC) resources was performed in addition to the methodology for collecting the quantities of interest before finally postprocessing the results and refining the results into easy-to-digest figures to easily determine how well simulation aligns with the experimental result, along with any methodology corrections to data collection.

- **Chapter 6** presents the results of the pipeline. Within this chapter, the verification of the methodology is completed and demonstrated for the custom boundary condition code, grid convergence studies, and parallel scaling performance across mesh refinement levels. Once the verification of the pipeline has been conducted and sufficiently demonstrated, the simulated results will be compared against the experimental results to demonstrate to what extent the results of the simulation pipeline can be considered properly validated.

- **Chapter 7** summarizes the findings from the material in Chapter 6 and how well met the goals set at the beginning of this thesis were met. In this chapter, limitations of the methodology, pipeline, and experimental data will be discussed to better understand

the discrepancies between the simulated results and experimental data. Lastly, the chapter will conclude by suggesting directions for future work and how the limitations previously stated may be overcome or otherwise how the work presented in this thesis may be improved upon in the future.

Chapter 2

Literature Review

This chapter reviews literature relevant to the present work. Topics covered are: the fundamentals of natural circulation and how temperature variations in thermophysical properties complicates modeling it. Next, part of the experimental record for molten salt natural circulation loops, including the TAMU facility this work targets, along with the thermophysical property data needed to interpret it. That is followed by a selection of previous computational studies of such loops, along with meshing choices that control discretization error. Then, literature outlining the reduction from full conjugate heat transfer to thermal boundary conditions followed by the V&V methodology adopted for this work along with a selection of digital twin applications that consume validated models. Lastly, the chapter concludes by stating the reseach gaps this thesis addresses.

2.1 Natural Circulation Fundamentals

Before examining what's required to model and simulate natural circulation, the quirks of molten salt, or how to best determine if a simulated result is even correct, it is important to first understand the fundamentals of natural circulation.

Natural circulation is the transport of a fluid resulting from heating and cooling that fluid at different locations within a closed system. Heating causes the fluid to expand, lowering its

local density, whereas cooling causes it to contract, raising its local density. The consequence of those density changes can be seen by considering a fluid parcel of volume V and density ρ surrounded by fluid of density ρ_0. By Archimedes' principle, the surrounding fluid exerts an upward buoyant force equal to the weight of the displaced fluid,

$$F_b = \rho_0 \, g \, V, \tag{2.1}$$

where g is the gravitational acceleration, while the parcel itself weighs $\rho g V$, leaving a net vertical force

$$F_{\text{net}} = (\rho_0 - \rho) \, g \, V. \tag{2.2}$$

In a loop that heats its fluid at a low elevation and cools it at a higher one, this imbalance can drive a continuous, self-sustained flow that requires no external pumping power, which is what we called natural circulation [13]. Since the density of a fluid is a function of its temperature, engineering analyses over modest temperature ranges often linearize the density about a reference state (ρ_0, T_0) as

$$\rho(T) \approx \rho_0 + (\partial \rho / \partial T)_p \, (T - T_0), \tag{2.3}$$

which is conventionally rewritten in terms of the volumetric thermal expansion coefficient, $\beta = -\rho_0^{-1} (\partial \rho / \partial T)_p$, as

$$\rho(T) \approx \rho_0 \, [\, 1 - \beta \, (T - T_0) \,] \, . \tag{2.4}$$

Equations 2.3 and 2.4 for the basis of the Boussinesq approximation. The classic approximation was formalized by Spiegel and Veronis [14] and later tightened by Gray and Giorgini [15]. The approximation treat density as constant in the unsteady and convective terms and its linear variation is kept only in the gravitational body force while each other physical property is held constant. Gray and Giorgini quantified when that treatment is appropriate only when properties vary from their reference value by less than 10%, which for

water, limits the allowable temperature difference near room temperature to approximately 1.25 °C. Consistent with those bounds, Bückle and Perić [16] found in simulations of buoyant convection of ethanol within a square cavity, that Boussinesq-based solutions deviated appreciably from both variable-property solutions and experimental observations once the temperature difference reached approximately 10 °C, and Krishnani and Basu [17] demonstrated for natural circulation loops specifically that the approximation can alter predicted transient behavior even at modest temperature differences. The experiments studied in this thesis do not come close to satisfying these bounds. The published bulk fluid temperature differences span $1.29 - 1.67$ °C for the water cases and $5.63 - 11.16$ °C for the salt cases (Tables 4.3 and 4.2), so even the mildest water case sits at the water bound of Gray and Giorgini. The temperature range the fluid properties must actually be evaluated over an evem wider range. Heater-adjacent wall thermocouples read approximately 11 °C (water) and 28 °C (salt) above the coolest fluid probe in the highest-power cases, and across the full test matrices the wetted surfaces span approximately $39 - 71$ °C for water and $167 - 231$ °C for salt. This temperature span means no single reference state for either fluid can serve the whole matrix fluid. That, paired with the desire for a unified modeling approach across both working fluids, led to the decision for the present work to retain full temperature dependence of every fluid property (Chapter 5).

NCLs have been studied for decades. For such a system to achieve steady-state circulation, the buoyancy head is balanced by the frictional losses integrated along the loop's path. From that idea, works such as Vijayan and Austregesilo [18] developed generalized scaling laws spanning multiple flow regimes and Vijayan later also published trends of stability for NCLs [19]. Analyses such as these are established workhorses of loop design and analysis, but also remain contingent on the applicability of the boussinesq approximation and the assumption that fully developed flow correlations hold. When performing preliminary entry length calculations, the laminar hydrodynamic and thermal entry lengths scale as $L_h \approx 0.03\ \mathrm{Re}_D\ D$ and $L_t \approx 0.043\ \mathrm{Re}_D\ \mathrm{Pr}\ D$ [20] to be within 5% of full development. So,

for the TAMU NCL and using the published data (reiterated in Appendix C), we find that hydrodynamic ranges for water span 0.504m - 1.280m and 0.053m - 0.113m for salt and the thermal entry lengths range from 3.058m - 5.431m for water and 4.128m - 5.124m. Given the loop's legs are less than 1m each and the total path length of the loop is roughly 4m (Section **??**), this finding undermines analysis done with the assumption of fully-developed flow and precludes the safe use of analytic solutions such as for the Nusselt number for laminar flow, common in engineering analysis. This limitation is revisted in Chapters 3 and 5.

2.2 Molten Salt Natural Circulation Experiments

Experimental data for molten salt natural circulation is limited by the practical challenges of high-temperature operation, corrosion, and optical access, though efforts toward characterizing the thermal-hydraulic behavior of salt loops span more than a decade. Srivastava et al. [21] performed one of the first combined experiment/code validation exercises on a molten nitrate salt loop, studying the transient evolution and subsequent steady state of the loop's flow and temperature distributions. Their thermal-hydraulic code captured the majority of the experimentally observed trends, with the greatest deviation occurring in the heater-trip transient, where over-predicted heat losses cooled the modeled system considerably faster than was observed experimentally, which they theorize as being resultant from heat absorbed into the structure of the loop that was not captured in the model. Most modern experiments similarly use nitrate salts, whose low melting points make them far easier to work with than the fluoride salts favored for nuclear applications. That rarity makes the natural circulation FLiBe loop operated at the University of Wisconsin-Madison by Britsch et al. [22] particularly notable. One finding of theirs deserves emphasis: no simple correlation between the loop's temperature readings (ΔT, $1/\Delta T$, T_{avg}, or similar) and the measured flow velocity could be established, with the best single predictor among their thermocouples achieving only $R^2 = 0.56$. This complexity is counter-intuitive relative

to the one-dimensional theory of Section 2.1, and it mirrors experience in the present work (Chapter 6).

The facility most relevant to the present work is the TAMU natural circulation loop operated by Reis, Seo, and Hassan [9]. It occupies a unique place in the literature: it studied water and HITEC salt within an identical geometry, and its transparent quartz test section allows the use of particle image velocimetry (PIV), providing radially resolved velocity profiles without perturbing the flow the way a physical flow meter would. Companion publications document argon injection experiments [23], the consequences of an inadvertent salt solidification event [24], and the dataset has since served as a validation target for the SAM system code at high Prandtl number conducted by the experimentalists themselves [25]. The facility, its instrumentation, and its test matrix are described in detail in Chapter 4.

Interpreting any of these experiments, and especially attempting to simulate them, rests on the fidelity of the working fluid's thermophysical properties. Beyond the density variation that generates the buoyant force itself important properties such as the specific heat determines the temperatures at which a given heating and cooling configuration equilibrates, and viscosity governs how readily the fluid circulates. Comprehensive single-source property collections for molten salts are rare. Sohal et al. [7] compiled an engineering database of liquid salt properties at Idaho National Laboratory, but its entries for HITEC ($NaNO_3$-$NaNO_2$-KNO_3, approximately 7-49-44 by mole and 7-40-53 by weight) trace back to the compilation of Janz and Tomkins [26] and, for viscosity and specific heat, have been superseded by more recent measurements. The review supporting this claim and the modern property suite selected for this work, are presented in Chapter 4 and Appendix B. For water, its properties are well-characterized by the IAPWS-IF97 with marginal uncertainties for temperatures in the range of 273.15 K - 1073.15 K and pressures up to 100 MPa [27, 28]. The polynomial fits used in this work reproduce reference values within 0.50 % over the experimental temperature range (Appendix B).

2.3 CFD Simulation of Natural Circulation Loops

Several groups have applied OpenFOAM to natural circulation simulation. Luzzi et al. [29] validated both a one-dimensional, semi-analytical model and a CFD model made using OpenFOAM V4 against a rectangular loop that heated its bottom leg and cooled its top leg. It was found in both the experiments and the corresponding simulations that not all configurations exhibited stable behavior. This observation of occasional instability is not an uncommon one for symmetric rectangular and circular geometries. As unstable configurations tend to be oscillatory, Pini et al. [30] exploited OpenFOAM's open-source nature to couple simulation data directly to MATLAB for eigenvalue-based prediction of oscillatory behavior in rectangular loops. Battistini et al. [31] extended the toolset to Large Eddy Simulation (LES) of the DYNASTY facility, predicting flow reversals and re-laminarization, phenomena typical of unstable natural circulation configurations, that had been absent from previous Reynolds-averaged Navier-Stokes (RANS) models of the facility.

Reviewing these OpenFOAM studies, readers can begin to see how consequential the choice of how to handle turbulence modeling can be. For modeling natural circulation systems that are already sensitive to material property inputs, adding the historically tricky problem of turbulence modeling complicates matters. Nalbandyan et al. [32] demonstrated this directly, testing three different turbulence models (namely the $k-\omega$SST, $k-\omega$SSTLM, and realizable $k-\epsilon$ models) applied to a 3D CFD against their 1-D model of the DYNASTY loop filled with water. Depending on the turbulence model, their relative errors versus the experiment for mass flow rate varied by $7.2\% - 16.2\%$ and their temperature difference through the cooler section of their loop was off by $1.6K - 2.3K$. For molten salt loops specifically, Simamora and Lee [33] used ANSYS Fluent to compare laminar and transitional (k-ω SSTLM) treatments of a two-dimensional nitrate salt loop model. The laminar model underpredicted the circulation velocity by roughly 30%, the transitional model by only 4.2%. The authors attribute the improvement largely to the improved mixing resultant from the introduction of turbulence, which allowed the temperature to better penetrate to

the flow's centerline, allowing the temperature sensitive viscosity to be more representative of the experiment and thereby better capture the experimental flow profile. This study's two-dimensional planar geometry and its reliance on an energy balance, serve as a notable counterpart to flow features exclusive to three-dimensional modeling that will be discussed later in Chapter 6.

For any of these modeling choices to be assessed fairly, discretization error must be controlled, and that control begins with the mesh. Truncation error analysis of the finite-volume method shows that hexahedral cells produce the smallest truncation errors of the common cell shapes for a given cell size since they have the smallest possible number of face pairs and thereby have the least amount of truncation for a given cell count in a given volume, with tetrahedral cells having the largest error [34]. Hernández-Pérez, Abdulkadir, and Azzopardi [35] demonstrated that hexahedral topologies likewise produce superior volume-of-fluid (VoF) results compared to tetrahedral or polar meshes for circular pipe cross-sections. Despite being out of scopre for this work, the applicability to VoF simulations is important because the meshing infrastructure built for this thesis is intended to outlive it; extending beyond single-phase flow to two-phase VoF simulation is planned within the digital twin effort at the University of Texas at Austin (Chapter 1) and both considerations motivated the purely hexahedral, structured meshes generated for the present work (Chapter 5).

2.4 Conjugate Heat Transfer and Thermal Boundary Conditions

As the flow within an NCL is nothing more than the fluid's response to its thermal boundaries, properly representing those boundaries is of the utmost importance. Fluid-only CFD simulations often fail to capture the effect that solid materials on a system can have on the working fluid, especially in transient models with large volumes of material with large thermal time constants such as many layers of insulation being applied to the external pipe

walls. Capturing these effects generally requires explicit simulation of the solid domains at the same time, typically referred to as conjugate heat transfer (CHT).

The many complexities and computational demands of CHT largely disappear, however, when only the system's steady state is of interest. For steady-state simulation, the full CHT approach can be replaced by a mixed (Robin) boundary condition that represents the wall's thermal resistances analytically. Ljulj et al. [36] provides a rigorous proof that the effects of the pipe's wall are not always negligible and can captured by effective Robin-type conditions. Their analysis was for a similar case to this work, steady-state cylindrical pipe flow, but differed by using a single layer and omitting radiation. This work extends the same construction to potentially thick walls composed of more than one layer and an external radiation term. This custom boundary condition, `rcExternalTemperature` is described in Chapter 5 and verified in Chapter 6.

2.5 Verification and Validation

The credibility of the results in this thesis depend upon the V&V framework applied to it. Roache [37] is one of the foundational texts for V&V, making the distinction between *verification*, which determines if the intended equations are being solved properly, and *validation*, determining if the correct equations are being solved. Oberkampf and Roy [38] extended the framework into uncertainty quantification and distinguishing between irreducible uncertainties and reducable uncertainties, which must be treated differently and ultimately factor in to how we determine if the results of this work can be considered validated, and if so, with what uncertainties attached to it?

For solution verification, the present work uses the Grid Convergence Index (GCI) procedure set by Celiḱ et al. [12]. The prodcedure is a five-step method that uses Roache's GCI to convert a Richardson extrapolation [39] of a set of solutions for a systematically refined mesh into a standardized uncertainty band. The validity of the GCI is only valid if the solution

lie within the asympototic range of the extrapolated solution and converge monotonically and the present work was designed to satisfy these conditions, detailed in Chapter 3. For validation, this work follows ASME V&V 20 [40], the standard for CFD and heat transfer, further detailed in Chapter 3. Beyond the validation uncertainty of the results themselves, there are portions of the experimet that had to be reconstructed for this work and carry their own uncertainty, which combine following the procedures of Coleman and Steele [41].

2.6 Digital Twins and Reduced-Order Models

The value of high-fidelity CFD simulations extends further beyond direct engineering analysis. One such application of particular interest in recent years is to use the data generated by high-fidelity models to inform the creation of reduced-order models (ROMs) or to train neural networks within digital twin frameworks. As for digital twin application to the nuclear industry, works such as Kochunas and Huan [10] outline how it is anticipated to improve both the efficiency and safety of operating nuclear systems. Later works such as Huang et al. [42] look at the maturation of the technology since Kochunas and Huan and propose a five-tier hierarchy of maturity for digitial twins in nuclear engineering spanning from guidance and training from simulation to fully autonomous system operation.

Almukhametov et al. [43] developed a graph neural network GNN-based digital twin for a case study to emulate concept reactors such as the Kairos Power FHR with a companion System Analysis Module (SAM) model manage to achieve an over 100x speed-up compared to the SAM model while retaining an R^2 value of up to 0.995. Hossain et al. [44] demonstrated DeepONet-based virtual sensors predicting thermal-hydraulic parameters 1,400 times faster than CFD, with accuracy sufficient for operational monitoring in most cases, but still experiencing errors of $\approx$ 10% or more for velocity, turbulent kinetic energy, and pressure outside of the best case scenario.

Efforts such as these demonstrate the need for rigorously validated CFD and continued

pursuits for advancing digital twin technology further motivate the pursuit of high-fidelity CFD. A validated CFD model of the TAMU loop could serve as the foundation for a digital twin of similar molten salt natural circulation systems.

2.7 Summary and Research Gap

This survey of the literature reveals several key gaps that this thesis addresses:

1. **Limited CFD Validation of Molten Salt Systems:** The experimental facility at Texas A&M provides the most comprehensive velocity profile dataset for molten salt natural circulation loops along with it's thermocouple readings, yet no independent, three-dimensional CFD simulation has been systemically Verified and Validated against this data.

2. **Insufficient Boundary Conditions short of full CHT:** OpenFOAM's stock boundary condition `externalTemperature` assumes a slab geometry for heat flux through surfaces assigned that boundary condition and for cylindrical pipes with thick insulation, the area ratio between the inner and outer surfaces makes this a poor approximation that would otherwise need to be resolved via more complex CHT modeling.

3. **Difficulty with Energy Balance Closure:** Previous experiments such as Reis et al. have large uncertainties associated with the heater power applied to the system not being equal to the calculated rate of heat removal by the system's cooler, making it impossible to accurately model the full system without better representing all the heat losses throughout the system.

4. **Accounting for Entry Length Effects:** Depending on the geometry of the system and the working fluid studied, there can at times be cases where the fluid never fully develops in the classical sense for neither its thermal or momentum entry lengths. These thermal and momentum entry lengths being fully developed is a typical assumption

of engineering calculations such as analytical Nusselt numbers used for heat transfer calculations of laminar flow and the physical perimeter of a loop being less than these entry lengths can undermine the accuracy of downstream calculations.

Chapter 3

Verification and Validation Methodology

3.1 Overview

The V&V process is structured as a sequence of increasingly demanding tests:

1. **Code verification:** first, it must be demonstrated that the custom-written code for this project, the `rcExternalTemperature` boundary condition, reproduces an independent analytical reference solution and that the discretization error vanishes at the expected rate under mesh refinement.

2. **Solution verification:** next, with that verified boundary condition, we need to estimate the numerical uncertainty of the full-loop simulations for the quantities of interest (QoI) by applying Richardson extrapolation and the Grid Convergence Index across a family of systematically refined meshes.

3. **Validation:** lastly, once the simulations have been verified, we compare how the simulation matches up against the experimental measurements of PIV velocity profiles, centerline and wall temperatures, and the calculated mass flow rate within the system.

From there, we judge each comparison against the combined numerical and experimental uncertainty.

This deliberate ordering is because an attempt at validation of the solution is only meaningful once the numerical error is understood and bounded; otherwise, an apparently good or bad agreement with the experiment cannot be separated from grid-dependence. [37]. With this process outlined, the remainder of this chapter details what is involved in the methodology to conduct the V&V process.

3.2 Richardson Extrapolation

Both code verification and solution verification hinge on the same idea that for a consistent, p^{th} order discretization a QoI, ϕ, is computed on a mesh with a representative spacing h following the form

$$\phi(h) = \phi_{\text{exact}} + g_p h^p + O(h^{p+1}), \tag{3.1}$$

such that the discrete solutions approach the exact solution as $h \to 0$. Given three solutions on systematically refined meshes with spacings $h_1 < h_2 < h_3$, and refinement ratios, r_{21} and r_{32}, where $r_{21} = h_2/h_1$ and $r_{32} = h_3/h_2$. The successive differences are defined as

$$\varepsilon_{21} = \phi_2 - \phi_1, \qquad \varepsilon_{32} = \phi_3 - \phi_2. \tag{3.2}$$

The observed order of accuracy then follows from Equation (3.1) to the general form of

$$p_{\text{obs}} = \frac{1}{\ln r_{21}} \left| \ln \left| \frac{\varepsilon_{32}}{\varepsilon_{21}} \right| + q(p_{\text{obs}}) \right|, \tag{3.3}$$

where $q(p_{\text{obs}})$ is uses $s = \text{sign}(\varepsilon_{32}/\varepsilon_{21})$ and is defined as

$$q(p_{\text{obs}}) = \ln \left(\frac{r_{21}^{p_{\text{obs}}} - s}{r_{32}^{p_{\text{obs}}} - s} \right). \tag{3.4}$$

So, if the solutions are in the asymptotic range, for a constant refinement ration $r = r_{32} = r_{21}$, the term $q(p_{\text{obs}}) = 0$ and Equation 3.3 reduces to the simplified form

$$p_{\text{obs}} = \frac{1}{\ln r} \ln \left| \frac{\varepsilon_{32}}{\varepsilon_{21}} \right|. \tag{3.5}$$

With this, the Richardson-extrapolated estimate of the QoI's exact value takes the form

$$\phi_{\text{exact}} = \frac{r^{p_{\text{obs}}} \phi_1 - \phi_2}{r^{p_{\text{obs}}} - 1}. \tag{3.6}$$

3.3 Grid Convergence Index

The Grid Convergence Index, introduced by Roache [37], provides a standardized measure of discretization uncertainty expressed as a percentage error band. The standardized, five-step procedure for determining the GCI as determined in Celiḱ et al. [12].

1. Define three systemically refined meshes with refinement ratio $r \geq 1.3$.
2. Compute the QoIs on each mesh.
3. Calculate the observed order of convergence, p_{obs} using Equation 3.5.
4. Calculate the extrapolated values for each QoI using Equation 3.6
5. Calculate and report the following error estimates, the observed order of convergence, and the GCI for the fine mesh,

$$\text{GCI}_{\text{fine}} = \frac{F_s \; e_a^{21}}{r_{21}^{p_{\text{obs}}} - 1} \tag{3.7}$$

Where F_s is the safety factor, which has the recommended value of 1.25 for studies

with three or more meshes, e_a^{21} is the approximate relative error,

$$e_a^{21} = \left| \frac{\phi_1^{12} - \phi_2}{\phi_1^{12}} \right| \tag{3.8}$$

and e_{ext} is the extrapolated relative error,

$$e_{\text{ext}}^{21} = \left| \frac{\phi_{\text{ext}}^{12} - \phi_1}{\phi_{\text{ext}}^{12}} \right| \tag{3.9}$$

With those five steps complete, we can then determine if the solutions are in the asymptotic range where the leading-order error term dominates uncertainty by calculating the asymptotic ratio (AR) using the following equation

$$\text{AR} = \frac{\text{GCI}_{\text{coarse}}}{r_{21}^{p_{\text{obs}}} \text{GCI}_{\text{fine}}} \approx 1. \tag{3.10}$$

Where an asymptotic ratio near 1 denotes that the solutions are within the asymptotic range and that the GCI provides a reliable estimate of uncertainty.

3.3.1 Convergence Classification

The ratio $R = \varepsilon_{21}/\varepsilon_{32}$ classifies the convergence behavior:

- $0 < R < 1$: **Monotonic convergence**, where the changes in the solution shrink in response to mesh refinement.
- $R < 0$: **Oscillatory convergence**, where the signs of the changes in the solution alternate in response to mesh refinement.
- $R > 1$: **Divergence**, where the changes in the solution grow in response to mesh refinement

Only when the solutions converge monotonically is clean Richardson extrapolation using the GCI estimate permissible. Oscillatory or divergent behavior classifications may signal

that the solutions are not yet within the asymptotic range or that there may be other issues within the solution and that calculated uncertainty estimates for solution sets with these behavior classifications should be treated with caution or using specialized methods. This work's solutions converge monotonically under mesh refinement so the methods within this chapter are discussed and later applied with monotonic convergence in mind.

3.3.2 Mesh Refinement Strategy

With the methodology established and in-line with the first step of determining the GCI discussed above, we can discuss the mesh refinement strategy used within this work.

The full-loop meshes form a three-level family by uniformly scaling the boundary-layer and bulk cell size in a hybridized approach to preserve block topology and ensure that a topologically blind refinement strategy would not muddy refinement where it was important as in the boundary layer or to overly refine the bulk flow regions since the full-loop mesh is expensive to compute, especially since pseudo-transient timestepping was needed, which will be further elaborated on Chapter 5.

For the BC-verification family, a similar approach was used to maintain a consistent methodology, however, the BC-verification family of meshes were smaller and much faster to compute overall. This relaxation of computing resources allowed expanding the code-verification study to a four-level approach using a refinement ratio of $r = 2$ on a simpler geometry which permitted better testing of the asymptotic order. For a 3D case, a single value for a representative cell size, h_{rep} can be found using the equation,

$$h = \left[\frac{1}{N} \sum_{i=1}^{N} (\Delta V_i) \right]^{1/3} \tag{3.11}$$

Where h is the representative cell size, ΔV_i is the volume of the ith cell, and N is the total number of simulated cells.

Table 3.1: Mesh refinement families. The full-loop family ($r = 1.5$) is used for solution verification and validation; the BC-verification family ($r = 2$) is used for code verification.

	Full-loop family ($r = 1.5$)			BC-verification family ($r = 2$)			
	Coarse	Medium	Fine	Coarse	Medium	Fine	Xfine
Bulk cell size [mm]	2.25	1.50	1.00	2.80	1.40	0.70	0.35
First cell size [mm]	0.225	0.150	0.100	0.600	0.300	0.150	0.075

3.4 Boundary Condition Verification

The full-loop's wall is represented by neither a simple fixed-temperature nor fixed-flux boundary condition, but instead uses a custom boundary condition, `rcExternalTemperature`, which solves a one-dimensional, cylindrical conduction problem from the pipe's inner surface through the composite wall layers to find the external temperature that is then used to solve for external losses from convective and radiative heat transfer. This was done for a combination of two reasons: the first was that early on, before the work had been scoped for a thesis-grade problem, the problem was assumed to be simple and that it would not need to model the full, solid domain and associated conjugate heat transfer. So, to not scrap the high-quality fluid domain mesh or significantly retrofit the working pipeline, and since we are only interested in the steady-state solution, a computed thermal circuit could prove sufficient. The second reason was, with the first established, OpenFOAM V13 did not have a stock boundary condition that would properly model the heat transfer for the system of interest as the closest match, `externalTemperature`, assumed a slab geometry for it's thermal circuit model, which under-predicts losses through the boundary due to the greatly increased surface area resultant from the thick insulation adorning the facility's exterior.

3.4.1 Verification Geometry

Boundary condition code verification was performed on a simple, two-segmented cylinder with an O-grid mesh and deliberately constructed for comparison to an analytical solution. The domain was comprised of an inner radius $r_0 = 10.0$ mm, an SS-316 wall (1.0 mm, $k =$

16.3 W/(m K)), and a ceramic insulation layer (50.0 mm, $k = 0.144$ W/(m K)), resulting in an outer radius $r_2 = 61.0$ mm. The fluid and insulation use constant properties evaluated at 440 K to isolate the boundary-condition discretization error from any temperature dependent property effects.

To best focus on the verification of the heat-transfer capabilities of this, the first section of the studied mesh is a 500 mm, adiabatic entry length preceding a 100 mm section with the `rcExternalTemperature` boundary condition applied to ensure that the flow's momentum boundary layer was fully developed before applying calculations to the system that rely upon that assumption for their accuracy, ensuring that the comparison to the analytic solution is well-posed. The hydrodynamic entry length for this flow to be 99% developed is $L_h \approx 0.05\ \mathrm{Re}\ D = 334\mathrm{mm} < 500\mathrm{mm}$, so the velocity profile is fully developed before the flow is subjected to any heat flux. Given this flow is transitioning from an adiabatic segment to one experiencing heat flux, we must also account for the thermal entry length, which to be 99% developed is expressed by $L_t \approx 0.05\ \mathrm{Re}\ \mathrm{Pr}\ D = 17.8\ \mathrm{m} \gg 100$ mm so the flow is thermally developing throughout the entire section of interest. The similarity in the equations hints towards the large impact a high Pr fluid has on the thermal entry length of a liquid; this long thermal entry length poses a problem for our system and attempting to apply engineering calculations that, many of which, assume the pipe flow to be hydrodynamically and thermally fully developed, though this will be expanded upon later in Chapter 5. What is important to note at this point though, is that capturing this thermal development is essential for an accurate comparison and that the internal convective resistance must be evaluated with a developing-flow correlation rather than a correlation for fully-developed flow or using an analytic Nusselt number.

3.4.2 Analytical Reference Solution

To be thorough in the verification of the boundary condition code, three thermal modes were each tested individually to ward against the possibility of error cancellations masking

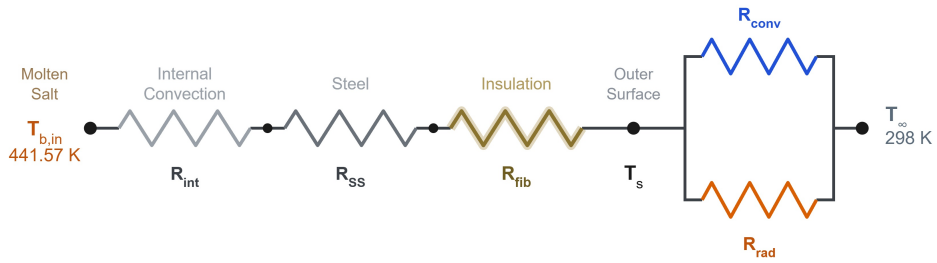

Figure 3.1: Equivalent cylindrical thermal circuit used to compute the analytical reference heat flow for Mode C (Mixed mode, $h_\infty > 0$, $\varepsilon > 0$).

mistakes in the code. The three thermal modes were:

1. Mode A: External convection only, $(h_\infty > 0,\ \varepsilon = 0)$
2. Mode B: Thermal radiation only, $(h_\infty = 0,\ \varepsilon > 0)$
3. Mode C: Combined convection and radiation $(h_\infty > 0,\ \varepsilon > 0)$

For each mode, the analytical reference heat flow is computed using a thermal circuit approach (Figure 3.1) for which the verification case was designed to best-match the assumptions a thermal circuit would use. Those assumptions are:

- **Axisymmetric Geometry:** both sections of pipe are straight, co-linear, and cylindrical.
- **Constant Centerline Temperature:** the non-adiabatic region being examined is short and the flow is fast, resulting in a negligible ΔT from the region's inlet to outlet.
- **Steady-State Flow:** with constant boundary conditions, there is no surviving d/dt after the flow develops
- **Constant Material Properties:** the simulation and circuit use constant properties.
- **Constant Environmental Conditions:** the ambient temperature T_∞, convective heat transfer coefficient h_∞ and environmental emissivity (assumed equal to 1), are set as constants.

- **No Axial Conduction:** `rcExternalTemperature` does not account for axial conduction and neither does the thermal circuit model.

With the assumptions stated, the simplest, linear case of mode A can be expressed algebraically with the followed equation:

$$Q = \frac{T_\infty - T_{b,\text{in}}}{\frac{1}{h_s\ (2\pi r_0 L)} + \frac{\ln(r_1/r_0)}{k_{ss}\ (2\pi L)} + \frac{\ln(r_2/r_1)}{k_{ins}\ (2\pi L)} + \frac{1}{h_\infty\ (2\pi r_2 L)}}. \tag{3.12}$$

To find the proper value of the internal convective heat transfer coefficient h_s, for the hydrodynamically fully developed flow but thermally undeveloped flow, we can correct the Nusselt number using S. R. Sellars, M. Tribus, and J. S. Klein's fit equation [45], reported in Lienhard and Lienhard's A Heat Transfer Textbook [20] for heat transfer from laminar flow in pipes with a thermal entry region,

$$\overline{\text{Nu}}_D = \frac{3.657}{\tanh\left(2.264\ \text{Gz}^{-1/3} + 1.7\ \text{Gz}^{-2/3}\right)} + 0.0499\ \text{Gz} \tanh\left(\text{Gz}^{-1}\right). \tag{3.13}$$

For modes B and C with the inclusion of the non-linear radiative heat transfer, the themal circuit must be solved using an iterative method, which was done using Newton's method.

3.5 Quantities of Interest

The following QoIs correspond to the experimental data reported by the experimentalists and as such are extracted from each simulation at the location detailed by the experimentalists. The geometry and instrumentation detail are shown in Figure 5.1.

- **Centerline temperatures** at six probe locations (TP1-TP6), corresponding to the experimental thermocouples sharing the same names and positioned in the center of the pipe's cross section.

- **Wall temperatures** at eleven locations (TW1-TW11), corresponding to thermocouples mounted on the external surface of the pipe walls.

- **Velocity profiles** within the field of view (FOV) of the experimental PIV window. The primary targets here are the published fluid velocity profiles at three normalized heights, $Y/H = 0.25,\ 0.50,$ and 0.75, where Y is the distance from the bottom of the PIV FOV and H is the height of the PIV FOV.

- **Mass flow rate** is computed two ways. The first way is from integrating cross-sectional mass flux from multiple, axially normal planes around the loop to ensure consistency. The second way accounts for the methodology used by Reis et al. [9], which used the grid average of the middle-plane velocity field data provided by the PIV.

3.6 Validation Error Assessment

Using the ASME standard for Verification and Validation, ASME V&V 20-2009 [40], the validation comparison error is defined as:

$$E = S - D, \tag{3.14}$$

Where S is the simulation prediction and D is the experimental measurement. The combined validation uncertainty is:

$$u_{\text{val}} = \sqrt{u_{\text{num}}^2 + u_{\text{input}}^2 + u_D^2}, \tag{3.15}$$

where u_{num} is the numerical uncertainty determined from the GCI, u_{input} is the uncertainty for the simulation input parameters, and u_D is the experimental uncertainty.

If $|E| \leq u_{\text{val}}$, the validation comparison is within the the representative uncertainty resultant from the combination of other uncertainties. If $|E| > u_{\text{val}}$, it is indicative of a modeling deficiency.

3.6.1 Error Metrics for Velocity Profiles

For single-terms error metrics, the two most commonly used are the L_2 and L_∞ norms, where the the metrics are calculated from the following equations:

$$L_2 = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(S_i - D_i)^2} \tag{3.16}$$

$$L_\infty = \max|S_i - D_i|. \tag{3.17}$$

The L_2 error norm gives a greater weight to larger errors compared to something like a simple average (also known as the L_1 norm) and the L_∞ norm only reports the maximum error. Applying these norms to the thermocouple data is fairly simple since they map 1-to-1 to the experimental system. However, it is important to note that the location of the thermocouples in space has its own uncertainty, especially for the wall thermocouples that lie outside the fluid domain, which will be discussed later in Chapter 6 along side spatial sensitivity of the centerline probes in high-temperature gradient regions of the facility. For the velocity profiles, a bit more care is needed since the experimental data is taken at discrete points in space and the simulation is essentially continuous. The key to accurately applying the error metrics for a case like this is to sample the simulated data at the equivalent points in space for the same number of entries so that the calculations can be performed appropriately.

Chapter 4

Experimental Setup

This chapter describes the TAMU natural circulation loop facility as operated by Reis, Seo, and Hassan [9], which provides the experimental data used for the validation of the models within this thesis. The experimental facility, working fluids, procedures, and test matrix are presented.

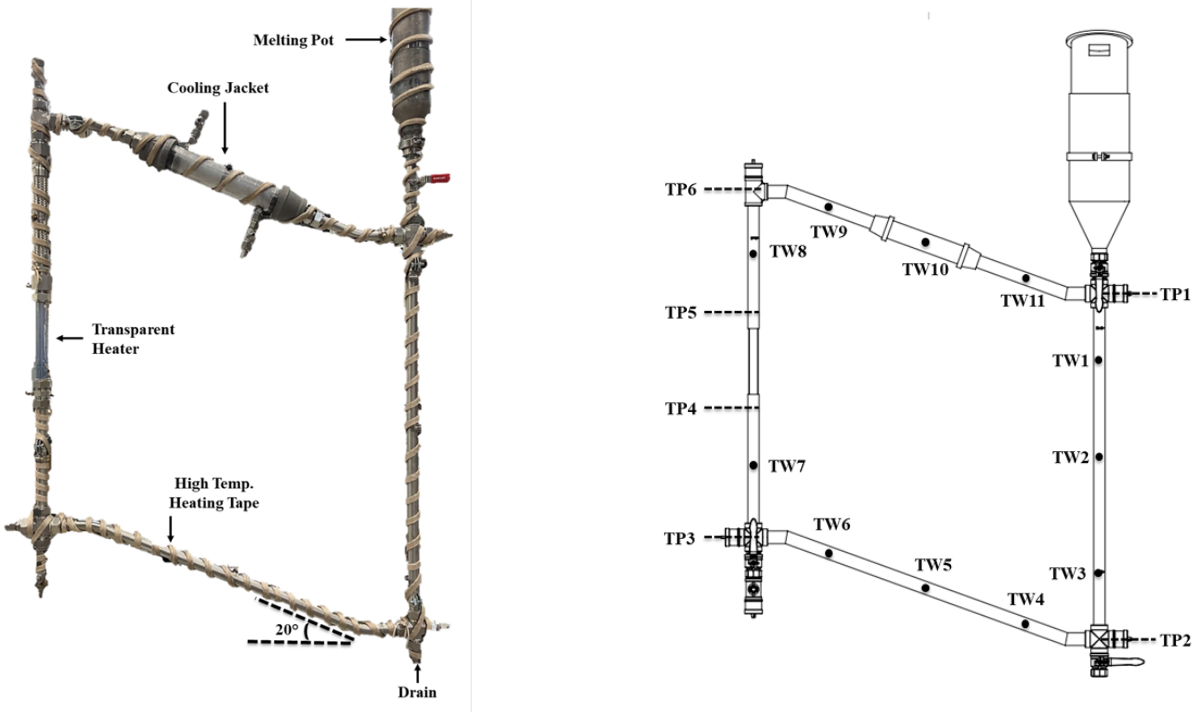

Figure 4.1: Facility (Left) and Designed Instrumentation Locations (Right) [9]

4.1 TAMU Natural Circulation Loop

Within the literature of NCL systems, loops with 4 legs are common, and referred to as "rectangular" on account of these 4-sided systems having a 90-degree rotation between each leg. The TAMU NCL is special in the sense that has a pair of legs set at an incline rather than being parallel to the ground in order to encourage a clockwise flow to develop within the system and avoid some of the instabilities that can result in an inverted flow. This inverted flow can occur when a rectangular loop is configured with its heater on the bottom leg and a cooler on its upper leg since the system is symmetric in such a case, the flow can develop in either a clockwise or counter-clockwise direction [5]. To better differentiate TAMU's inclined loop from typical rectangular loops, the facility referenced in this thesis will be referred to as a "rhombic" loop on account on it's inclination from parallel and equal-length legs. The loop operates with two working fluids, water and HITEC molten salt, and uses an identical pipe configuration for both fluids.

4.1.1 Loop Geometry

The loop's geometry consists of four legs, each connected at a 90 degree angle with cross or tee-fittings. Each leg of the loop is nominally 3 feet in length. The top and bottom legs were bent symmetrically to a nominal incline of 20 degrees with a radius of curvature of approximately 1 inch, starting roughly 4.325 inches from either end of the pipe. Each pipe leg had a nominal outer diameter (OD) of 1 inch and a nominal inner diameter (ID) of 0.93 inches and was composed of SS-316 grade stainless steel.

A brief aside is worth mentioning before continuing with the description of the loop's geometry; the reported dimensions are mixed between imperial and metrics units, which is left as reported by the experimentalists instead of all being converted in an effort to avoid miscommunications and conflicted reports that arose in the conduction of the work within this thesis.

The four legs are:

- **Lower leg (heater):** this leg is wrapped with 120V AC, resistance heaters (McMaster: 4550T123) capable of outputting 312 W each that served as the primary heat input for the system. Three of these heaters were connected in series to wrap the whole of the bottom leg with a pitch between wraps of 1/2" to 1" apart.

- **Left leg (riser):** most notably, this leg contains the primary focus of the experimental configuration, a quartz glass test section measuring 186 mm of exposed length. The test section had a nominal ID of 21 mm and OD of 1". The quartz test section was fitted into the middle of the left leg after the 3' pipe was cut into two segments that would allow the test section to be installed such that it was centered along the leg's length and that the total length of the leg was equal in length to the other three legs of the loop. The quartz test section's transparency is what allows PIV measurements to be taken for the experiment. It is worth noting that the ID of 21 mm is approximately 0.83 inches, which represents a step reduction in ID of 0.10 inches and an approximate 21% reduction in cross sectional area.

- **Upper leg (cooler):** contains a 1 foot long cooling jacket that serves as a concentric, counter-flow air-cooled heat exchanger with reducer fittings on either side to mount the cooling jacket in the center of the upper leg. This heat exchanger is intended as the primary vector of heat removal from the system as the counterpart of the lower leg's heating tape in order to create the temperature difference that drives the system's natural circulation. The upper leg is nominally bent in the same way at the lower leg, but it should be noted that the fittings the cooling jacket is mounted to are essentially through-wall pipe fittings, resulting in the upper leg being segmented and connected by the in-line pipe fittings that double as the heat exchanger's mounting fixture.

- **Right leg (downcomer):** A standard, vertical pipe connecting the lower right junction and upper right junctions together without any additional fittings, segmentation,

or heaters during operation.

As previously mentioned, the loop's four legs are connected to each other by four fittings. Three of the four junctions are cross-fittings that connect the legs of the loop. These cross-fittings are either capped off on their unused ports or have pipe extensions that serve some other experimental purpose; the top right junction connects to the melting pot for inputting the working fluid, the lower right junction's lower port connects to the drain tank for emptying the loop, and the lower left junction's bottom port was fitted with a bubble injection valve for experiments outside the scope of this thesis. The remaining junction in in the upper left corner was a tee-junction whose upper port served as an off-gas vent for the loop for the aforementioned bubble injection experiments. Each of these junctions have an expanded outer diameter to the accommodate the standard pipe, with an inner diameter of 25.4 mm (1 in) for their ports and an for the interior region of the junction, each fitting had an inner diameter of 22.225 mm (7/8 in) for each of the intersecting cylindrical regions of the junction.

4.1.2 Thermal Insulation

For each experiment, the loop is insulated with ceramic fiber insulation sourced from McMaster-Carr (code 93315K18) to reduce passive heat losses to the environment. This insulation was configured differently between the salt and water experiments due to the different operating temperatures, with extra insulation applied for the salt tests until the surface temperature of the insulation was deemed safe. The exact configuration for both the water and salt tests has been lost and is a large source of uncertainty that is difficult to quantify. More information on how this uncertainty was handled can be found in Chapter 5. The test section (quartz glass) is uninsulated to maintain optical access for the PIV measurements, which results in the second largest source of heat loss after the heat exchanger and is heated in the higher-temperature molten salt cases to counteract these losses.

4.2 Instrumentation

The following section outlines the instrumentation that was used on the loop to collect the data for this V&V study.

4.2.1 Temperature Measurements

The thermocouples that were used were K-type thermocouples that had a reported uncertainty of 1.1 °C for individual measurements, as reported by the manufacturer, and an uncertainty of 1.56 °C for differences in temperature as calculated by the experimentalists and reported in Reis et al. [9].

- **Fluid probes (TP1-TP6):** The centerline temperature probes of the "TPX" series were nominally centered within the pipe's cross section at the reported locations in each of the four corner junctions of the loop plus two more in the fittings that mounted the top and bottom of the test section to the rest of the loop.
- **Wall thermocouples (TW1-TW11):** The wall temperature probes of the "TWX" series were placed on the outside of the loop approximately equidistant from each other along each of the four legs. The exact axial and azimuthal locations have been lost since the conduction of the experiment, so the comparisons conducted in this study are for their nominal axial locations and record data at each of the cardinal directions at the nominal axial point in space. Notably, TW10 is an exception to the trend of the other wall thermocouples, as it was mounted on the exterior of the heat exchanger jacket rather than the primary loop's pipe wall.

4.2.2 Velocity Measurements

The velocity measurements used for this study and reported in Reis et al. [9] were conducted using Particle Image Velocimetry. This non-intrusive method of measuring the flow used

tracer particles that were introduced to the flow and were visualized using high-speed imaging to capture the instantaneous positions of the tracer particles within a 1 mm, radially centered laser sheet to record the velocity profile passing through the transparent test section.

Two different cameras were used to record theses velocity fields, one targeting the full-field velocity and another targeting the near-wall velocity field. The full-field PIV camera recorded 6000 images at 250 frames per second and the near-wall PIV camera recorded 8000 images at 1000 frames per second for datasets of roughly 24 and 8 seconds respectively, assuming all images were used.

4.2.3 Cooling System Measurements

The cooling system seen on the top leg of the loop is a concentric counter-flow heat exchanger, using room temperature air as the coolant for the experimental facility. The heat exchanger ran at a constant volumetric flow rate of 188 L/min for the water cases and 37 L/min for the salt cases. The heat exchanger jacket was 1 foot long and used two reducer fittings (McMaster: 44705K661) on either side of the jacket to mount the cooling jacket onto the loop.

To determine the heat removed by the heat exchanger, temperature measurements were taken by the same K-type thermocouples at the coolant's inlet and outlet and then Reis et al. determined the energy loss from the heat exchanger by evaluating the properties of air at the average temperature of the inlet and outlet and using the equation

$$Q = \dot{m} C_p (T_{out} - T_{in}). \tag{4.1}$$

4.3 Working Fluids

4.3.1 HITEC Molten Salt

Reis et al. [9] used thermophysical properties as reported by Idaho National Lab's (INL) molten salt database, but a review of the literature for this work found fault in the reported equations for both the dynamic viscosity and the specific heat capacity of HITEC's ternary eutectic mixture. INL itself cites Janz et al. citejanz1981 from 1981 for each of its equations describing HITEC salt. Janz et al. itself is another repository of equations reported by other works and cites Kirst et al. [46] from 1940 for the table of viscosity data it fits and cites Voskresenskay et al. [47] for the equations used to describe the heat capacity of the molten salt mixture, who used drop-calorimetry to collect their data. Neither of the compositions reported in section 148.10 of Janz et al. are for the eutectic ternary mixture and the reported equation's uncertainty of 5% still far overshoots modern measurements and the manufacturer-reported values by nearly 50% for the temperatures studied in this work (Equation 4.4 produces a value of $c_p \approx 2301$ J/kg K vs the modern $c_p \approx 1423.47$ J/kg K at 200 $°C$). Drop-calorimetry was used to collect their data for heat capacity.

The equation of state reported for dynamic viscosity is a polynomial fit of an exponential equation that claims a greater range of validity than the original equation and data the fit was made to capture from Kirst et al. [46]

Reis et al. used the equations:

$$\rho = 2293.6 - 0.7497\ T \tag{4.2}$$

$$\mu = 0.4737 - 2.297 \times 10^{-3}\ T + 3.731 \times 10^{-6}\ T^2 - 2.019 \times 10^{-9}\ T^3 \tag{4.3}$$

$$c_p = 5806 - 10.833\ T + 7.2413 \times 10^{-3}\ T^2 \tag{4.4}$$

$$k = 0.78 - 1.25 \times 10^{-3}\ T + 1.6 \times 10^{-6}\ T^2 \tag{4.5}$$

In the publication, the equations for viscosity and specific heat are rounded for brevity.

Those roundings produce appreciably different values when used, but evaluating the full and abbreviated equations reveals that Reis et al. did in fact use the full, non-rounded equations for their analysis. While the scripts that generated the values reported in Reis et al. [9] could not be recovered, the determination was made by attempting to find a way to calculate the evaluation temperature used for the reported Prandtl number, Pr that classified each of the 4 experimental salt cases. The best method found was a simple average of the measured values of each TP probe (recorded in Celsius), before converting to Kelvin by adding 273 (as opposed to 273.15) for the evaluation. This method reproduced the values of Pr exactly to reported precision for salt tests 1 and 4, and were 0.01 higher than the published values for salt test 2 and 3. The rounded versions of the equations published in Reis et al. could not provide a simultaneous agreement across each of the 4 salt test cases superior to the full equations using any single TP reading, difference between TP readings, or using TW measurements in any mixed configuration either, so it was determined with confidence that the full equations above were used. Attempts at recreating the calculations behind published results to ensure that the same methodology was being followed within the present work in the absence of recoverable scripts are present at several points in this work; this one in particular hints that some of the published calculations were done by hand. The evidence for this comes up in several places: spreadsheets provided by the experimentalists having only numeric values without any formulas, the apparent use of +273 as the conversion factor from Celsius to Kelvin instead of +273.15 is the kind of truncation more likely done by hand, and the calculated values of Pr for salt test 2 and 3 being off by only 0.01 hints at some difference in rounding done during the calculation, but lacking direct confirmation, cannot be confirmed.

Table 4.1: Prandtl Number Reconstruction Comparisons

Evaluation method	Salt 1	Salt 2	Salt 3	Salt 4
Reported Value	54.29	50.81	41.65	31.74
Full Equations, $[\mathrm{TP}_{\mathrm{avg}}] + 273$	54.29	50.82	41.66	31.74
Full Equations, $[\mathrm{TP}_{\mathrm{avg}}] + 273.15$	54.16	50.69	41.55	31.66

While not an inherent flaw in the experimental results, it does introduce an additional source of uncertainty for modeling efforts such as this one and attempting to differentiate between methodology differences, rounding errors, or other unknowns when trying to recreate reporting methodologies exactly can be difficult.

The equations reported by Janz et al. for density and thermal conductivity do not have any clear, modern superiors so have been retained for the present analysis. Curiously, Sohal et al. states there are no correlations for the thermal conductivity of HITEC despite it being listed alongside the other properties and equations reported in Janz et al. [26]. Reis uses a correlation reported by Santini et al. 1984, which is also used for this work [48].

The modern suite of thermophysical properties chosen for this study is outlined in Chapter 5, but for comparison's sake, are evaluated below.

Key properties at the nominal operating temperature (200 °C, 493.15 K):

- Density: $\rho \approx 1938.88\,\mathrm{kg/m^3}$
- Dynamic viscosity: $\mu \approx 6.77 \times 10^{-3}\,\mathrm{Pa\,s}$
- Specific heat: $c_p \approx 1423.47\,\mathrm{J/(kg\,K)}$
- Thermal conductivity: $k \approx 0.547\,\mathrm{W/(m\,K)}$

4.3.2 Water

The properties used for water are from the IAPWS [27] and use high-order fits within Region 1 of their classification structure, where Region 1 is inclusive of liquid water at below its saturation temperature for pressures from 0 to 100 MPa. The exact coefficients used can be found in Appendix B.2

4.4 Experimental Test Matrix

Below are some relevant, tabulated values relating to the experimental provided by Reis et al. For the most complete detailing of the experiments and its results, reading of Reis et al. [9] is recommended.

Table 4.2: HITEC salt experimental test cases from Reis et al. [9].

Case	Q_{heater} [W]	Q_{cooler} [W]	Re_D	Pr	Ra_L	ΔT [°C]
Salt 1	232.3	55.58	74.86	54.29	4.27×10^{10}	11.16
Salt 2	265.7	56.34	85.19	50.81	4.04×10^{10}	10.09
Salt 3	297.5	60.55	107.37	41.65	3.66×10^{10}	7.95
Salt 4	337.6	65.98	158.92	31.74	3.12×10^{10}	5.63

Table 4.3: Water experimental test cases from Reis et al. [9].

Case	Q_{heater} [W]	Q_{cooler} [W]	Re_D	Pr	Ra_L	ΔT [°C]
Water 1	58.8	26.86	711.78	4.23	2.07×10^{10}	1.29
Water 2	77.6	35.63	953.19	3.78	2.51×10^{10}	1.43
Water 3	93.0	43.02	1228	3.5	2.88×10^{10}	1.54
Water 4	129.0	58.06	1806.52	2.96	3.54×10^{10}	1.67

4.5 Experimental Uncertainties

4.5.1 Energy Balance

From Reis et al.'s Equation 25, we can reproduce the full matrix's range of uncertainty for the net heat into the system within roughly 1% of the reported values for the maximally heated cases. An uncertainty of 20% was assumed for the heat loss calibration

$$\sigma_{Q_{net}} = Q_{net}\sqrt{\left(\frac{\sigma_{Q_h}}{Q_h}\right)^2 + \left(\frac{\sigma_{Q_{loss}}}{Q_{loss}}\right)^2} \tag{4.6}$$

$$Q_{net} = \dot{m}C_p(TP6 - TP2) \tag{4.7}$$

$$Q_{loss} = Q_h - Q_{net} \tag{4.8}$$

Table 4.4: Experimental Energy Balance from Experiments of Reis et al. [9].

Quantity [W]	Salt 1	Salt 2	Salt 3	Salt 4	Water 1	Water 2	Water 3	Water 4
ΔT [K]	11.16	10.09	7.95	5.63	1.29	1.43	1.54	1.67
Q_h	232.3	265.7	297.5	337.6	58.8	77.6	93	129
σ_{Q_h}	2.16	2.47	2.77	3.14	0.59	0.78	0.93	1.29
Q_{loss}	-120.54	-140.25	-28.66	80.05	13.96	17.67	15.50	23.06
$\sigma_{Q_{loss}}$	24.11	28.05	5.73	16.01	2.79	3.53	3.10	4.61
Q_{net}	352.84	405.95	326.16	257.55	44.84	59.93	77.50	105.94
$\sigma_{Q_{net}}$	70.6	81.3	65.3	51.6	9.0	12.0	15.5	21.2

From Table 4.4, two main items reveal themselves. First, when applied to the whole test matrix instead of just the highest power cases for both working fluids (as done in [9]), Equation 4.7 produces values exceeding the total heater power applied to the system. Second, the equation used to determine the uncertainty for the net heat does not factor in either the individual thermocouple uncertainty or their own reported uncertainty for temperature differences of 1.56 K [9]. Comparing against the values for ΔT in Table 4.4, this unpropagated uncertainty more than doubles the uncertainty for water cases 1, 2, and 3. This demonstrates that the need for better energy-balance understanding mentioned in Chapter 1 is even greater than previously thought.

4.5.2 Bulk Velocity

One of the largest difficulties related to the experimental data set used for this work's modeling is that of the bulk fluid velocity for each of the experimental cases. The reported bulk velocities from [25] and their associate mass flow rate are tabulated below.

The only published mention of how U_∞ was calculated is from [9] and states, "The bulk fluid velocity was found from an area-weighted average of the full-field velocity". This much is true, but more complicated than it seems. After much trial-and-error from both the author

Table 4.5: Bulk Velocity and Mass Flow Rate from Experiments of Reis et al. [9].

Quantity	Salt 1	Salt 2	Salt 3	Salt 4	Water 1	Water 2	Water 3	Water 4
U_∞ [m/s]	0.0185	0.0196	0.0205	0.0237	0.0190	0.0230	0.0277	0.0350
$\dot{m}$ [kg/s]	0.0158	0.0168	0.0175	0.0201	0.0083	0.0100	0.0120	0.0151

and the experimentalists, it was determined that the method used to produce the values for U_∞ seen in Table 4.5 was

$$U_{PIV} = \frac{2}{R^2} \int_0^R U(r)\ r\ dr, \tag{4.9}$$

$$U_\infty = U_{PIV} \left(\frac{D_{test}}{D_{pipe}} \right)^2. \tag{4.10}$$

For these calculations, the velocity data captured in the full-field PIV data was assumed to be symmetrical for each case, then the right half of the data was used for $U(r)$, where $r = 0$ is the centerline and $R = D_{test}/2$ is the right wall of the test section. D_{test} is the inner diameter of the test section (0.83") and D_{pipe} is the inner diameter of the stainless steel pipe used in the majority of the loop (0.93"). No post-processing scripts could be found by the experimentalists to confirm this exact implementation, but it reproduces the tabulated values for U_∞ within 1.3 % for the available data.

The "available data" is specified apart from all test cases because the full-field velocity data for each water case had been lost, deleted, or for some other reason was unrecoverable despite extensive efforts by the experimentalists to locate and provide that data. Thankfully, the full-field velocity data for the salt cases was recovered and subsequently provided, so we are able to test reconstruction methodologies for QoIs that require the full-field velocity data to bound the uncertainty attached to any given reconstruction. For instance, we are able to attempt an extraction of the full-field velocity data for each of the water cases based on Figure 14 from Reis et al. [9].

4.5.3 Velocity Data Extraction

Rather than being satisfied with only the full-field salt data and giving a best-guess at the performance of the simulated water cases, the extraction of three distinct velocity profile heights of $Y/H = 0.25$, 0.50, and 0.75 in Figure 14 of Reis et al. [9] was inspired by the experience of performing image pixel-measurements for the modeling effort described in Chapter 5.

Source Figures and Axis Calibration

First, Figure 12 and 14 from Reis et al. [9] were zoomed in on as much as possible to fully capture one of the four subfigures for each figure in-frame. From there, the "source" image is turned into a "calibration" image to ensure that the axes and origin for the figure are set correctly along side the alignment of vertical and horizontal gridlines with the major tickmarks along the x and y axes.

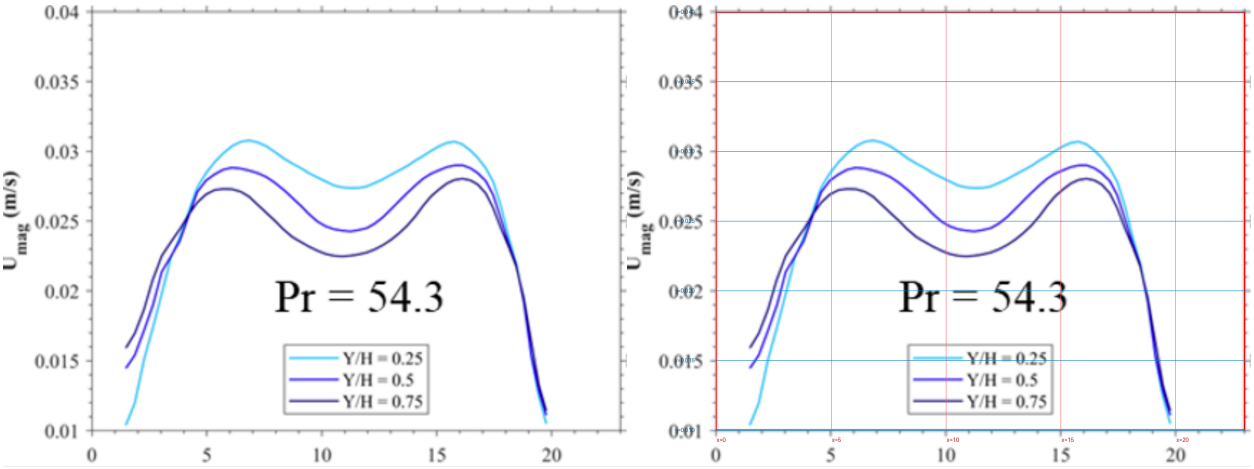

Figure 4.2: Source Image (left) and Calibration Image (right)

Curve Extraction

From there, the source image is used again with the calibrated coordinate frame and the addition of two "masking" frames to allow for the clean extraction of the line data without

pollution from the legend or Prandtl number annotation. With those bounds set, they can be safely ignored. Once the undesirable annotations are ignored, the three profiles can begin to be separated. Each non-white pixel is assigned to one of three brightness classes. Once each non-white pixel is assigned to one of the three groups, a median filter is applied. This filter looks at each column of pixels, and assigns a representative pixel coordinate for each of the three brightness groups by taking the median verical location of pixels in each group for that collumn. For places where curves cross or overlap, Gaussian smoothing is applied to improve the resultant profile.

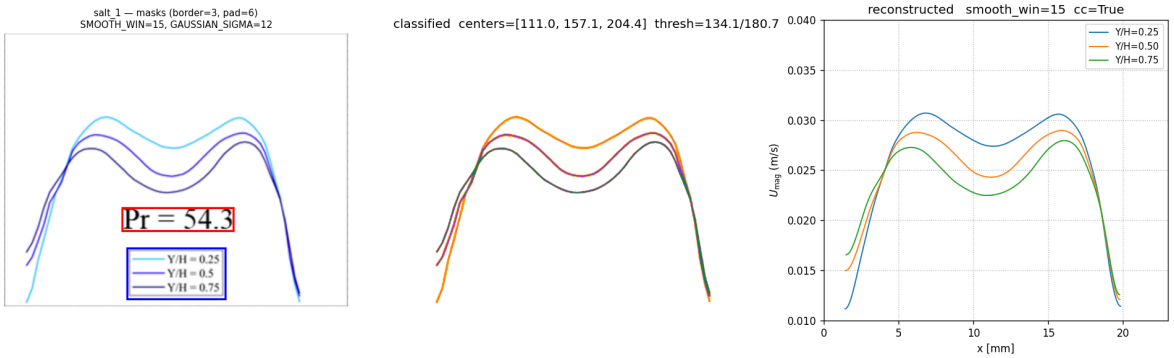

Figure 4.3: Masking and Extraction Images for Salt Test 1

To verify that the digitization is serviceable, the saved array of data is then overlaid back upon the source image using the calibrated coordinate frame to provide visual confirmation.

Error Quantification Against Salt Data

Since the salt tests have a raw full-field velocity dataset, the accuracy of the entire digitization process can be bounded for factoring in to the uncertainty analysis of the water full-field velocity data comparisons. For each salt test and height, the raw data was sliced at the corresponding Y/H row and the digitized curve was restricted to the overlapping horizontal window. The comparison involves roughly 80 data points across the board for each height comparison. L_2, bulk excludes an additional 1.25 mm from either side of the overlap window to provide a frame of reference for the accuracy of the digitization away from either tail of

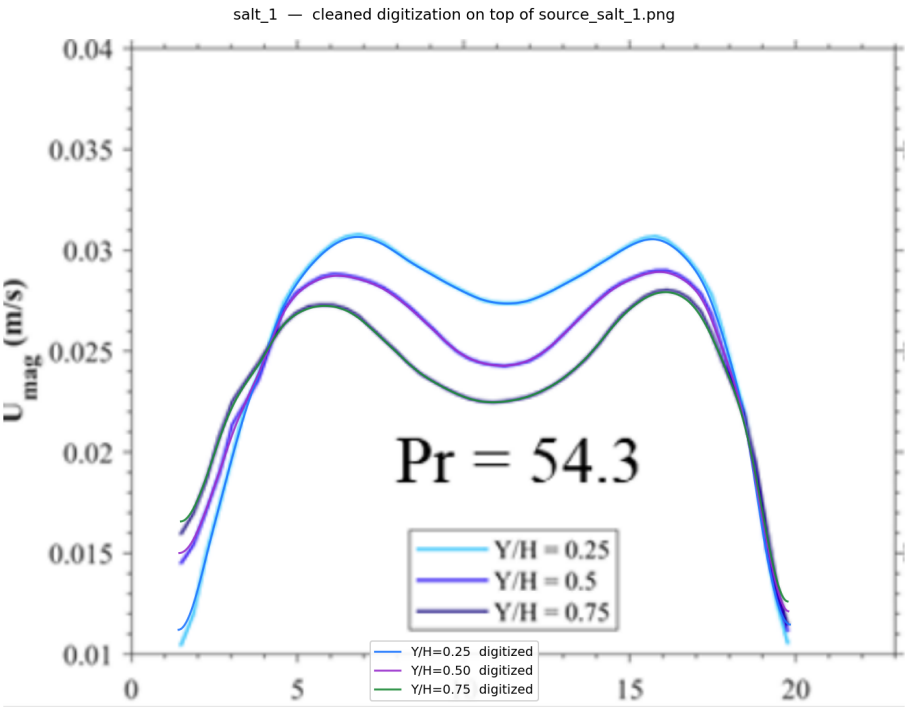

Figure 4.4: Digitized Velocity Profile of Salt Test 1 Overlaid on Source Image

the profile, as the ends of the profiles have the greatest error associated with them.

From the results in Table 4.6, the digitization uncertainty assigned to the water velocity profiles is rounded up from the tabulated L_2 percent error of 1.36% to the assigned value of

$$u_{dig} = 1.5\% \tag{4.11}$$

and is intended to compound upon the experimentally reported PIV uncertainty for each water case for calculations dependent on the full-field velocity data, such as the calculated value for mass flow rate in each experiment. This will be further discussed in Chapter 6

4.5.4 Data Uncertainty Summary

To conclude this chapter, we can summarize each contribution to u_D in the table below. The entry u_{PIV} is the uncertainty prescribed to each of the full-field PIV datasets reported by Reis et al. in [9]. The comparisons performed in Chapter 6 are mapped to each experimental

Table 4.6: Digitized VS Raw Residuals for Salt Tests

Test	Y/H	L_2 [10^{-4} m/s]	L_2 [%]	L_2, bulk [%]	L_∞ [%]
Salt 1	0.25	1.44	0.55	0.27	2.09
	0.50	3.17	1.26	0.46	6.83
	0.75	1.92	0.81	0.36	4.73
Salt 2	0.25	1.59	0.57	0.38	2.20
	0.50	2.84	1.06	0.89	4.20
	0.75	1.55	0.60	0.31	3.50
Salt 3	0.25	3.33	1.13	0.31	6.96
	0.50	10.39	3.65	0.85	14.10
	0.75	2.60	0.95	0.66	3.48
Salt 4	0.25	2.33	0.72	0.39	2.96
	0.50	2.78	0.89	0.24	4.07
	0.75	2.04	0.67	0.30	2.45
All Tests	All Heights	3.79	1.36	0.50	14.10

thermocouple, so Reis's ΔT uncertainty of 1.56 K is not included, favored instead by the 1.1 K uncertainty for each thermocouple (referred to as u_{tc} below). The row for u_{meth} is the RMS of the methodology using Equations 4.9 and 4.10, rounded up from 0.85% to 0.9% and still below the conservative upper bound of 1.3%. Then, u_{dig} is the uncertainty associated with the full-field PIV data for the water cases resultant from the digitization effort; which has a value of zero for each salt case since that data did not need to be extracted.

Table 4.7: Contributions to Data Uncertainty (u_D)

Quantity	Salt 1	Salt 2	Salt 3	Salt 4	Water 1	Water 2	Water 3	Water 4
u_{PIV} [%]	2.33	2.17	2.06	1.91	2.11	1.80	1.60	1.38
u_{tc} [K]	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
u_{meth} [%]	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
u_{dig} [%]	0	0	0	0	1.5	1.5	1.5	1.5

Chapter 5

Computational Modeling

This chapter describes the two computational models developed for this thesis: a first-order energy balance model for closing the experimental energy budget, and the three-dimensional OpenFOAM CFD simulation of the full loop.

5.1 First-Order Energy Balance Model

5.1.1 Purpose and Approach

As seen in Chapter 4, there is considerable uncertainty with respect to the global energy balance of the experimental facility. Lacking information such as detailed insulation configurations, external thermal imaging, or insulation temperature measurements, the first step that needed to be taken in order to model the system was to improve our understanding of the energy flow into and out of the system. Before tackling that though, it was deemed prudent to ensure the geometry of the system was well understood as while the publication lists some helpful dimensions, they are insufficient for simulated reproduction of the experimental results. The exact construct of the facility was not available; neither a computer model, detailed dimensions, nor Bill of Materials could be provided. Furthermore, the instrumentation diagram did not align with as-constructed images of the experimental facility.

The overlay seen in Figure 5.1 was performed by importing both images into Blender 5.1.2, setting both image origins to the center of the lower left fitting, and scaling the image of the instrumentation diagram until both left legs were overlaid while ensuring the aspect ratios of both images was kept throughout the process.

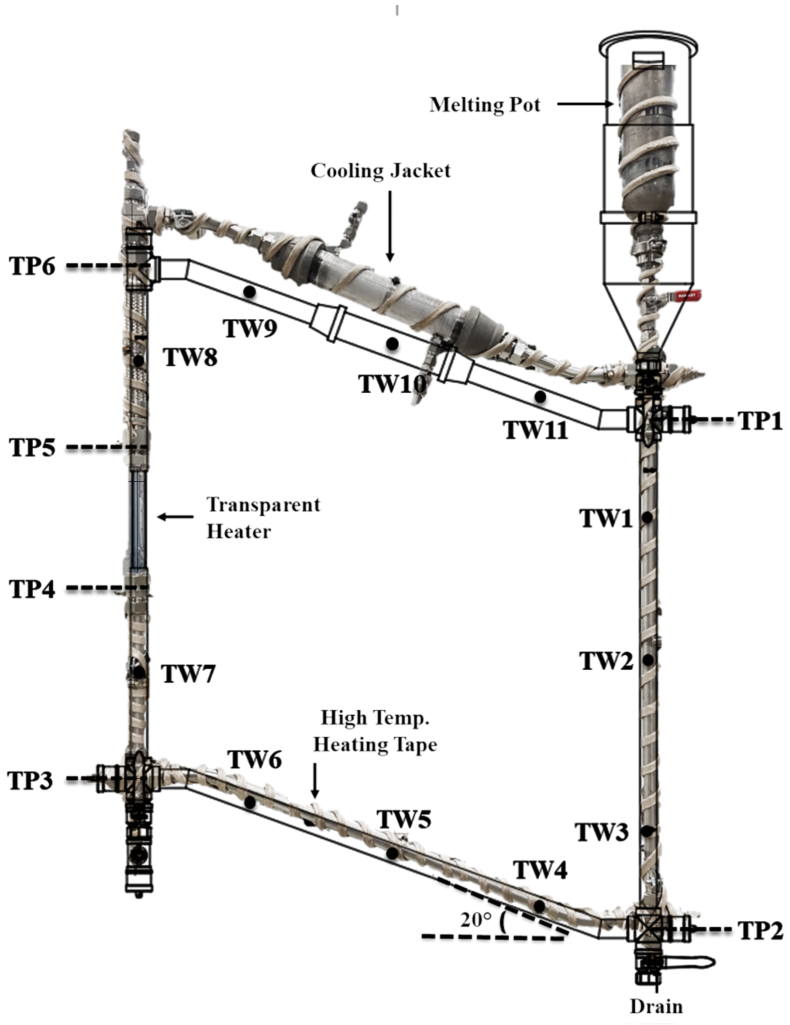

Figure 5.1: Overlaid Instrumentation Diagram with Constructed Facility

While a full Bill of Materials was not available for how the loop was constructed, collabo-

ration with the experimentalists help determine the exact fittings purchased from McMaster-Carr for the corners of the loop and the reducers used to mount the heat exchanger to the facility. Using this information, a CAD model was reverse-engineered from compatible parts listed on McMaster-Carr.

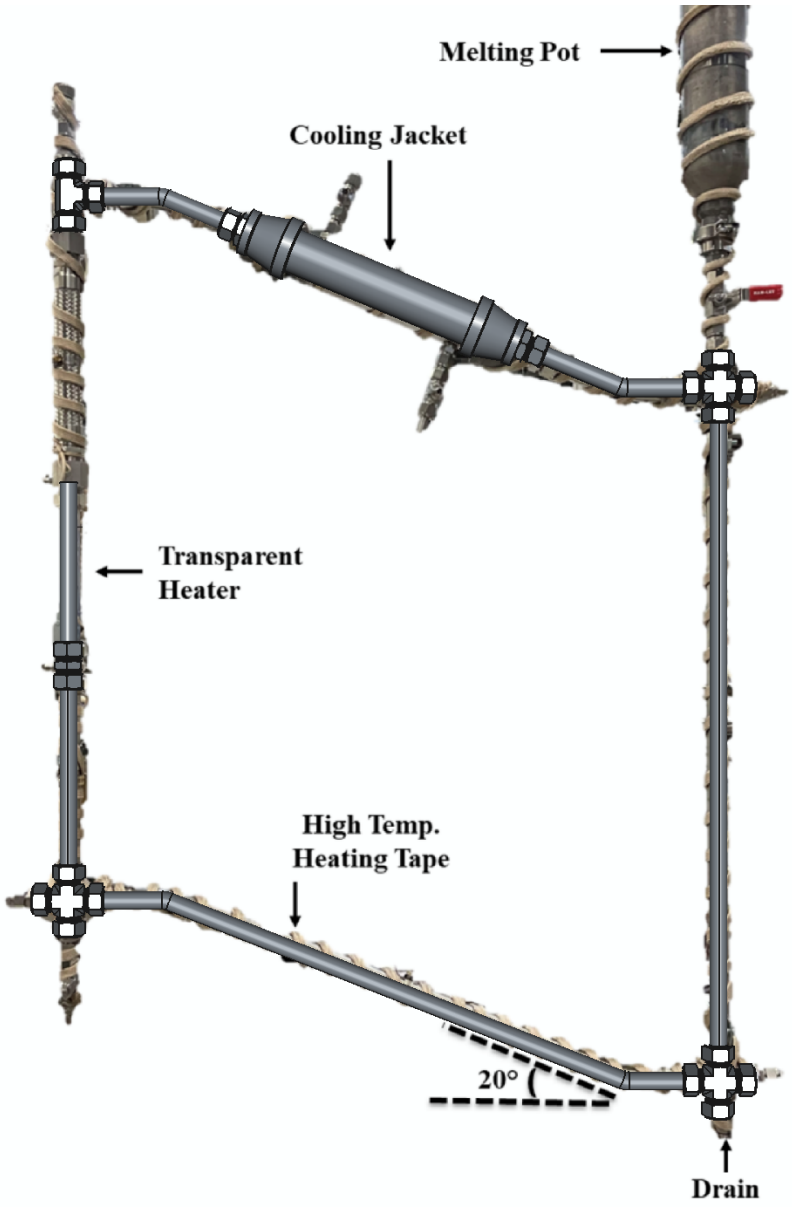

Figure 5.2: Reverse Engineered CAD Overlaid on Image of Experimental Facility.

From this effort to better understand the geometry of the facility before continuing to contingent studies such as the first order model and CFD simulation, a few items are worth

noting. First and most obviously from Figure 5.2, a high-confidence part for the upper fitting of the test section and the upper segment of piping for the left leg could not be determined, and is henceforth modeled assuming the same fitting used on the lower test section for the undetermined upper fitting and that the left leg's upper segment has the same cross section and material as the standard piping used in the rest of the loop for the remaining length between the test section's upper fitting and the top left tee-junction of the facility. Second, the nominal incline of the loop was better matched in the CAD reconstruction using an incline of 21.5 degrees for the lower leg. Third, enforcing the simultaneous constraints of the left leg's total length being equal to the right leg's total length and having the upper leg use the same incline as the lower leg produced poor results in trying to match the upper left region of the CAD with primary published image of the experimental facility. Follow-on correspondence with the experimentalists confirmed that, based on the distance from the top face of the bottom left fitting to the bottom face of the top left fitting, the left leg of the loop is approximately 1.35 cm longer than the right leg. Applying this information to the CAD produced the result seen in Figure 5.2

To ensure that the 3D CAD's apparent matching to the image of the experimental facility was not an artifact of poor image scaling, the image was scaled into physical dimensions based on pixel measurements between known points, primarily relying on the distance from the center-faces of the right-side fittings. This primary reference was chosen based on having the exact fittings known, combined with the comparably unremarkable right leg, nominally being a stock, unbent, 3 ft pipe that was assumed to be fully mated within both fittings. It is also worth mentioning that the pairs of fittings on both the left and right sides are not pixel-perfectly aligned vertically, hinting at either slight bending in the straight legs, lens distortion within the original image, or a combination of both, but this misalignment was no greater than a single-digit number of pixels for either side and is neglected.

From here, 39 segments were identified and subsequently implemented to the first order model based on their geometry, layer configuration, material, and proximity to instrumen-

tation. The first three criteria are obvious reasons for segmentation, but the divisions by proximity to instrumentation warrant explaination. This was done because even if the geometry was similar to an adjacent section of piping, a division would be made so that the new segment would use the experimental data for that region of the model closest to the experimentally collected value. For example, the right leg of the loop is self-similar along it's axial length, but is divided into three equal parts, each centered on TW1, TW2, and TW3 in an attempt to model subtle differences in losses from different lengths of pipe. For each segment, a one-dimensional radial conduction problem is solved through the composite wall layers (steel, insulation, heater tape, quartz) with the measured surface temperature from the experimental probes serve as the inner boundary condition and the external boundary condition is a mix of natural convection and thermal radiation heat transfer.

5.1.2 Radial Conduction Solver

For each segment, the radial conduction solver accepts a `LoopSegment` object defined with a unique name, segment length, and inner radius, followed by an arbitrary amount of `RadialLayer` objects. Those `RadialLayer` objects are defined by a material, thickness, and power for segments such as the primary heater on the lower leg and test section that are electrically heated during operation.

`LoopSegment` objects themselves are purely annular models, lacking an internal fluid volume and associated internal convective heat transfer coefficient. This approach was chosen for two main reasons, those being: first, to avoid using standard correlations of fully or partially developed laminar flow while prilimiary calculations showed that the both the hydrodynamic and thermal entry lengths of both fluids was near or exceeding the total path length around the loop, in some cases exceeding by several times the loop's perimeter, as mentioned in Chapter 2. Secondly, most of the temperature data that was collected, and especially for typical cylindrical lengths of pipes, was collected using thermocouples on the external walls of the loop. This combined with the 0.035" pipe walls of stainless steel led to

the model's approximation that the internal surface temperature was equal to the external surface temperature of the stainless steel recorded by the TW thermocouple series. A caveat to that is the first order model *does* still include the annular contribution of the stainless steel pipe wall, effectively double counting it's contribution of thermal resistance to the model, but this c ontribution is negligible; retained by the model in favor of continued, accurate accounting of each segment's solid domain throughout the simulation pipeline, particularly for ensuring the final outer radius is modeled faithfully to model the intended external surface area of the surrounding insulation.

From there, the heat flow for each segment was solved using a tridiagonal matrix implemented with:

- **Inner boundary:** Fixed temperature from thermocouple reading.
- **Outer boundary:** Natural convection using the Churchill-Chu correlation [49] blended for pipe orientation (horizontal to vertical), plus Stefan-Boltzmann radiation:

$$\overline{\mathrm{Nu}}_L = \left\{ 0.825 + 0.387 \left[\frac{\mathrm{Ra}_L}{\left[1 + (0.492/\mathrm{Pr})^{9/16}\right]^{16/9}} \right]^{1/6} \right\}^2 \tag{5.1}$$

$$\overline{\mathrm{Nu}}_D = \left\{ 0.60 + 0.387 \left[\frac{\mathrm{Ra}_D}{\left[1 + (0.559/\mathrm{Pr})^{9/16}\right]^{16/9}} \right]^{1/6} \right\}^2 \tag{5.2}$$

$$h_{\mathrm{ext}} = \frac{\overline{\mathrm{Nu}}_{eff}\ k}{D} \tag{5.3}$$

$$q_{\mathrm{outer}} = h_{\mathrm{ext}}(T_s - T_\infty) + \varepsilon\sigma(T_s^4 - T_\infty^4), \tag{5.4}$$

The two Nusselt numbers were blended using a weight, w, determined by the segment's orientation with respect to the ground. This weight is calculated by $w = (\sin\theta)^p$, where p is effectively the strength of the blending such that larger values of p result in larger bias towards the horizontal equation for inclined sections while still allowing for vertical

segments to use only the vertical Nusselt number equation. This work uses $p = 2$.

$$h_{eff} = (1 - w)\ h_{horizontal} + w\ h_{vertical} \tag{5.5}$$

where ε is the surface emissivity and σ is the Stefan-Boltzmann constant.

- **Material properties:** Temperature-dependent, evaluated at cell temperature via polynomial fits. Conductivities at cell faces use the harmonic mean: $k_{\text{face}} = 2k_L k_R/(k_L + k_R)$.

The outer surface temperature and heat flux are coupled through the external boundary condition, requiring an iterative solution. Under-relaxation (factor 0.5) is applied, with convergence declared when both the surface temperature change is below 10^{-3} K and the heat flux change is below 10^{-2} W/m^2, which is approximately 10^{-3}% iterative error for both the temperature and heat flux.

5.1.3 Cooler Heat Removal

The cooling power is computed using an energy balance on the air side:

$$Q_{\text{cooler}} = \dot{m}_{\text{air}} \left[c_p(T_{\text{out}}) T_{\text{out}} - c_p(T_{\text{in}}) T_{\text{in}} \right], \tag{5.6}$$

where $c_p(T)$ is evaluated separately at inlet and outlet temperatures using polynomial fits for air. This enthalpy balance approach is superior to the average-temperature method ($Q = \dot{m}\bar{c}_p \Delta T$) used in the original experimental analysis, as it correctly accounts for the nonlinearity of $c_p(T)$ over the large air temperature rise (e.g., 25 °C to 113 °C for salt test 1.)

5.1.4 Composite Fin Model

At each of the corners around the loop, there are structural components that are necessary to account for in order to close the energy balance of the system. These components are: three caps for unconnected legs of cross junctions, the drain valve, the melting pot valve, the bubble injection valve used for other experiments, and the off-gas valve. No temperature data exsists for any of these regions, so the first order model approximates each of them as composite, cylindrical fins with a fixed-temperature base. Using this approach, the solid "fins" away from the system account for between 9-16% of the energy leaving the system.

The seven cylindrical composite fins were modeled using a two-dimensional axisymmetric solver:

- **Geometry:** Stainless steel core pipe surrounded by insulation sheath of the same thickness as the insulation used on the rest of the loop, with an isothermal base held at the temperature of the nearest thermocouple reading.
- **Physics:** Radial and axial conduction, side-surface natural convection using the same cylindrical correlations as above plus radiation, and using horizontal/vertical plate correlation for the far-end surface depending on fin orientation.
- **Discretization:** Adaptive radial/axial grid (target cell size 0.8 mm, minimum 25 cells per direction).
- **Solver:** Sparse matrix with linearized radiation, up to 400 iterations, using the same tolerances as the radial conduction solver detailed above.

The total fin heat loss summed over all seven fins is included in the energy balance closure. The solid-core modeling overestimates the heat transfer for non-solid fins, such as the off-gas and drain tank valves, but lacking a full-fidelity geometry and additional data in those regions, this compomise has been made.

5.1.5 Energy Balance Results

For each test case, the energy balance takes the form:

$$Q_{\text{heater}} + Q_{\text{test section}} - Q_{\text{cooler}} - Q_{\text{passive}} - Q_{\text{fins}} = Q_{\text{unaccounted}}, \tag{5.7}$$

where Q_{passive} is the total convection and radiation loss from the each of the pipe segments, and Q_{fins} is the total fin loss.

The first order model solves each of the 8 experimental cases at once to verify the model's applicability to the test matrix. Initially, this universal approach shared insulation configurations between all 8 cases as there are no are no mentions of changes in insulation being done between the performance of the water and the salt cases. Under this constraint, the global energy balance of both the water and salt cases could not be acceptably closed simultaneously, leading to the conclusion that there *must* have been different insulation configurations between the water suite and salt suite of experiments, which was subsequently confirmed by the experimentalists.

With that confirmation, and the use of two different insulation thicknesses, the model achieves acceptable energy closure for all test cases, with the exception of salt test 1. The anamoloies regarding salt test 1 will be further elaborated on in Chapter 6.

5.2 OpenFOAM Simulation

5.2.1 Governing Equations

The simulations solve the steady-state incompressible Navier-Stokes equations with buoyancy and temperature-dependent properties. The continuity, momentum, and energy equations in the PIMPLE (merged PISO-SIMPLE) framework are: conservation of mass, momentum, and energy.

Conservation of mass is

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{U}) = 0. \tag{5.8}$$

Conservation of momentum is

$$\frac{\partial (\rho \boldsymbol{U})}{\partial t} + \nabla \cdot (\rho \boldsymbol{U} \otimes \boldsymbol{U}) = -\nabla p_{\text{rgh}} - (\boldsymbol{g} \cdot \boldsymbol{r}) \nabla \rho + \nabla \cdot \boldsymbol{\tau}_{\text{eff}}, \tag{5.9}$$

where $\boldsymbol{r}$ is the position vector, $\boldsymbol{g}$ is the gravitational acceleration, and the pressure is decomposed as

$$p_{\text{rgh}} \equiv p - \rho \, \boldsymbol{g} \cdot \boldsymbol{r}. \tag{5.10}$$

This decomposition removes the hydrostatic head from the pressure variable, so that buoyancy enters Equation (5.9) solely through the $(\boldsymbol{g} \cdot \boldsymbol{r}) \nabla \rho$ term. So, for a single-phase, isothermal loop, the $\nabla \rho$ term would reduce to zero. This is exactly what we would expect for a system with a uniform density gratient: no natural circulation.

The effective stress tensor is

$$\boldsymbol{\tau}_{\text{eff}} = \mu_{\text{eff}} \left[\nabla \boldsymbol{U} + (\nabla \boldsymbol{U})^{\mathsf{T}} - \frac{2}{3} (\nabla \cdot \boldsymbol{U}) \boldsymbol{I} \right], \qquad \mu_{\text{eff}} = \mu + \mu_t, \tag{5.11}$$

in which $\mu = \mu(T)$ is the molecular dynamic viscosity and μ_t is the turbulent viscosity supplied by the closure model (Section 5.2.3). The transpose and dilatation terms are retained because μ varies appreciably over the operating temperature range and $\nabla \cdot \boldsymbol{U} \neq 0$.

Conservation of energy is solved in enthalpy form,

$$\frac{\partial (\rho h)}{\partial t} + \nabla \cdot (\rho \boldsymbol{U} h) + \frac{\partial (\rho K)}{\partial t} + \nabla \cdot (\rho \boldsymbol{U} K) - \frac{\partial p}{\partial t} = \nabla \cdot (\alpha_{\text{eff}} \nabla h) + \rho \, \boldsymbol{U} \cdot \boldsymbol{g}, \tag{5.12}$$

where h is the specific enthalpy, $K = |\boldsymbol{U}|^2/2$ is the specific kinetic energy, and

$$\alpha_{\text{eff}} = \frac{\mu}{\text{Pr}} + \frac{\mu_t}{\text{Pr}_t} \tag{5.13}$$

5.2.2 Thermophysical Model

Within OpenFOAM, the thermophysical model uses the `heRhoThermo` class with the `icoPolynomial` equation of state, which evaluates density as a polynomial function of temperature:

$$\rho(T) = \sum_{i=0}^{N} a_i T^i. \tag{5.14}$$

Transport properties (μ, k) and specific heat (c_p) are similarly represented as temperature-dependent polynomials. This avoids the Boussinesq approximation entirely since temperature variations are captured in all terms of the governing equations.

$$\rho[\text{kg/m}^3] = 2293.6 - 0.7497\ T \tag{5.15}$$

$$\mu[\text{Pa s}] = 1.149 \times 10^{-3} \exp\left(-810.896/T + 7.806 \times 10^5/\ T^2\right) \tag{5.16}$$

$$c_p[\text{J/kg K}] \approx 1423.47 \tag{5.17}$$

$$k[\text{W/m K}] = 0.78 - 1.25 \times 10^{-3}\ T + 1.6 \times 10^{-6}\ T^2 \tag{5.18}$$

Each of the above equations are evaluated using T in Kelvin. The equation for density comes from Janz et al. [26] and has a 2% uncertainty within the applicable temperature range of 470-870 K. The equation for dynamic viscosity comes from Jin et al. [50] and has a 2.5% uncertainty within the applicable temperature range of 420-695 K. The equation for specific heat capacity provided by Parida et al. [51] does not have a quantified uncertainty, but does report the specific heat range found within the literature they found. Using that as the bounds for uncertainty, the uncertainty is approximately ±4% for the liquid phase and is weakly dependent on temperature, so has been modeled as a constant property. The equation for thermal conductivity comes from Santini et al. [48] and has an reported uncertainty and an applicable temperature range of 420-770 K.

For water, the equations used higher-order fits against the IAPWS-IF97 standard. The

exact coefficients are listed in Appendix B.

5.2.3 Turbulence Modeling

Initially, all the simulations in this work were scoped to only examine laminar modeling of the experiments as the pipe Reynolds numbers (Re_D = 75-1110) are well below the critical value for turbulent transition ($\mathrm{Re}_D \approx 2300$) in typical pipe flow. However, early simulation attempts did not recreate the skewed velocity profile shape seen in water tests 3 and 4 and the flucuations present in the water tests were much greater than the salt test simulations. The idea that the salt tests were "more laminar" than the water tests led to the idea that some amount of turbulence was being incurred despite being below the typical critical Reynolds number. As mentioned in Section 2.1 the long entry lengths delay or prevent full-velocity development, which would give creedance to shifting the characteristic length of the flow from the typical dimension of the pipe's inner diameter, to the height of the loop. Calculating the Rayleigh number of the system using this characteristic length produces values exceeding 10^{10}, which is known as the critical threshold for turbulence in cases involving natural circulation. Typically in a pipe flow such as this, the transition to turbulence predicted by a critical Rayleigh number would be suppressed by the full development of the laminar velocity profile before the transition could occur, but because of our provably long hydrodynamic entry lengths, we can predict to see some level of turbulence, especially in water cases 3 and 4; which are exactly the two cases that whose velocity profiles are better captured by turbulence modeling than laminar modeling as seen in Chapter 6.

As any turbulence present was theorized to be in the midst of transitioning from laminar flow, the transitional turbulence model, $k-\omega$ SSTLM was chosen as the primary modeling choice. Secondarily, when researching the $k-\omega$ SSTLM model that OpenFOAM uses, it is the standard formulation that assumes the production of turbulent kinetic energy from buoyant effects is negligible. So, a `codedFvModel` was made to reintroduce the buoyant production of turbulent kinetic energy when the extension "*_Gb" was found on a simulation case name.

This added term is:

$$G_b = -\,C_g\,\nu_t\,(\mathbf{g}\cdot\nabla\rho)\,, \qquad C_g = \frac{1}{Pr_t} = 1.0 \tag{5.19}$$

5.3 Mesh Generation

5.3.1 Overview

The computational mesh is generated procedurally using `classy_blocks` [52], a Python wrapper around OpenFOAM's `blockMesh` utility. The mesh is fully structured hexahedral, avoiding the numerical diffusion and poor convergence rates associated with tetrahedral or polyhedral meshes in pipe-flow geometries [34].

5.3.2 Cross-Section Topology

Every pipe segment uses either a 12-block or 20-block O-grid cross-section. The standard 12-block topology is composed of:

- **4 core blocks** form a central square, avoiding the singular axis point of polar meshes.
- **8 shell blocks** form the annular region between the core and the pipe wall.

The 20-block topology is the same as the 12-block topology with an additional 8 blocks composing a second annular layer when step transitions are required throughout the loop's geometry, as when fitting pipes together.

This topological approach produces bounded aspect ratios across the entire cross-section and ensures smooth boundary layer resolution at the wall. The core-to-pipe radius ratio is derived from the boundary layer grading parameters to ensure a smooth transition from the graded boundary layer region to the uniform core.

5.3.3 Boundary Layer Inflation

Wall-normal mesh grading is controlled by three input parameters:

Parameter	Description
`first_cell_size`	Wall-adjacent cell height
`bulk_cell_size`	Interior cell target size
`c2c_expansion`	Cell-to-cell expansion ratio

From these, the number of boundary layer cells, total boundary layer thickness, core ratio, and transition ratio are derived automatically:

$$n = \left\lfloor \frac{\ln(\Delta_{\text{bulk}}/\Delta_1)}{\ln(r_{\text{exp}})} \right\rfloor + 1, \tag{5.20}$$

$$\delta_{\text{BL}} = \Delta_1 \frac{r_{\text{exp}}^n - 1}{r_{\text{exp}} - 1}, \tag{5.21}$$

$$\text{core_ratio} = 1 - \delta_{\text{BL}}/R_{\text{pipe}}. \tag{5.22}$$

This scheme ensures consistent mesh refinement across grid convergence studies: scaling Δ_1 and Δ_{bulk} by the refinement ratio changes the mesh spacing while preserving topology.

5.3.4 Junction Topology

The four corner junctions use a kernel-cylinder architecture:

- **Kernel:** 8 hexahedral Loft blocks forming a near-spherical core, with a `kernel_factor` (default 0.7) controlling the kernel radius as a fraction of the pipe radius and a `kernel_blend` (default 0.5) interpolating between cubic and spherical shapes.
- **Cylindrical legs:** One O-grid cylinder per pipe direction extending from the kernel to mate with other cylindrical pipe segments.
- **Cap and bridge blocks:** Loft blocks connecting kernel faces to cylindrical legs.

Vertex pairing between the kernel and cylinder legs is resolved by brute-force search over 16 orientation combinations, with right-hand-rule volume verification to ensure valid hexahedra.

5.3.5 Bend Segments

The loop contains four pipe bends (two in the lower heater leg, two in the upper cooler leg). These bends are constructed using Lofts to first connect one face to another and create the blocks in between before being assigned Arc edges to control the radius of curvature, which uses a value of 1 inch.

5.3.6 Expanded Cross-Sections

Junction extensions and valve bodies have an expanded outer diameter (12.7 mm radius vs. the standard 11.811 mm radius). Eight additional concentric ring blocks extend the O-grid from the bore to the expanded radius, with the annular step face patched as a wall. The mesh remains fully conformal at these transitions.

5.4 Boundary Conditions

5.4.1 Velocity and Pressure

The loop is a closed domain with no inlet or outlet, complicating the set-up of the simulation with the lack of an inlet or an outlet.

- **Velocity ($\boldsymbol{U}$):** No-slip (`noSlip`) at all wall patches.
- **Pressure (p_{rgh}):** `calculated`, the absolute pressure is given an initial value of atmospheric pressure at 101325 Pascals and is calculated at the boundary as the simulation evolves.
- **Hyrdostatic Pressure (p_{rgh}):** `fixedFluxPressure` at walls, with the reference pressure set to zero at a point in the downcomer (right vertical leg) at mid-height, away from thermal and flow disturbances. As the system is closed, this is a guage pressure.

5.4.2 Temperature: `rcExternalTemperature`

A custom boundary condition, `rcExternalTemperature`, was developed to model the thermal resistance of cylindrical pipe walls with external convection and radiation. This replaces OpenFOAM's stock `externalTemperature` boundary condition, which assumes a slab geometry for the wall conduction resistance.

For the TAMU loop, the thickly insulated pipes can have an outer-to-inner radius ratio in excess of 5, making the slab approximation inaccurate.

Physics Model

The boundary condition solves a one-dimensional cylindrical conduction problem from the pipe inner surface to the outer surface, coupled with external heat transfer:

1. **Cylindrical wall conduction:** The thermal resistance (per unit inner surface area) through N concentric layers is:

$$R_{\text{ps}} = \sum_{i=1}^{N} \frac{r_0 \ln(r_i/r_{i-1})}{k_i}, \tag{5.23}$$

   where r_0 is the pipe inner radius, r_{i-1} and r_i are the inner and outer radii of layer i (so r_N is the outermost surface), and k_i is the layer conductivity.

2. **External heat transfer:** At the outer surface, convection and radiation are applied:

$$q_{\text{out}} = \frac{A_{\text{outer}}}{A_{\text{inner}}} \left[h_{\text{ext}}(T_s - T_a) + \varepsilon\sigma(T_s^4 - T_{\text{sur}}^4) \right], \tag{5.24}$$

   where $A_{\text{outer}}/A_{\text{inner}} = r_{\text{outer}}/r_0$ is the area ratio for a cylinder.

3. **Energy balance:** The heat flux from the fluid into the wall equals the outward heat loss:

$$\frac{T_p - T_s}{R_{\text{ps}}} = q_{\text{out}}, \tag{5.25}$$

where T_p is the fluid-side temperature and T_s is the outer surface temperature. This equation is solved iteratively for T_s and the resulting gradient is applied as a `fixedGradient` boundary condition.

Diagnostic Fields

The boundary condition computes and stores per-face diagnostic fields:

- q_{src}: heat source contribution (from heater power Q)
- q_{conv}: external convection heat flux
- q_{rad}: external radiation heat flux
- q_{fluid}: heat flux from fluid into wall
- q_{balance}: residual (should approach zero at steady state)

5.5 Solver Configuration

5.5.1 Solution Strategy

The simulations use a pseudo-transient approach iteration, using adjustable time-stepping to stay within the range of stability below a maximum Courant number of 5. This configuration sacrfices temporal accuracy for computation speed towards steady state, but for an instance such as this where we are only interested in the steady-state of the simulation, there is no loss in simulation fidelity. Attempts were made to use a strictly steady-state solver approach, but were ultimately thrwarted by oscillatory behavior resultant from the sensitive nature of natural convection and the stagnant initial condition, a better initialized or pseudo-transient-seeded initial condition may result in stable convergence using a steady-approach, but has not been tested at this time. These pseudo-transient simulations were run until they sufficiently converged to steady state. A case was determined to be at steady-state when, over the last

100 simulated seconds, the change in the average loop temperature over is less than 10^{-4} %, the variance of the temperature probes changes by less than 0.1 %, (higher threshold set for water cases with relatively strong, but still statistically stationary fluctuations) and there is less than a 0.01 % change in mass flow rate. Most cases are an order of magnitude or more converged beyond these criteria.

The PIMPLE algorithm is configured with:

Parameter	Value
`nOuterCorrectors`	1
`nCorrectors`	2
`nNonOrthogonalCorrectors`	1

5.5.2 Discretization Schemes

Term	Scheme
Time (`ddt`)	`backward`
Gradient (`grad`)	`Gauss linear`
Divergence ($\boldsymbol{U}$, h)	`Gauss limitedLinear 1.0`
Laplacian	`Gauss linear corrected`

The `limitedLinear` scheme is a second-order scheme that provides stability for buoyancy-driven flows while maintaining second-order accuracy in smooth regions.

5.5.3 Linear Solvers

Field	Solver	Preconditioner	Tolerance
p_{rgh}	GAMG	DIC Gauss-Seidel	10^{-6}
$\boldsymbol{U}$, h	PBiCGStab	DILU	10^{-6}

5.5.4 Initial Conditions

The temperature field is initialized uniformly at the mean of the experimental thermocouple readings for each case. The velocity and pressure fields are initialized to zero. Gravity is $\boldsymbol{g} = (0, -9.81, 0)$ m/s^2. The simulated loop itself lies within the x-y plane, where views within this such as Figures 5.1 and 5.2 observe with the z axis oriented towards the camera.

5.5.5 Parallel Execution

Domain decomposition uses the `scotch` algorithm. The `-copyUniform` flag is required during `decomposePar` to ensure the custom boundary condition can access uniform field data in parallel. Simulations were run using the Texas Advanced Computer Center (TACC) and specifically their LoneStar6 computer cluster, simulations were ran with up to 128 cores. Scaling performance is reported in Section **??**.

5.6 Execution Pipeline

The simulation pipeline is fully automated through the custom `foam-vv` Python framework, which orchestrates the following steps for each case:

1. **Case generation:** Read YAML configuration files (geometry, physics, study definition), create a blockMeshDict via `classy_blocks` for subsequent mesh generation, then rendering OpenFOAM dictionaries from Jinja2 templates.

2. **Mesh generation:** Run `blockMesh` → `transformPoints` (mm → m) → `topoSet` (to assist with postprocessing objects such as the faceZones for collecting mass flow rate data).

3. **Domain decomposition:** `decomposePar -copyUniform`.

4. **Simulation:** `foamRun` in parallel.

5. **Post-processing:** Extract probe values, velocity profiles, wall heat fluxes, mass flow rates, compute error metrics, before finally generating a report PDF file.

5.7 Input Uncertainty

As there are several sources of uncertainty and not all contributions are universally applicable to each of the examined QoI, the components used for the calculation of the overall value of u_{input} are outlined as follows. Each input was assigned a bound from the data that was used to create the model. For the uncertainty distributions used, each quantity is assumed to be a rectangular distribution, with the exception of the thermocouple and heater power uncertainty, which are represented by a standard Gaussian distribution. The u column of Table 5.1 was determined by ($u = a/\sqrt{3}$ for a rectangular distribution, where a is the half-width). The uncertain inputs and their adopted bounds are collected in Table 5.1.

Table 5.1: Uncertain Simulation Inputs and Their Standard Uncertainties

Input	Modeled Value	Bounds	u
Loop pipe ID, D_{pipe}	0.93 in	±0.005 in	0.31%
Test section bore, D_{test}	0.83 in	±0.005 in	0.35%
Loop length, L	4.353 m	±2 cm	0.27%
Insulation thickness (salt)	2.00 in	[1.65, 2.35] in	0.202 in
Insulation thickness (water)	0.40 in	[0.30, 0.50] in	0.058 in
Ambient temperature, T_{amb}	measured, per case	—	1.1 K
Heater power, P_{heater}	measured, per case	—	0.93% or 1%
External convection, h_{ext}	Churchill-Chu	±11.1%	6.4%
Insulation emissivity, ε_{ins}	0.95	±0.05	0.029
Aluminum emissivity, ε_{Al}	0.10	±0.05	0.029

The diameters reflect the significant figures attributed to the reported dimensions for diameter. The bounds set for the loop length are an estimate that comes from the CAD reconstruction in Figure 5.2 and the approximate 1 cm misalignment of the left, riser leg of the loop between the CAD and image, multiplied by two for the length uncertainty to account for the top and bottom leg. The insulation bounds initially come from pixel measurements

of Figure 3 from Reis et al. [9]. Estimates from that image against the nominal outer diameter of the quartz test section being 1" place the insulation thickness at roughly 1.65". As mentioned in Section 4.5.1, attempting to close the energy balance across all 8 cases with this thickness applied to the first order model was unsuccessful, leading to the two different insulation uncertainties above. The bounds for those insulation thicknesses were chosen ultimately because of the conflicting information from the reported McMaster entry (nominally 2" thick of ceramic fiber per layer) and Reis et al. [9] claiming to use 3 layers of fiberglass insulation; ultimately settling to model the salt cases using the provided McMaster item and using the difference between the stated and estimated as the thickness bounds. For water, after establishing that the water cases used a different insulation configuration, the thinnest variant of the reported insulation was used (a distinctly different part on McMaster-Carr). The insulation was also sold in thicknesses of 2", 1", and 0.5" under different part numbers, so the idea for modeling the water's insulation was an unrecorded purchase of the thinnest variation. With this though, the first order model and follow-on CFD results ran too hot, being better served with a modeled thickness of 0.40". So, barring better information, the modeled 0.40" configuration was used as the center of the distribution with the purchasable 0.5" thickness serving as the upper bound, and a symmetric bound being set at 0.3" as a possible "effective insulation thickness" in the event the insulation was poorly mounted and caused the system to lose more heat than would otherwise be predicted.

The ambient temperatures use the same ± 1.1 K thermocouple uncertainty for the gaussian representing the ambient temperature confidence. There is no data for the ambient conditions of the experimental facility, so the estimate of the inlet temperature of the heat exchanger was used. For the external convection coefficient, the Churchill-Chu bounds are centered between 0.12 and 0.15 for the value of A used in their published fits, where 0.12 was derived for air and 0.15 was used for the general fit. The heater power was reported by the experimentalists to have a standard uncertainty of 0.93% for powers between 100 and 999 W and an uncertainty of 1% for powers between 1 and 100 W.

For the emissivities, engineering values were picked from Table 10.1 in Lienhard and Lienhard [20] as a baseline for aluminum, quartz glass, and insulation emissivities. The entry for quartz glass is very near to the experimental cases, so has been neglected for uncertainty analysis. Similar uncertainty was not provided for the heat exchanger's flowrate, so has been excluded from the uncertainty analysis here.

With each input expressed in terms of the QoI, we can summarize their contributions in Table 5.2. The entries u_{geo}, u_{ins}, u_{mat} and u_{env} are the geometric, insulation, material, and environmental contributions above, with the propagated QoI given in the subscript. The geometric uncertainty is calculated from the uncertainties in both test-section and loop-pipe diameters as well as the loop length. To propagate u_{geo} to the QoIs of velocity and mass flow rate, Poiseuille pipe flow scaling laws were assumed applicable; an assumption that is undermined by the under-developed flow discussed in Section 2.1, but is retained here as an attempt to bound input uncertainty.

The insulation uncertainty is determined from the first order model being tested with ± 0.05 inch insulation perturbations from the modeled thickness. By maintaining the same boundary conditions, the different insulation configurations produce a different energy flow necessary to uphold those conditions condtions, allowing us to quantify the change of energy flow as a function of insulation thickness. From this, we can multiply by the value of u_t to recover an energy flow uncertainty. From here, we can then convert that result to the uncertainty introduced to the QoI. For temperature, we return the model to the nominal configuration and uniformly perturb the system's input temperatures by ± 1 K, which can tell us how the energy flow changes in response to temperature. Dividing our previous energy flow uncertainty by this result gives us the propagated uncertainty for temperature for this input. The mass flow rate uncertainty similarly comes from the energy flow uncertainty but needs to account for the feedback of the fluid viscosity's response to the predicted change in temperature for such a predicted change in energy flow in addition to the typical scaling of mass flow rate with resonse to a change in net energy. Beacause the velocity's response

within a fixed geometry is equal to $U = \dot{m} \,/\, \rho\, A$, where A here is the cross-sectional area, the uncertainty associated with velocity is considered equal to the mass flow rate uncertainty.

The material uncertainty does not include uncertainty for any solid materials contained within the `rcExternalTemperature`. The insulation uses properties from the manufacturer's datasheet fit with a second order polynomial. Only five data points exist and only one is within the applicable range of experimental temperatures, but without more data and the R^2 value of the fit exceeding 0.99, its associated uncertainty has been ignored despite being the most likely source of unaccounted error. The other solids modeled (SS-316 and quartz) have negligible thermal resistances by comparison and are generally well-characterized, so their uncertainty is also neglected. Water properties also being well-posed, leaves the properties of HITEC as the only remainging uncertainty; with standard uncertainties from Section 5.2.2 that have been propagated with the exception of density, whose uncertainty is already carried by the back-calculation of the experimental mass flow from the PIV data in the previous chapter. Poiseuille pipe flow scaling laws are also used for this propagation to velocity and mass flow rate. For the propagation to temperature uncertainty, the property uncertainties are applied to the heat transfer equation $Q = \dot{m} c_p \Delta T$.

The environmental uncertainty shares a similar propagation path, but rather than accounting for the insulation's uncertainty, uses the first order model to characterize the uncertainty from the ambient temperature, external convective heat transfer coefficient, the emissivity of the insulation and heat exchanger assembly's aluminum exterior, as well as the measured uncertainty associated with the facility's heater power. These quantities are combined with a root-sum-square before being quantified below and further propagated.

The final three rows combine each contribution into the root-sum-square for the value of u_{input} to be used for each QoI for each of the eight experimental cases. The temperature entries are applied equally to both centerline and wall temperature probes.

Table 5.2: Contributions to Modeling Input Uncertainty (u_{input})

Quantity	Salt 1	Salt 2	Salt 3	Salt 4	Water 1	Water 2	Water 3	Water 4
$u_{geo,\dot{m}}$ [%]	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63
$u_{geo,U}$ [%]	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.94
$u_{ins,U} = u_{ins,\dot{m}}$ [%]	1.72	1.60	1.57	1.56	2.42	2.59	2.62	2.64
$u_{ins,T}$ [K]	2.30	2.36	2.49	2.68	0.84	1.11	1.29	1.73
$u_{mat,U} = u_{mat,\dot{m}}$ [%]	2.36	2.36	2.36	2.36	0	0	0	0
$u_{mat,T}$ [K]	0.47	0.42	0.33	0.24	0	0	0	0
$u_{env,U} = u_{env,\dot{m}}$ [%]	2.08	1.90	1.89	1.88	3.20	2.65	2.34	1.84
$u_{env,T}$ [K]	2.78	2.80	2.99	3.22	1.11	1.13	1.16	1.21
$u_{\text{input}}(\dot{m})$ [%]	3.64	3.48	3.47	3.46	4.06	3.76	3.56	3.28
$u_{\text{input}}(T)$ [K]	3.63	3.68	3.91	4.20	1.39	1.58	1.73	2.11
$u_{\text{input}}(U)$ [%]	3.70	3.55	3.54	3.53	4.12	3.82	3.63	3.36

Chapter 6

Results

This chapter presents the verification and validation results. Following the V&V methodology of Chapter 3, verification results are presented first (Sections 6.1-6.2), followed by validation against experimental data (Section 6.4).

6.1 Boundary Condition Verification

The custom `rcExternalTemperature` boundary condition was verified on a simplified single-pipe geometry by comparing the integrated wall heat flux against analytical reference values.

6.1.1 Verification Geometry

The verification domain is a vertical cylinder with an O-grid mesh:

- Inner radius: $r_0 = 10.0$ mm, diameter $D = 20.0$ mm
- Wall layers: SS-316 (1.0 mm, $k = 16.3$ W / m K) and insulation (50.0 mm, $k = 0.144$ W / m K)
- Outer radius: $r_2 = 61.0$ mm, area ratio $r_2/r_0 = 6.1$
- Adiabatic entry length: $L_{\text{adiab}} = 500.0$ mm

- Heated test section: $L_{\text{test}} = 100.0$ mm

Both the simulation and the analytical reference use constant fluid properties evaluated at based on the equations of state used by Reis et al. [9] covered in Chapter 4. T = 440 K, ($\rho = 1964$ kg / m^3, $\mu = 0.01176$Pa s, $c_p = 2442$J / kg K, $k = 0.540$W / m K, $\varepsilon = 0.85$, Pr = 53.2, Re = 334). This isolates the boundary condition discretization error from any effects of temperature-dependent property evaluation.

The adiabatic entry length was set after performing a preliminary hydrodynamic entry length calculation as mentioned in Section 2.1 to ensure that assumptions of full hydrodynamic development were applicable. The hydrodynamic entry length for laminar pipe flow is approximately:

$$L_h \approx 0.05\,\mathrm{Re}\,D = 0.05 \times 334 \times 0.02 = 334\text{mm}. \tag{6.1}$$

Since $L_{\text{adiab}} = 500\text{mm} > L_h$, the velocity profile is fully developed before the heated section begins.

The thermal entry length, by contrast, is:

$$L_t \approx 0.05\,\mathrm{Re}\,\mathrm{Pr}\,D = 0.05 \times 334 \times 53.2 \times 0.02 = 17.8\text{m}, \tag{6.2}$$

so the flow is thermally developing throughout the entire 100 mm test section ($L_{\text{test}} \ll L_t$). This Graetz-corrected thermal development is accounted for in the analytical reference (Section 6.1.2).

6.1.2 Analytical Reference

The analytical heat flow was computed from a two-node thermal resistance network. For Mode A (convection only), the total resistance is the sum of three series resistances (all per unit inner surface area):

$$Q = \frac{(T_{\text{amb}} - T_f)\,A_{\text{inner}}}{R_{\text{int}} + R_{\text{ps}} + R_{\text{ext}}}. \tag{6.3}$$

The internal film resistance accounts for the thermally developing boundary layer between the bulk fluid and the pipe inner wall. With $\mathrm{Gz} = \mathrm{Re}\,\mathrm{Pr}\,D/L_{\mathrm{test}} = 334 \times 53.2 \times 0.02/0.1 = 3554$, the correlation of Equation (3.13) gives $\overline{\mathrm{Nu}}_D = 26.67$, yielding an internal convection coefficient of $h_{\mathrm{int}} = \overline{\mathrm{Nu}}_D\, k_f/D = 720\mathrm{W}\ /\ \mathrm{m}^2\mathrm{K}$. The three resistances are:

- $R_{\mathrm{int}} = 1/h_{\mathrm{int}} = 0.00139\ \mathrm{m}^2\mathrm{K}\ /\ \mathrm{W}$ (internal film; only 0.93% of total)
- $R_{\mathrm{ps}} = 0.11902\ \mathrm{m}^2\mathrm{K/W}$ (cylindrical wall conduction, dominated by the 50 mm insulation layer)
- $R_{\mathrm{ext}} = 1/(a_r\ h_{\mathrm{ext}}) = 0.02939\mathrm{m}^2\mathrm{K/W}$ (external convection, where $a_r = r_2/r_0 = 6.1$ is the area ratio)

While the internal film resistance contributes less than 1% of the total, it is still necessary to account for, otherwise the GCI will have a near 1% error that always remains. For Mode B (radiation only) and Mode C (mixed), the outer surface temperature was obtained by Newton iteration on the nonlinear energy balance at the outer surface, using $R_{\mathrm{eff}} = R_{\mathrm{int}} + R_{\mathrm{ps}}$ as the total resistance from bulk fluid to outer surface.

The analytical reference values are:

Mode	$Q_{\mathrm{analytical}}$ [W]	$q_{\mathrm{analytical}}$ [W/m^2]
A: Convection only	-6.022	-958.5
B: Radiation only	-6.076	-967.0
C: Convection + radiation	-6.673	-1062.1

6.1.3 Grid Convergence Results

Four systematically refined meshes were used (refinement ratio $r = 2$):

We can calculate the representative cell size using Equation 3.11 to allow us to use a single reference size for a mesh rather than use either the first cell or bulk cell size used in the grading of the mesh. When calculating the representative cell size in this way, the

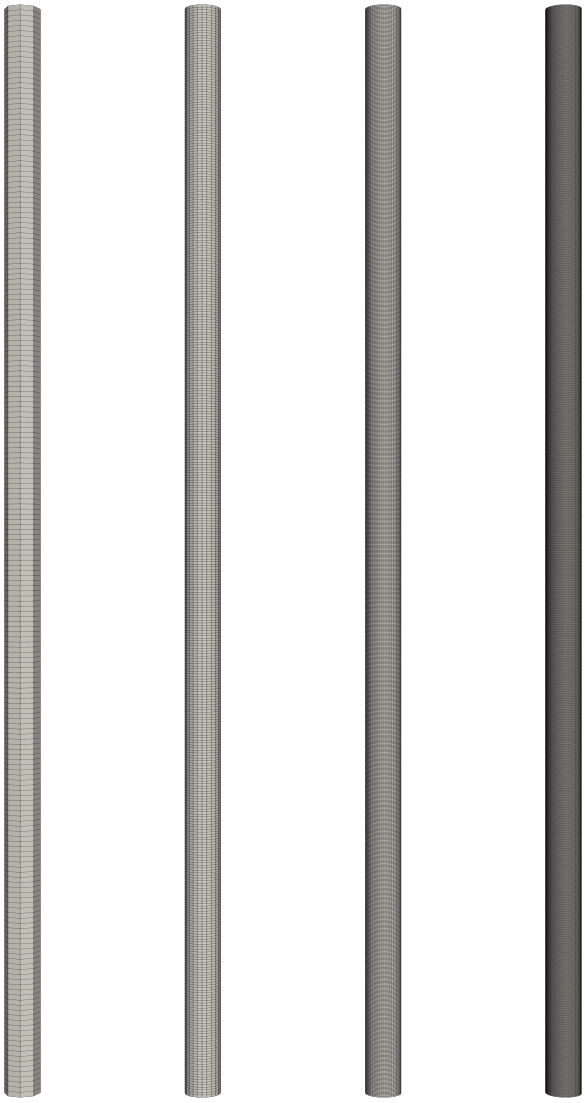

Figure 6.1: Side-by-side Axial Comparison

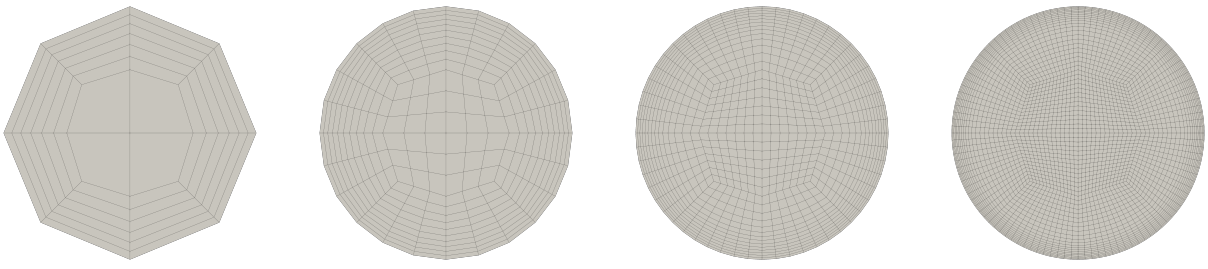

Figure 6.2: Side-by-side Radial Comparison

nominal refinement ratio of 2 is no longer applicable and needs to be recomputed for each successive refinement.

Table 6.1: Representative cell sizes for BC verification.

Level	First cell [mm]	Bulk cell [mm]	h [mm]	Refinement ratio (r)
Coarse	0.600	2.80	2.484	-
Medium	0.300	1.40	1.164	2.136
Fine	0.150	0.70	0.608	1.916
Xfine	0.075	0.35	0.327	1.856

Table 6.2: BC verification: Mode A (convection only).
$Q_{\text{analytical}} = -6.022\,\text{W}, \quad q''_{\text{analytical}} = -958.468\,\text{W/m}^2$

	Simulated		Rel. error [%]	
Level	Q [W]	q'' [W/m^2]	Q	q''
Coarse	−5.868	−958.388	2.559	0.008
Medium	−6.003	−958.106	0.323	0.038
Fine	−6.016	−958.047	0.099	0.047
Xfine	−6.018	−957.997	0.062	0.049

Table 6.3: BC verification: Mode B (radiation only).
$Q_{\text{analytical}} = -6.076\,\text{W}, \quad q''_{\text{analytical}} = -967.028\,\text{W/m}^2$

	Simulated		Rel. error [%]	
Level	Q [W]	q'' [W/m^2]	Q	q''
Coarse	−5.920	−966.931	2.560	0.010
Medium	−6.056	−966.659	0.323	0.038
Fine	−6.070	−966.570	0.100	0.047
Xfine	−6.072	−966.546	0.063	0.050

One thing notable about the convergence behavior for Tables 6.2 - 6.4 is that the relative error of Q appears to monotonically converge towards the analytic solution, while the relative error of q'' seems to *grow* monotonically. This may at first look like a sign of divergence, but when examining the the Richardson extrapolated value for both Q and q'' across each mode, their extrapolated values roughly share the same, small difference from the analytical solution. This is not surprising, as the Graetz number correction for developing flow has its own,

Table 6.4: BC verification: Mode C (mixed convection + radiation).
$Q_{\text{analytical}} = -6.673\,\text{W}, \quad q''_{\text{analytical}} = -1062.094\,\text{W/m}^2$

	Simulated		Rel. error [%]	
Level	Q [W]	q'' [W/m^2]	Q	q''
Coarse	−6.503	−1061.997	2.559	0.009
Medium	−6.652	−1061.653	0.327	0.042
Fine	−6.666	−1061.551	0.104	0.051
Xfine	−6.669	−1061.521	0.067	0.054

Table 6.5: GCI summary for BC verification, Modes A-C.
Realized refinement ratios $r_{21} = 1.8560$, $r_{32} = 1.9161$ (nominal 2.0).

				GCI [%]			
Mode	Grids	p_{obs}	f_{ext}	Fine	Coarse	Asym. ratio	δ_{ext} [%][a]
Surface-integral Q [W]							
A: Convection	M/F/X	2.7462	−6.0190	0.0103	0.0564	1.0004	−0.0541
	C/M/F	2.9769	−6.0185	0.0472	0.3281	1.0022	−0.0614
B: Radiation	M/F/X	2.7433	−6.0727	0.0104	0.0565	1.0004	−0.0546
	C/M/F	2.9794	−6.0722	0.0471	0.3276	1.0022	−0.0621
C: Mixed[b]	M/F/X	2.7592	−6.6694	0.0101	0.0557	1.0004	−0.0590
	C/M/F	2.9792	−6.6689	0.0470	0.3271	1.0022	−0.0659
Area-averaged flux q'' [W/m^2]							
A: Convection	M/F/X	1.9334	−957.9865	0.0013	0.0044	1.0000	−0.0503
	C/M/F	1.4630	−957.9667	0.0070	0.0182	0.9999	−0.0523
B: Radiation	M/F/X	1.9968	−966.5362	0.0013	0.0043	1.0000	−0.0508
	C/M/F	1.3556	−966.5068	0.0081	0.0196	0.9999	−0.0539
C: Mixed	M/F/X	1.8031	−1061.5059	0.0018	0.0054	1.0000	−0.0554
	C/M/F	1.4948	−1061.4896	0.0073	0.0193	0.9999	−0.0569

M/F/X = medium/fine/xfine; C/M/F = coarse/medium/fine.
[a] $\delta_{\text{ext}} = (f_{\text{ext}} - f_{\text{analytical}})/f_{\text{analytical}}$.

small uncertainty associated with it ($< 1\%$), so the relative error for Q and q'' approaching from opposite sides towards a similar extrapolated value percent difference from the analytical solution is encouraging for confidence in the boundary condition implementation.

6.1.4 Observed Convergence Order

On a similar note, the values of p_{obs} across the board are worth discussing. First, the values of p_{obs} for Q are all nearly a whole observed order higher than expected. This is another point

that warrants concern at first, but by looking at Figure 6.4, you can see that there is also a *geometric* convergence at play as the cylindrical mesh converges towards a truly cylindrical shape. This additional source of convergence results in an artifical inflation of the observed order higher than the numerical theory would predict. As for the values of p_{obs} attached to the area-averaged heat fluxes, two trends are notable. First, the values of p_{obs} for the finer triplet of meshes (M/F/X) are each ≈ 2. Second order convergence is expected for the temperature field and since heat flux is obtained from differentiating the temperature distribution, the conservative bounding of heat flux expects p_{obs}. Three factors contribute to the $p_{\text{obs}} \approx 2$ that we see in Table 6.5: the numerical schemes used in OpenFOAM for simulation are second order across the board, the analytic solution for heat flux with `rcExternalTemperature` directly using the second order convergent Temperature field, and the use of high-quality meshes contributing towards a smooth, asymptotic expansion of the temperature distribution limiting the impact of numerical discretization error.

To summarize, all three modes satisfy the acceptance criteria:

- $\text{GCI}_{\text{fine}} < 0.01\%$ (far below the 2% acceptance threshold).
- Asymptotic ratio ≈ 1.0 (confirming solutions are in the asymptotic range).
- Energy balance residual $q_{\text{balance}} < 10^{-12}$ W/m^2 (machine precision).
- Relative error on the finest mesh $< 0.1\%$.

With the acceptance criteria met, the verification cases each reproduce the analytical solution within 0.1% on the finest mesh and each mode converging strongly, the boundary condition `rcExternalTemperature` can be classified as successfully verified by the V&V methodology set in Chapter 3

6.2 Grid Convergence Study

Regretably, a three-mesh refinement study for the full-loop study could not be completed across the entire simulation matrix. The full-loop simulation pipeline leveraged the faster convergence of the coarse mesh to converge to a solution from the initial conditions before mapping the converged solution from the coarse mesh to the next finer mesh. After mapping the solution to the medium refinement mesh, the simulation experiences start-up instability before settling to the new converged state within the refined mesh. The decision ultimately boiled down to presenting a mix of settled and mostly settled fine-mesh cases with the full exclusion of the fine-mesh turbulence cases, or the two-refinement study that could present confident and honest results across the full matrix. The latter option was chosen. The cost of this includes Richardson extrapolation of the QoIs, discussion of monotonic vs potential oscillatory convergence, and a larger u_{num}, for each case, but earns a discussion of the performance of the turbulent models. This is particularly interesting following the discussion in Section 5.2.3 where we predict to see the turbulent model out-perform the laminar counterpart on account of the long hydrodynamic entry lengths for water cases 3 and 4.

Thankfully, Roache provides procedure for cases such as this; where, for performing GCI on two mesh refinements, the value of F_s used in Equation 3.7 takes the value of 3 instead of 1.25 [37].

For this GCI, the meshed system can be seen below in Fig 6.3, which is set up to provide a visualization for what section of the computational mesh corresponds to the quartz-glass test section and a representative laser PIV sheet in green.

Several mesh cross-sections are provided (Figures 6.4 - 6.7 to help visualize how the mesh is structured around geometric complexities such as bends and junctions, as well as the structured cross-sectional mesh. For more information on mesh quality, the statistics for the utilized meshes are in Appendix A. For the consideration of turbulence models, the maximum y^+ value recorded for the medium meshes was for Water 4 with the $k\omega$ SSTLM_Gb model

Figure 6.3: Experimental PIV Window Visualization

at just under $y^+ = 5$. This undermines turbulence modeling without wall functions, but it should be noted that the value of y^+ only goes so high because the transitions to-and-from step changes in a pipe's cross-section were chosen to not have smooth axial grading as walls normally do in the system. This was done because for a fully-conformal and fully-hexahedral mesh, the wall refinement would be topologically required to propagate to the pipe's bulk flow region. This would result in a significant slow-down for the simulation's convergence on account of the pseudo-transient approach bounded by the Courant limit of 5 for minimal

gain in turbulence modeling, so was seen as an acceptable trade off, especially since the values for $y^+ \approx 1$ or lower in the vast majority of the mesh.

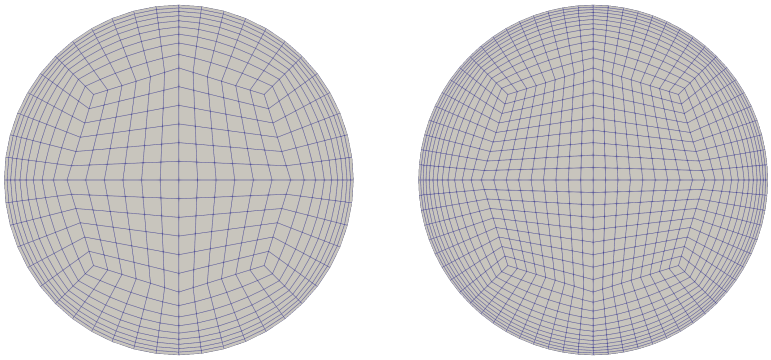

Figure 6.4: Loop Pipe Cross Sections

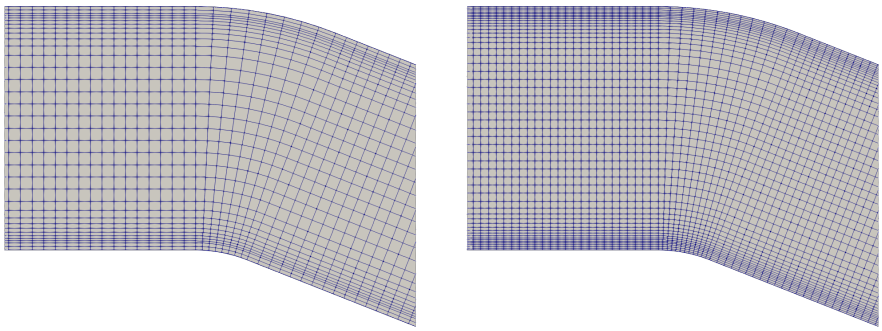

Figure 6.5: Loop Pipe Bend

With that bound set, we can compute the numerical uncertainty of the simulation results before finally determining the validation uncertain for each of the QoIs studied. The full matrix including the uncertainty for turbulent cases can be found in Appendix C. This is true for each table of uncertainties unless explicitly stated otherwise. Below, are the values for numerical uncertainty calculated from the two-mesh GCI. $\mathrm{TP}_{\mathrm{mean}}$ and $\mathrm{TW}_{\mathrm{mean}}$ relate to the average numerical uncertainty among the TP and TW series of probes of probes respectively. $\dot{m}_{\mathrm{flow}}$ is the numerical uncertainty associated with the cross-sectional mass flux through the

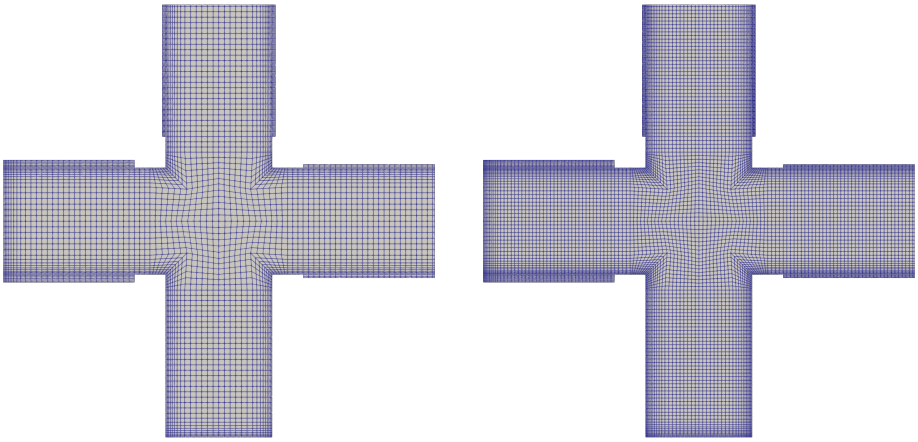

Figure 6.6: Loop Pipe Cross Junction

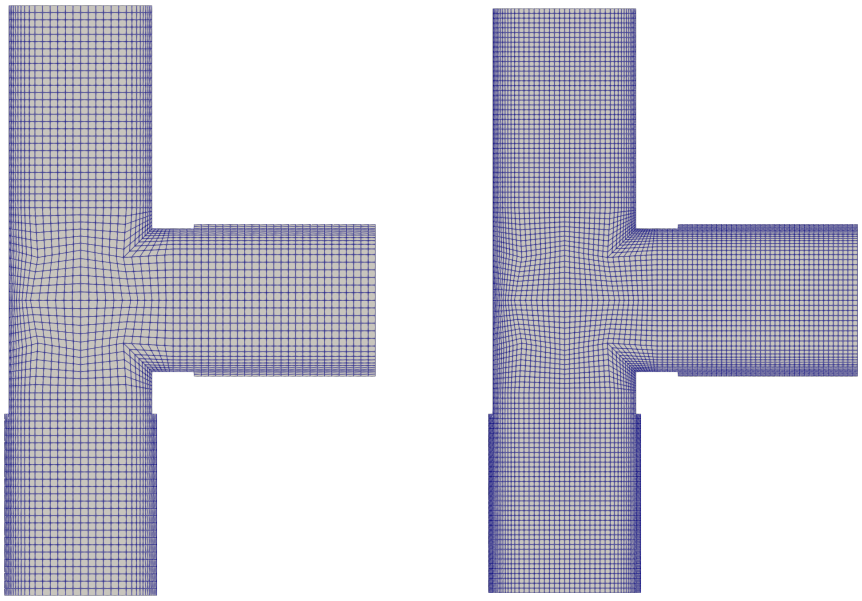

Figure 6.7: Loop Pipe Tee Junction

pipes in the typical form mass flow rate is calculated ($\dot{m} = U_{\text{avg}}\ \rho\ A$) whereas $\dot{m}_{\text{PIV}}$ is the mass flow rate as calculated using the experimentalists' PIV method from Reis et al. [9]. U_{PIV} is similarly calculated and U_{max} bounds the numerical error despite not being directly used for any QoI validation uncertainties.

Table 6.6: u_{num} for CFD Quantities of Interest (Laminar family, two-grid GCI)

QoI	Salt 1	Salt 2	Salt 3	Salt 4	Water 1	Water 2	Water 3	Water 4
TP_{mean} [K]	0.19	0.96	1.34	1.93	0.03	0.04	0.04	0.09
TW_{mean} [K]	0.28	0.30	0.31	0.22	0.02	0.05	0.03	0.03
$\dot{m}_{\text{flow}}$ [%]	2.70	3.55	4.62	5.22	0.95	1.01	0.88	0.83
$\dot{m}_{\text{PIV}}$ [%]	1.47	5.75	8.54	8.42	7.04	6.56	5.79	3.57
U_{PIV} [%]	1.46	5.79	8.59	8.50	7.04	6.56	5.79	3.57
U_{max} [%]	12.72	5.60	6.92	6.57	1.28	1.59	0.89	1.89

6.3 Validation Uncertainties

With the values of u_{num} established, we now have all three components necessary to compute u_{val} for each QoI. Doing so, the validation uncertainty is summarized in Tables 6.7 - 6.9

Table 6.7: Validation uncertainty for (TP_{mean}) and (TW_{mean}).

	TP_{mean} [K]				TW_{mean} [K]			
Case	u_{num}	u_{input}	u_D	u_{val}	u_{num}	u_{input}	u_D	u_{val}
Salt 1	0.19	3.63	1.10	3.80	0.28	3.63	1.10	3.81
Salt 2	0.96	3.68	1.10	3.96	0.30	3.68	1.10	3.85
Salt 3	1.34	3.91	1.10	4.28	0.31	3.91	1.10	4.07
Salt 4	1.93	4.20	1.10	4.75	0.22	4.20	1.10	4.35
Water 1	0.03	1.39	1.10	1.77	0.02	1.39	1.10	1.77
Water 2	0.04	1.58	1.10	1.93	0.05	1.58	1.10	1.93
Water 3	0.04	1.73	1.10	2.05	0.03	1.73	1.10	2.05
Water 4	0.09	2.11	1.10	2.38	0.03	2.11	1.10	2.38

For Tables 6.8 and 6.9, the uncertainty for $\dot{m}_{\text{flow}}$ and $\dot{m}_{\text{flow}}$ share u_D and u_{input}; only u_{num} differs between the calculations.

Table 6.8: Validation uncertainty for the mass flow rate: test-section cross-sectional flux ($\dot{m}_{\text{flow}}$) and reconstructed PIV methodology ($\dot{m}_{\text{PIV}}$). u_{input} and u_D are shared; only u_{num} differs between the variants.

Case	$u_{\text{num}}(\dot{m}_{\text{flow}})$ [%]	$u_{\text{num}}(\dot{m}_{\text{PIV}})$ [%]	u_{input} [%]	u_D [%]	$u_{\text{val}}(\dot{m}_{\text{flow}})$ [%]	$u_{\text{val}}(\dot{m}_{\text{PIV}})$ [%]
Salt 1	2.70	1.47	3.64	3.42	5.67	5.20
Salt 2	3.55	5.75	3.48	3.31	5.97	7.50
Salt 3	4.62	8.54	3.47	3.24	6.62	9.77
Salt 4	5.22	8.42	3.46	3.15	7.01	9.63
Water 1	0.95	7.04	4.06	2.59	4.91	8.53
Water 2	1.01	6.56	3.76	2.35	4.54	7.92
Water 3	0.88	5.79	3.56	2.20	4.28	7.15
Water 4	0.83	3.57	3.28	2.04	3.96	5.26

Table 6.9: Validation uncertainty for the bulk velocity: reconstructed PIV methodology (U_{PIV}) and profile peak ($U_{\max}$, $Y/H = 0.50$). u_{input} and u_D are shared; only u_{num} differs between the variants.

Case	$u_{\text{num}}(U_{\text{PIV}})$ [%]	$u_{\text{num}}(U_{\max})$ [%]	u_{input} [%]	u_D [%]	$u_{\text{val}}(U_{\text{PIV}})$ [%]	$u_{\text{val}}(U_{\max})$ [%]
Salt 1	1.46	12.72	3.70	2.33	4.61	13.45
Salt 2	5.79	5.60	3.55	2.17	7.13	6.98
Salt 3	8.59	6.92	3.54	2.06	9.51	8.04
Salt 4	8.50	6.57	3.53	1.91	9.40	7.70
Water 1	7.04	1.28	4.12	2.59	8.56	5.03
Water 2	6.56	1.59	3.82	2.34	7.95	4.76
Water 3	5.79	0.89	3.63	2.19	7.18	4.33
Water 4	3.57	1.89	3.36	2.04	5.31	4.36

6.4 Validation Against Experimental Data

For visually assessing the validation error as it applies to the temperature measurements, parity plots were chosen. These plot the experimentally recorded value for each temperature probe within a case; so, for the ideal case of perfectly agreeing predicted and measured values, the plot's diagonal line is the target. Along this line, two uncertainty bands are applied. The thinner, darker band represented only the uncertainty associated with the instrument recording the data. The broader, lighter line is the applied validation uncertainty to the target line and if the predicted value falls within that band, it can be classified as

"validated" in the sense the the comparison error cannot be distinguished from other error sources. The blue bands seen in Figure 6.8 and 6.9 correspond to the four orientation extremes (front/back top/bottom or left/right) for a given segment of pipe at the nominal axial point of the wall temperature measurement. With the azimuthal orientations of the wall thermocouples being lost, the bounds of those four points on the perimeter allow us to see the effects thermal stratification may have had on the experimental measurements, which at times is considerable. The temperature parity plots the two turbulence models are not considerably different from the laminar parity seen in Figure 6.9, so have been excluded for brevity.

To pre-empt the next series of plots, it is useful to see the temperature distribution typical of a molten salt test in the bottom left junction, following just after the heater leg and before the test section. When looking at the previous parity plots for salt, you may notice that TP3 is often an outlier with relation to nearly all other measurements and that this trend is not visible in the water cases. That is because: following the heater leg, the molten salt's high Prandtl number results in a steep temperature gradient as the freshly heated, relatively fast-flowing salt moves up the riser leg, slips past the largely stagnant fluid in the lower junction. Figure 6.10 shows the steep temperature gradient (specifically for salt test 4 on the medium mesh), along with a 0.1" radius sphere centered in the cross junction. From this visual, we can then quantify the potential impact on the temperature measurement in the event of a small bump or other perturbation away from the true center of the junction, which as we can see from Figures 6.11 and 6.12 that such a small change in location can impact the recorded temperature by over 10 K in some cases.

The final set of QoI figures represents the most difficult batch to acceptably match. In order to produce a serviceable prediction of the PIV data, the location of the PIV plane must be replicated appropriately. Beyond the PIV's location, the velocity profiles have historically been the most sensitive to changes throughout the modeling effort but the current work validates each test within uncertainty, with the glaring exception of salt 1 as the primary

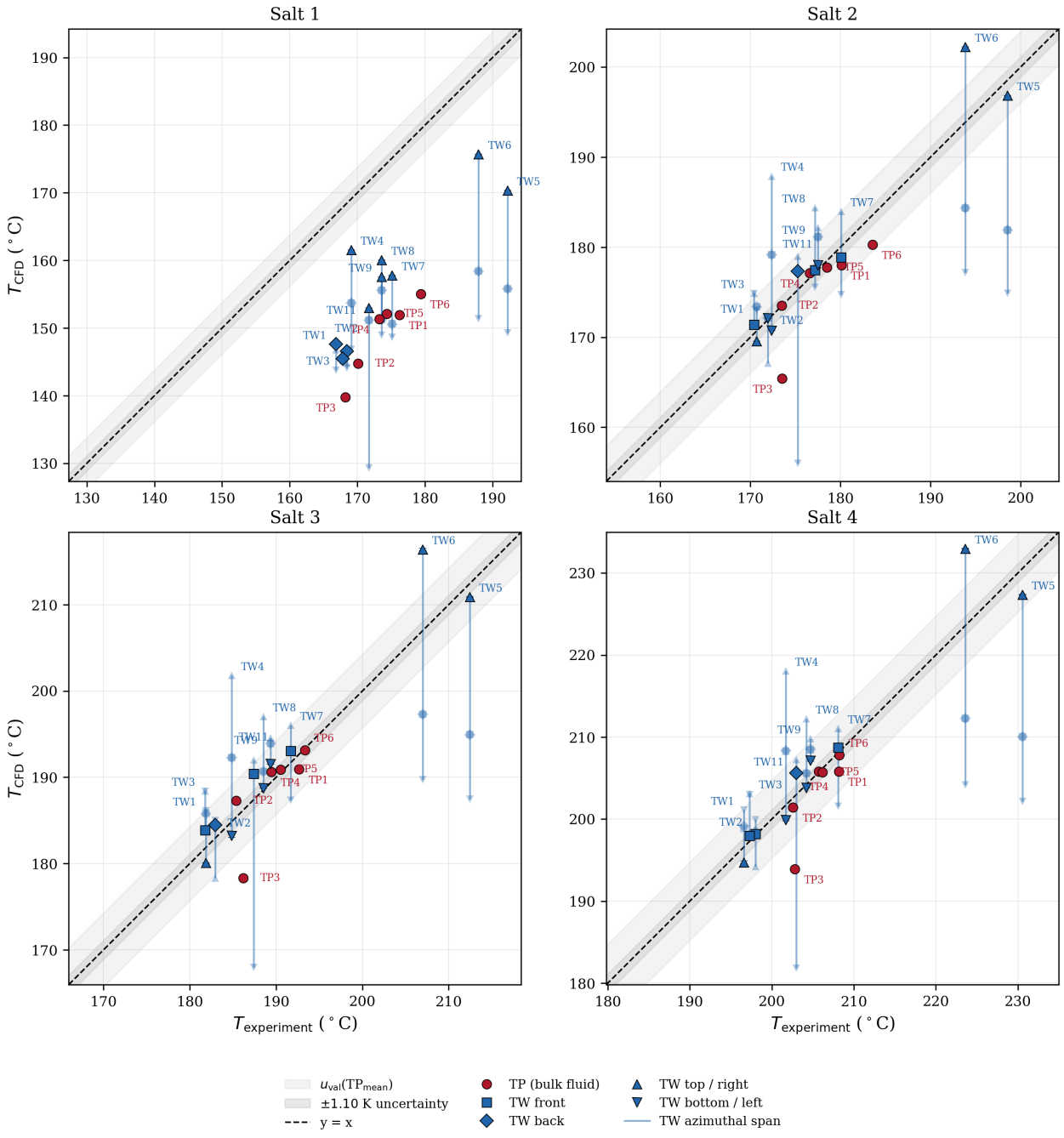

Figure 6.8: Temperature Parity for Salt Test Series

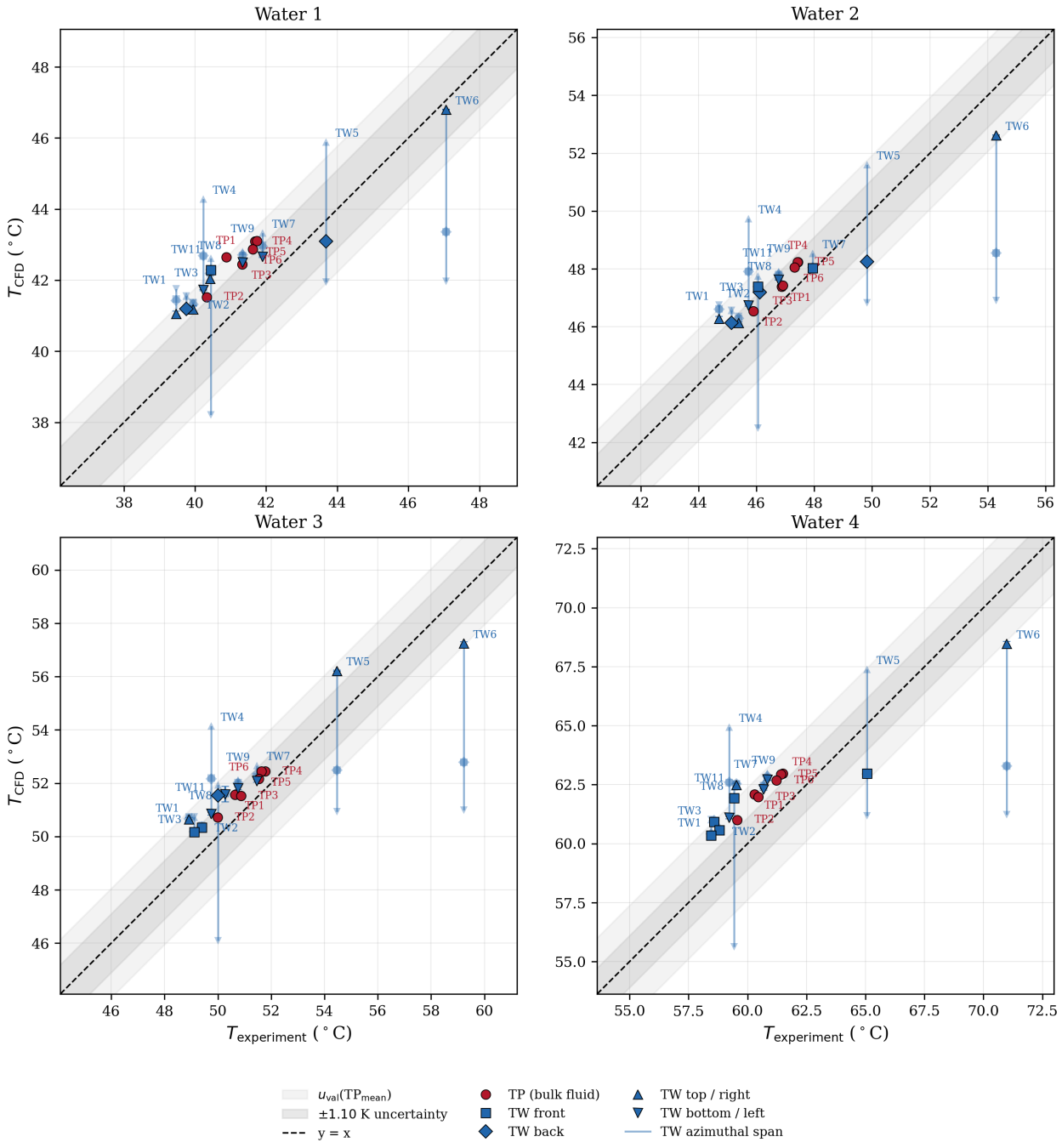

Figure 6.9: Temperature Parity for Laminar Water Test Series

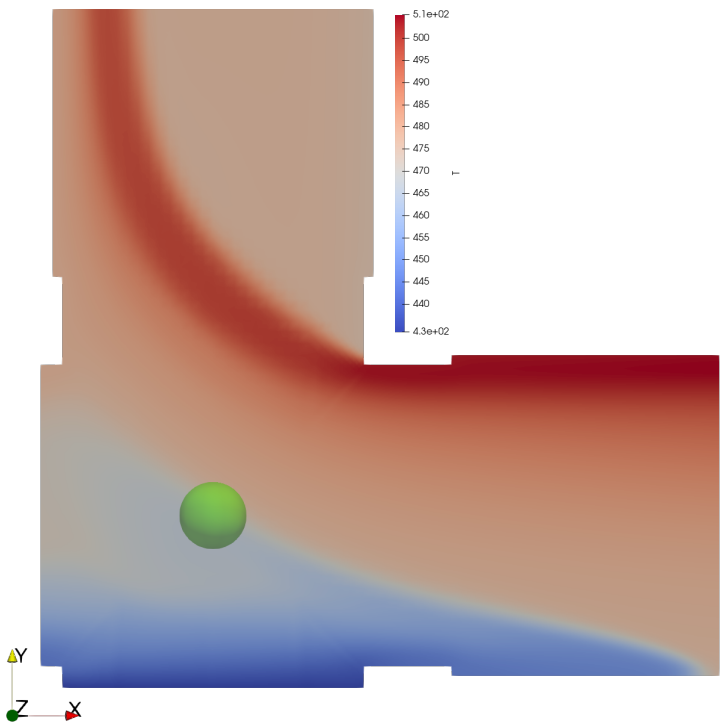

Figure 6.10: Temperature Distribution in Lower Left Junction of Salt 4 (medium refinement)

exception to be discussed momentarily. Water 2 and Water 4 have their velocity profiles best captured by turbulence models. The full set of water velocity profiles is in Appendix C. The standard SSTLM velocity profiles seemed to prefer to skew to the right, where the addition of buoyant production in the Gb variant shifted the skew to the left for most cases. Due to the odd shape of the recorded velocity profile for water 3, it is thought that the captured profile was in the midst of some fluctuation, but this idea is unquantified beyond visual inspection.

Among the most experimentally insightful discoveries of the modeling efforts within this work are the insights in to the three-dimensional flow profile within the system. From taking a slice of the loop, we can see that the flow is practically swept up the rise leg along the

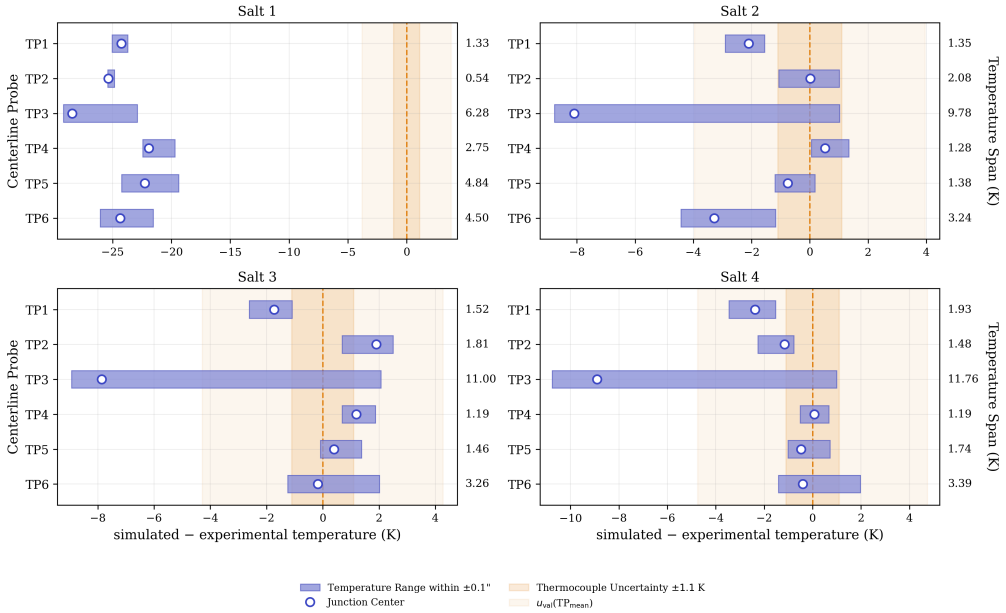

Figure 6.11: Position Sensitivity for Salt Centerline Probes

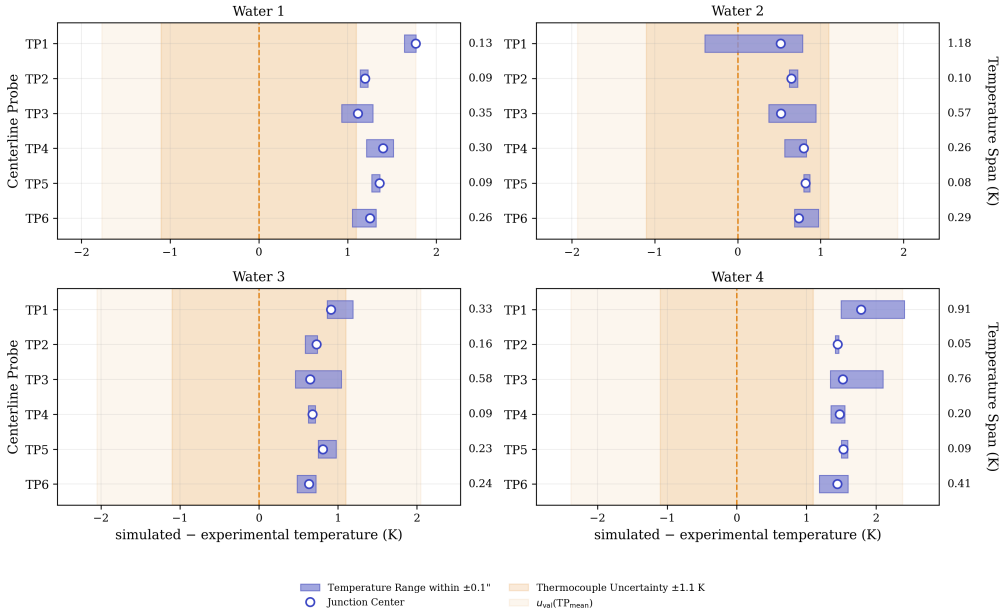

Figure 6.12: Position Sensitivity for Water Centerline Probes

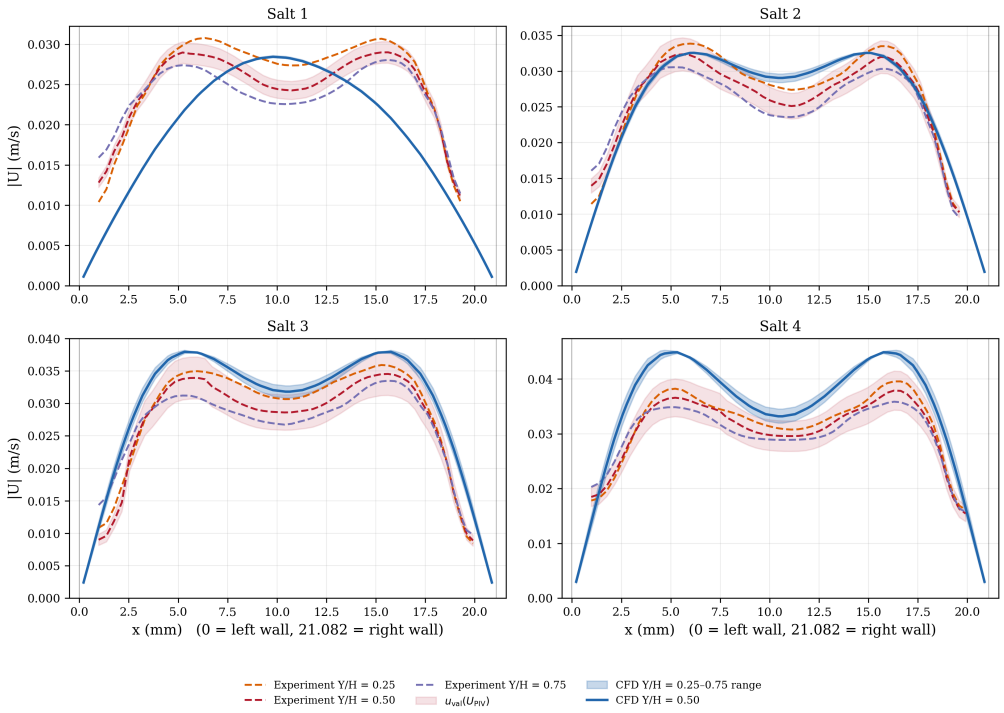

Figure 6.13: Velocity Profile Error Comparison for Salt

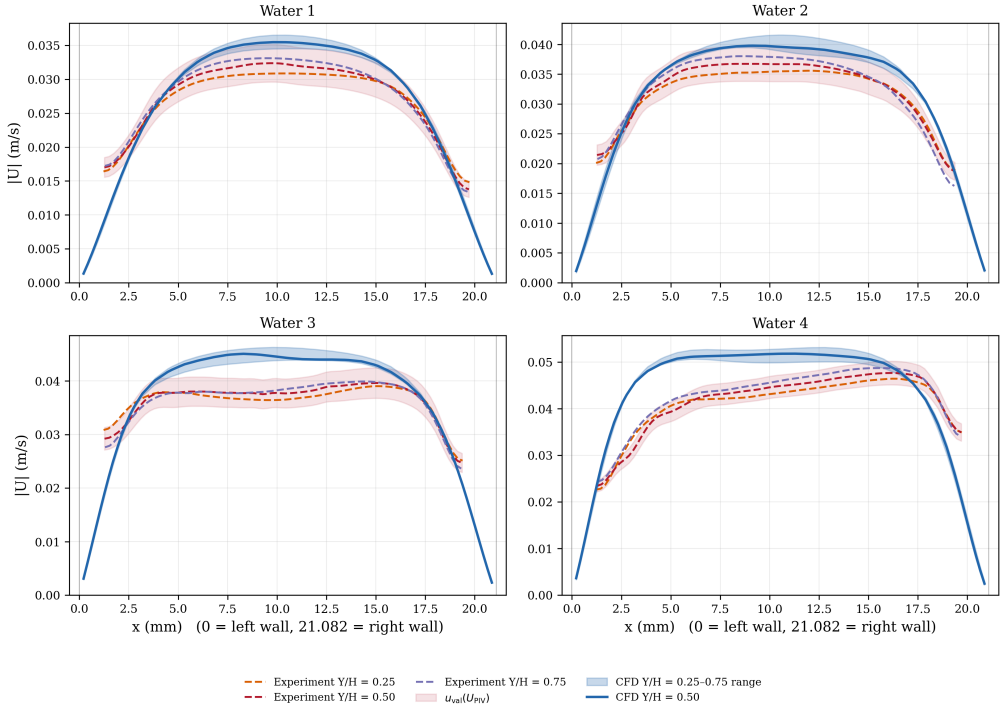

Figure 6.14: Velocity Profile Error Comparison for Laminar Water

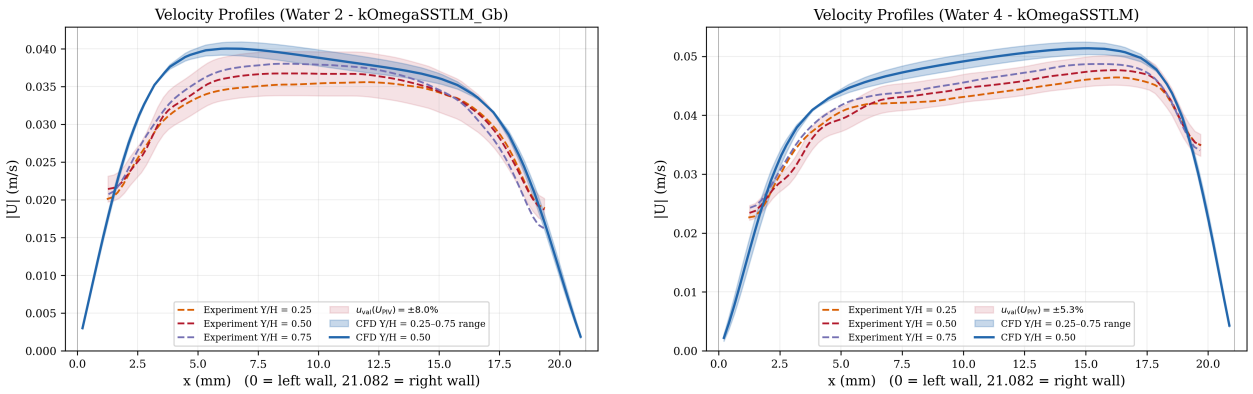

Figure 6.15: Velocity Profile Error Comparison for Best Water Fits

developed streamlines without hardly contributions to the lower-left stagnant regions. The effect here is two-fold; not only is there a fast stream developed along the leftmost wall of the loop, but that streamline creates a low pressure region at the elbow that results in an internal recirculation, seen by the negative vertical velocity components seen on the lower end of the legend for the right sub-figure within Figure 6.16

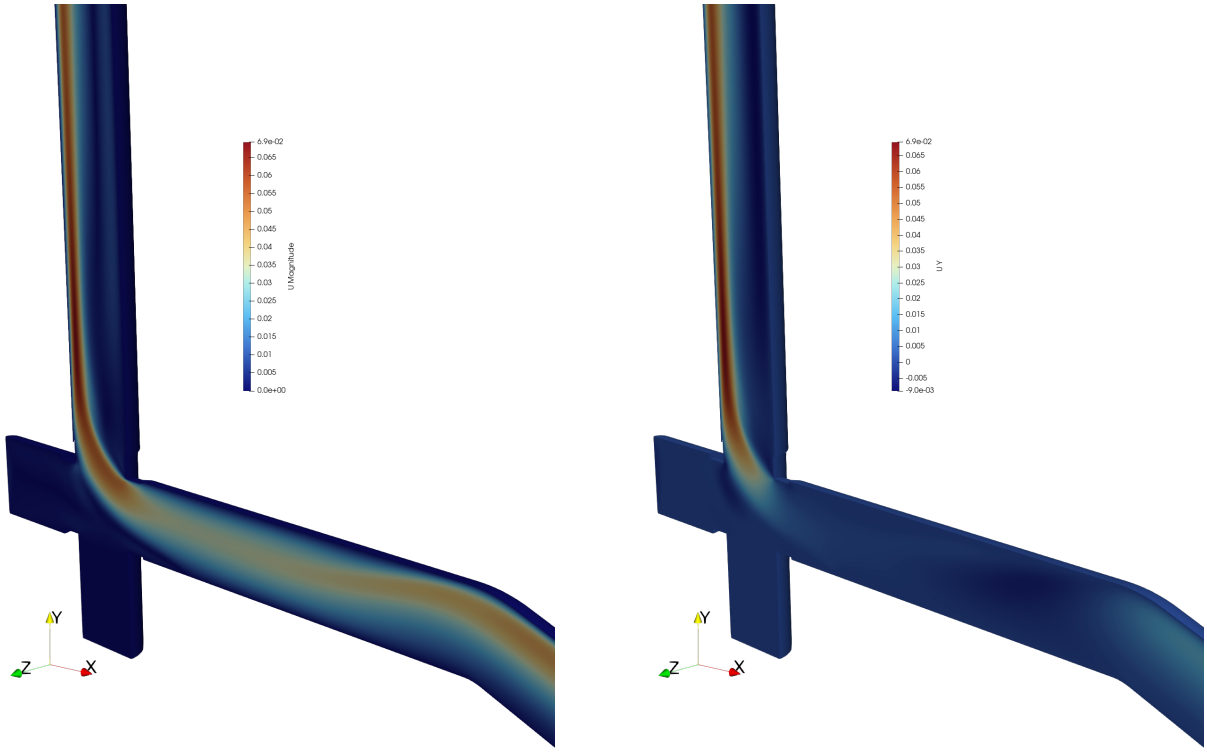

Figure 6.16: Lower Left Velocity Planar Cross Section

We can see that this extends all the way through the test section, as seen on the left of

Figure 6.18 and it actually continues all the way up to the top junction where it is split by creases in the tee-junction, seen in Figure 6.17. As an observation note, the apparent oddity in the upper-right corner of Figure 6.17 is the flow being deflected by the internal Steinmetz structure of the intersecting cylindrical boreholes.

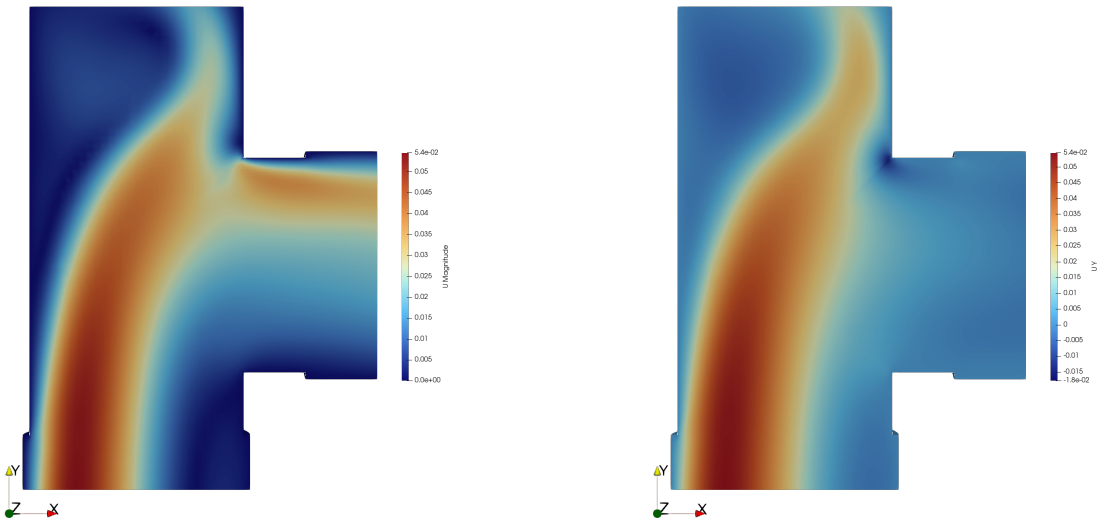

Figure 6.17: Upper Left Velocity Planar Cross Section

This reveals a consequence of the use of the PIV methodology. The experimentalists later confirmed that the internal recirculation was not an artifact of the CFD, but in-fact a real, physical behavior that managed to go unrecorded by the original experimental effort.

With the visualizations covered, the final summary of the the validations efforts success can be tabulated. Below are the results using the validation uncertainty for water 2 with kOmegaSSTLM_Gb turbulence modeling and using the validation uncertainty for water 4 with kOmegaSSTLM. Aside from those two exceptions, each case is compared against its laminar results before being classified as validated or not.

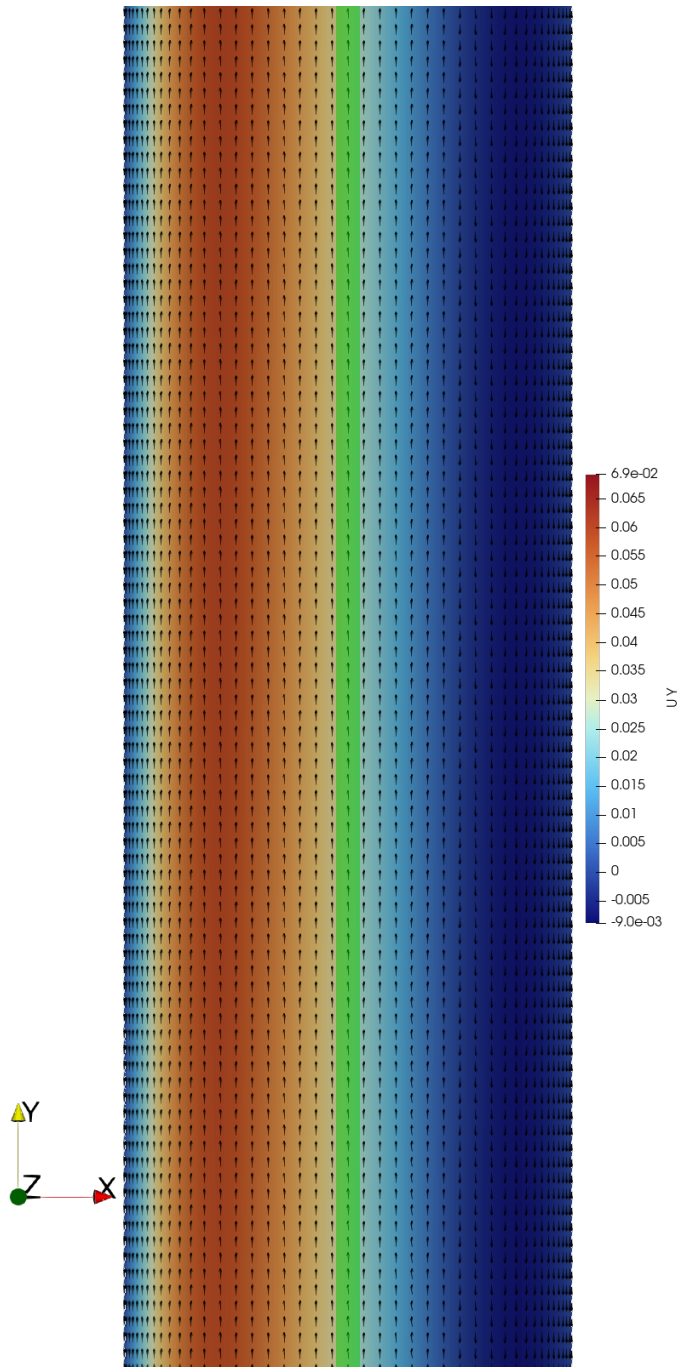

Figure 6.18: Test Section Velocity Planar Cross Section

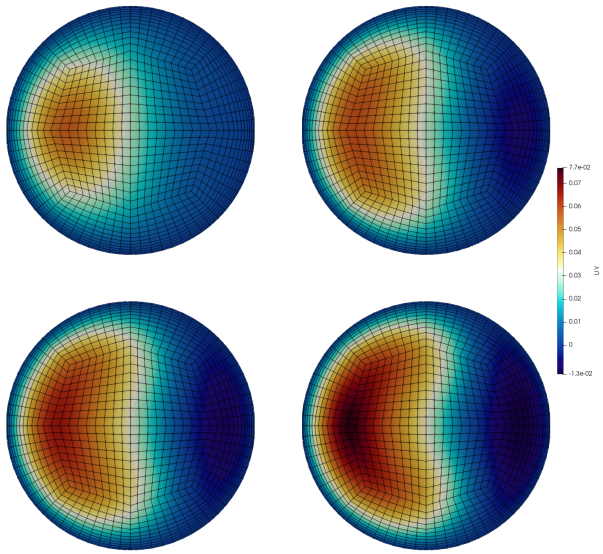

Figure 6.19: All Salt Test Section Axial Velocities (Full Range)

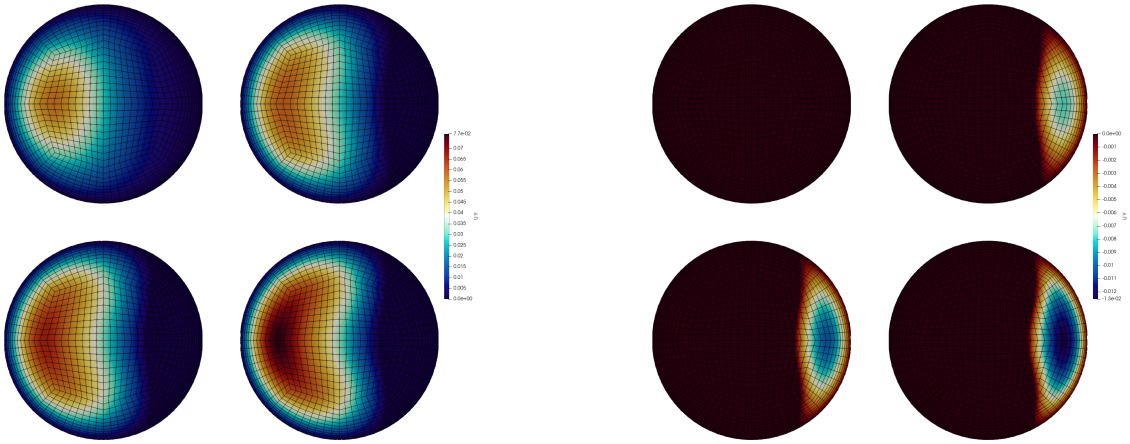

Figure 6.20: All Salt Test Section Axial Velocities (Positive-only[left] - Negative-only[right])

Table 6.10: Validation verdicts for the medium refinement level ($\dot{m}$ operator: `piv_method`).

	Hitec salt				Water			
Quantity	1	2	3	4	1	2	3	4
Mass flow rate								
$\dot{m}_{\text{sim}}$ [kg/s]	0.011347	0.016311	0.018466	0.021974	0.0085476	0.010188	0.012166	0.015055
$\dot{m}_{\text{rep}}$ [kg/s]	0.0158	0.0168	0.0175	0.0201	0.0083	0.0100	0.0120	0.0151
E [%]	−28.18	−2.91	+5.52	+9.32	+2.98	+1.88	+1.38	−0.3
u_{val} [%]	5.20	7.50	9.77	9.63	8.53	15.17	7.15	7.85
$\lvert E\rvert/u_{\text{val}}$	5.42	0.39	0.56	0.97	0.35	0.12	0.19	0.04
Validated	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Velocity								
U_{sim} [m/s]	0.013096	0.019014	0.021634	0.025893	0.019679	0.023503	0.028117	0.034969
U_{rep} [m/s]	0.0185	0.0196	0.0205	0.0237	0.0190	0.0230	0.0277	0.0350
E [%]	−29.21	−2.99	+5.53	+9.25	+3.57	+2.19	+1.51	−0.09
u_{val} [%]	4.61	7.13	9.51	9.40	8.56	15.18	7.18	7.89
$\lvert E\rvert/u_{\text{val}}$	6.33	0.42	0.58	0.98	0.42	0.14	0.21	0.01
Validated	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Centerline temperature (mean of TP1 - TP6)								
T_{sim} [°C]	149.16	175.35	188.54	203.42	42.62	47.58	51.81	62.11
T_{exp} [°C]	173.59	177.63	189.57	205.62	41.26	46.98	51.07	60.74
E [K]	−24.43	−2.27	−1.04	−2.20	+1.36	+0.60	+0.74	+1.37
u_{val} [K]	3.80	3.96	4.28	4.75	1.77	1.93	2.05	2.38
$\lvert E\rvert/u_{\text{val}}$	6.43	0.57	0.24	0.46	0.76	0.31	0.36	0.58
Validated	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wall temperature (best-aligned azimuth)								
T_{sim} [°C]	157.57	179.46	192.26	207.67	42.46	47.76	52.27	62.75
T_{exp} [°C]	174.64	178.78	190.78	206.78	41.43	47.20	51.34	61.16
E [K]	−17.07	+0.67	+1.48	+0.88	+1.03	+0.56	+0.93	+1.59
u_{val} [K]	3.80	3.84	4.06	4.34	1.77	1.93	2.05	2.38
$\lvert E\rvert/u_{\text{val}}$	4.50	0.18	0.37	0.20	0.58	0.29	0.45	0.67
Validated	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes

6.5 Discrepancies and Possible Explanations

First, the dataset associated with salt test 1 has been determined as faulty in some unknown way, but with several corresponding reasons. The first aberration within the dataset can be seen in the Appendix C Table C.1. Specifically, Salt 1's record of TW10 being greater than the value recorded for Salt 2; given that these measurements were asserted to be taken at steady-state, and Salt 2's power input is greater than Salt 1, it should be impossible to record a higher temperature baring significantly different environmental effects on account of the second reason. Second, given the nominal input heater powers, differing by roughly 30-40 W in each successive salt case, the expected increase in temperature between case 2 and 3 and then 3 and 4 does not apply between the nominal power increase between salt 1 and salt 2. Third, for the salt data that was given by the experimentalists, the numbering ran from 2 to 9, citing a failed first test with no usable data among the other unpublished cases. For those reasons, magnified by the strength of the V&V effort applied to the remaining 7 cases, it is believed that Salt 1 suffered from data mislabeling, entry error, or other not-understood misalignment from the presented configuration that was not able to be resolved even with the help of the experimentalists and has thusly been disqualified from determining the success of the current work's Verification and Validation for this dataset.

Second, despite being technically validated, the large differences in the magnitudes of $U_{\max}$ for salt test 4 leave room for improvement and on a related note, the difference in flow development between successive profile heights was much greater for the experimental case than predicted in this work. This is most likely attributable to the difference in test-section heating. Reis et al. [24] Figure 8 demonstrates the heat applied to the test section was concentrated at either end of the quartz glass, whereas the quartz within this model assumed a uniform distribution of the heat flux. The heat flux being applied through distinct rings likely resulted in a stronger impulse of buoyant force resultant by the more concentrated heating at the beginning and end of the test section. The effect of this was potentially further amplified by the high Prandtl number of molten HITEC not permitting heat to penetrate

to the centerline of the salt, causing a larger gradient in the temperature sensitive viscosity term of molten HITEC.

Chapter 7

Conclusions

7.1 Summary

This thesis presented a comprehensive verification and validation study of three-dimensional CFD simulations of the TAMU single-phase natural circulation loop, targeting the experimental data of Reis, Seo, and Hassan [9]. The principal contributions are:

1. An automated CFD modeling pipeline for the TAMU natural circulation loop capable of running and validating a model resultant from input configuration files.

2. An integrated first-order model within the modeling pipeline capable of significantly reducing uncertainty associated with the energy accounting of the original experiment.

3. Successfully verifiied the implementation of the custom `rcExternalTemperature` bounadry condtion as a suitable reduced-order model for cylindrical pipe heat losses when compared to the effort and computational expense of a full CHT simulation.

4. Performed systemic analysis to successfully follow the American Society of Mechanical Engineer's standard for Verification and Validation and vaidate the simulation pipeline for modeling the experimental effort detailed in Reis et al. [9].

5. Created a model capable of serving as a suitable starting point for further reduced-order models and machine learning training with quantified uncertainty validated within a 99% confidence interval.

7.2 Future Work

Future work for this project could include any or all of the following aspects:

1. Extensive turbulence analysis using other RANS models to create a detailed account of the modeling choice's impact on the previously validated simulations and/or expansion of the turbulence study to Large Eddy Simulation for higher-resolution of the development of turbulence within the system.

2. Detailed transient analysis beyond the steady-state modeling performed in this work.

3. Further improvements using reduced-order techniques to further reduce uncertainty and improve the quality of the model with potential applicability towards true measurement predictions rather than simulated predictions performed far after the experiment was conducted.

4. Implementation of the validated model into a Digital Twin framework, potentially serving as feedback to inform future experimental design.

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Appendix A

Mesh Details

This appendix provides additional details on the mesh topology and generation procedure described in Section 5.3.

A.1 Loop Geometry Specification

Table A.1 lists all pipe segments in the loop path, including segment type, length, and any non-default pipe radius.

Table A.1: Loop path segments. All dimensions in mm. Default pipe radius is 11.811 mm.

Segment	Type	Length / Angle	Radius	Notes
Lower leg (heater region)				
pipeleg_lower_01_fitting	straight	31.5	—	Junction patch
pipeleg_lower_02_straight	straight	78.4	—	
pipeleg_lower_03_bend	bend	$-21.5°$, $r = 25.4$	—	Negative bend
pipeleg_lower_04_straight	straight	220.0	—	Heater zone
pipeleg_lower_05_straight	straight	220.0	—	Heater zone
pipeleg_lower_06_straight	straight	220.0	—	Heater zone

Segment	Type	Length / Angle	Radius	Notes
pipeleg_lower_07_bend	bend	$+21.5°$, $r = 25.4$	—	
pipeleg_lower_08_straight	straight	78.4	—	
pipeleg_lower_09_fitting	straight	31.5	—	Junction patch
Right leg (riser)				
pipeleg_right_01_lower	straight	304.8	—	
pipeleg_right_02_middle	straight	304.8	—	
pipeleg_right_03_upper	straight	304.8	—	
Upper leg (cooler region)				
pipeleg_upper_01_straight	straight	109.9	—	
pipeleg_upper_02_bend	bend	$-21.5°$, $r = 25.4$	—	Negative bend
pipeleg_upper_03_straight	straight	115.4	—	
pipeleg_upper_04_reducer	straight	83.4	—	Reducer inlet
pipeleg_upper_05_cooler	straight	262.5	—	Cooling jacket
pipeleg_upper_06_reducer	straight	83.4	—	Reducer outlet
pipeleg_upper_07_straight	straight	115.4	—	
pipeleg_upper_08_bend	bend	$+21.5°$, $r = 25.4$	—	
pipeleg_upper_09_straight	straight	109.9	—	
Left leg (test section)				
pipeleg_left_01_upper	straight	356.0	—	
pipeleg_left_02_connector	straight	2.3	—	
pipeleg_left_03_fitting	straight	31.5	10.541	NCC upstream
pipeleg_left_04_test_section	straight	186.0	10.541	Quartz tube
pipeleg_left_05_fitting	straight	31.5	10.541	NCC downstream
pipeleg_left_06_connector	straight	2.3	—	

pipeleg_left_07_lower	straight	304.8	—

A.2 Junction Specifications

Four junctions connect the loop legs. Each junction is configured with active legs (connecting to pipe segments), stub legs (capped extensions for probes or valves), and optional expanded-radius extensions.

Junction	Active legs	Stub legs	Extensions
Lower left	$+x$, $+y$	$-x$, $-y$	$-x$ (expanded, 27.4 mm), $-y$ (27.4 mm)
Lower right	$-x$, $+y$	$+x$, $-y$	$+x$ (expanded, 27.4 mm), $-y$ (50.8 mm)
Upper right	$-x$, $-y$	$+x$, $+y$	$+x$ (expanded, 27.4 mm), $+y$ (76.2 mm)
Upper left	$+x$, $-y$	$+y$	$+y$ (152.4 mm)

The junction expanded radius for valve extensions is 12.700 mm, compared to the standard pipe radius of 11.049 mm.

A.3 Mesh Refinement Levels

Table **??** summarizes the mesh parameters for each refinement level used in this work.

Table A.2: BC Mesh refinement levels.

Level	First cell [mm]	Bulk cell [mm]	$c2c$ ratio
Coarse (GCI)	0.600	2.800	1.20
Medium (GCI)	0.300	1.400	1.15
Fine (GCI)	0.150	0.700	1.10
xFine (GCI)	0.075	0.350	1.10

Table **??** summarizes the mesh parameters for each refinement level used in this work.

Table A.3: Mesh refinement levels.

Level	First cell [mm]	Bulk cell [mm]	$c2c$ ratio	Kernel factor	Kernel blend
Coarse (GCI)	0.225	1.125	1.20	0.7	0.5
Medium (GCI)	0.150	0.750	1.15	0.7	0.5
Fine (GCI)	0.100	0.500	1.10	0.7	0.5

Table A.4: Loop Mesh Statistics

Metric	Coarse	Medium	Fine
Number of cells	2551824	8241228	28614208
Max Aspect Ratio	10.4841165	10.035558	9.95386505
Max Non-Orthogonality [°]	45.1214825	46.2884874	47.1364132
Average Non-Orthogonality [°]	6.97850602	7.23676813	7.50647319
Max Skewness	1.22050041	1.13456854	1.18472537
Minimum Determinant	0.0846038198	0.0728796419	0.0934034923
Total Volume [m^3]	0.00180937379	0.00181222443	0.00181354409
Representative Cell Size, h [mm]	0.89171456	0.60358729	0.39870235

Table A.5: BC Verification Mesh Statistics

Metric	Coarse	Medium	Fine	xFine
Number of cells	11076	118128	838880	5371968
Max Aspect Ratio	4.73809805	4.54713617	4.9654134	5.66992619
Max Non-Orthogonality [°]	18.7464067	24.4333571	30.0332729	32.2305369
Average Non-Orthogonality [°]	4.33947838	5.40885246	6.63247473	7.67114781
Max Skewness	1.24398863	0.881361531	0.501279714	0.527133118
Minimum Determinant	0.12868525	0.236797053	0.295047283	0.307140013
Total Volume [m^3]	0.000169705627	0.000186349713	0.00018810032	0.000188396703
Representative Cell Size, h [mm]	2.4837303	1.1641045	0.60752371	0.32732559

Appendix B

Thermophysical Properties

This appendix documents the thermophysical property polynomial coefficients used in the OpenFOAM simulations. All properties are expressed as polynomials in temperature T [K]:

$$\phi(T) = \sum_{i=0}^{N} a_i T^i. \tag{B.1}$$

B.1 HITEC Molten Salt

HITEC is a ternary eutectic of $NaNO_3$-$NaNO_2$-KNO_3 (molar ratio 7-49-44) with a melting point of approximately 142 °C. Property correlations are based on Janz and Tomkins [26].

Table B.1: HITEC salt polynomial coefficients. T in K.

Property	Units	a_0	a_1	a_2	a_3	a_4
ρ	kg/m^3	2293.6	-0.7497	0	0	0
c_p	J/(kg K)	5806	-10.83	7.24×10^{-3}	0	0
μ	Pa s	0.4737	-2.30×10^{-3}	3.73×10^{-6}	-2.02×10^{-9}	0
k	W/(m K)	0.7800	-1.25×10^{-3}	1.60×10^{-6}	0	0

The Prandtl number at the nominal salt operating temperature (493.15 K) is $\mathrm{Pr} \approx 54$.

B.2 Water

Water properties are higher-order polynomial fits against the IAPWS-IF97 standard at atmospheric pressure ($P = 0.101\,325\,\mathrm{MPa}$). Table B.2 lists the coefficients, which are evaluated in T [K] by Horner's method, lowest degree first. As with air, the table is transposed so that the coefficients read down the rows.

Table B.2: Water polynomial coefficients. T in K.

	ρ [kg/m^3]	c_p [J/(kg K)]	μ [Pa s]	k [W/(m K)]
a_0	$-5.177\,304\,557\,5 \times 10^{3}$	$1.561\,986\,071\,5 \times 10^{5}$	$2.699\,518\,305\,9 \times 10^{1}$	$-2.440\,486\,494\,9 \times 10^{1}$
a_1	$8.724\,711\,081\,9 \times 10^{1}$	$-2.209\,442\,670\,0 \times 10^{3}$	$-5.613\,937\,819\,4 \times 10^{-1}$	$3.614\,581\,742\,2 \times 10^{-1}$
a_2	$-4.947\,159\,533\,7 \times 10^{-1}$	$1.282\,754\,456\,8 \times 10^{1}$	$5.009\,877\,049\,3 \times 10^{-3}$	$-2.131\,614\,410\,0 \times 10^{-3}$
a_3	$1.414\,050\,649\,7 \times 10^{-3}$	$-3.718\,593\,766\,5 \times 10^{-2}$	$-2.485\,779\,288\,8 \times 10^{-5}$	$6.369\,271\,600\,3 \times 10^{-6}$
a_4	$-2.040\,793\,472\,7 \times 10^{-6}$	$5.381\,001\,345\,5 \times 10^{-5}$	$7.403\,729\,372\,1 \times 10^{-8}$	$-9.575\,003\,855\,5 \times 10^{-9}$
a_5	$1.183\,221\,677\,6 \times 10^{-9}$	$-3.107\,088\,988\,4 \times 10^{-8}$	$-1.323\,364\,664\,3 \times 10^{-10}$	$5.765\,209\,796\,7 \times 10^{-12}$
a_6	0	0	$1.314\,130\,798\,3 \times 10^{-13}$	0
a_7	0	0	$-5.591\,825\,306\,9 \times 10^{-17}$	0

Residuals were verified against IAPWS-IF97 using the `iapws` Python library over 1 °C to 99 °C in 0.5 °C steps. Table B.3 summarizes the per-property accuracy. The operational range for the TAMU loop water tests is 35 °C to 65 °C, with an initial temperature of 40 °C.

Table B.3: Water fit residuals relative to IAPWS-IF97. Temperatures in °C, errors in percent.

Property	Band with $\lvert\varepsilon\rvert \le 0.05$	Max $\lvert\varepsilon\rvert$ over 35–65	ε at 40
ρ	1–99	0.0007	-0.0005
μ	1–94	0.0393	-0.0071
k	1–41	0.3842	-0.0427
c_p	1–26	0.4767	$+0.2032$

Density and viscosity track IAPWS-IF97 to well below the standard's own ±0.05% uncertainty band across the full liquid window. Thermal conductivity and specific heat drift upward at higher temperature, reaching roughly 0.4–0.5% at the top of the operational range, which indicates their fits were tightest near ambient. Both remain well inside the experimental PIV uncertainty (1.4–2.1%) for every operating condition in this study.

B.3 Air Properties

Air properties are used in the first-order energy balance model for computing external natural convection coefficients and cooling jacket heat removal. Table B.4 lists the coefficients. The table is transposed relative to Table B.1: because the air fits are eighth order, the coefficients are listed as rows and the properties as columns so that the table fits the page width.

Table B.4: Air polynomial coefficients. T in K.

	ρ [kg/m^3]	c_p [J/(kg K)]	μ [Pa s]	k [W/(m K)]
a_0	$1.261\,35 \times 10^{1}$	1.0613×10^{3}	$-1.007\,206\,91 \times 10^{-6}$	$-1.089\,854\,26 \times 10^{-3}$
a_1	$-1.883\,358\,21 \times 10^{-1}$	-4.3282×10^{-1}	$8.680\,869\,91 \times 10^{-8}$	$8.527\,478\,29 \times 10^{-5}$
a_2	$1.536\,474\,83 \times 10^{-3}$	1.0234×10^{-3}	$-6.058\,340\,48 \times 10^{-11}$	$2.114\,407\,37 \times 10^{-7}$
a_3	$-7.506\,288\,89 \times 10^{-6}$	-6.4747×10^{-7}	$-2.541\,751\,06 \times 10^{-13}$	$-1.440\,177\,02 \times 10^{-9}$
a_4	$2.254\,628\,98 \times 10^{-8}$	1.3864×10^{-10}	$1.315\,893\,28 \times 10^{-15}$	$4.406\,695\,56 \times 10^{-12}$
a_5	$-4.077\,920\,11 \times 10^{-11}$	0	$-2.811\,117\,82 \times 10^{-18}$	$-7.793\,838\,78 \times 10^{-15}$
a_6	$4.072\,733\,12 \times 10^{-14}$	0	$3.023\,643\,53 \times 10^{-21}$	$7.577\,156\,31 \times 10^{-18}$
a_7	$-1.724\,572\,68 \times 10^{-17}$	0	$-1.331\,240\,28 \times 10^{-24}$	$-3.131\,389\,90 \times 10^{-21}$

B.4 Solid Material Properties

Table B.5 lists the thermal properties of solid materials used in the wall boundary condition and first-order model.

Table B.5: Solid material thermal properties.

Material	k [W/(m K)]	ε	Application
SS-316	16.3	0.25	Pipe walls
Insulation	-	0.95	Insulation
Fused quartz	1.4	0.95	Test section
Aluminum	237	0.15	Cooler shell

The Cerablanket (64) thermal conductivities were plugged into Excel with the temperatures converted to Kelvin, then fitted with a Excel's trendline feature using a second order polynomial. The following fits the manufacturer's data with an R^2 value of 0.998, but comes with the caveat of not having data near room temperature, so must be extrapolated to that

range. The polynomial does become negative at any point and does not suffer spurious oscillations so does not appear victim to overfitting.

$$k(T) = 6.421 \cdot 10^{-2} - 1.959 \cdot 10^{-4}\ T + 3.929 \cdot 10^{-7}\ T^2$$

$$k(T) = 6.4209E - 0.2 - 1.9593E - 04\ T + 3.9286E - 07\ T^2$$

Appendix C

Additional Data

This appendix provides supplementary data tables referenced in the main text.

C.1 Experimental Temperature Data

Tables C.1 and C.2 list the complete thermocouple readings for all eight test cases.

Table C.1: Experimental thermocouple readings [°C] for salt test cases.

Probe	Salt 1	Salt 2	Salt 3	Salt 4
TP1	176.22	180.13	192.70	208.23
TP2	170.12	173.89	187.10	204.21
TP3	168.22	172.05	185.53	202.95
TP4	173.26	177.24	190.76	206.71
TP5	174.35	178.28	191.28	207.14
TP6	179.38	183.86	196.18	211.01
TW1	166.85	170.71	181.85	196.62
TW2	168.42	171.93	182.93	198.00
TW3	167.82	170.42	181.75	197.27
TW4	169.11	172.34	184.85	201.71
TW5	192.23	198.54	212.44	230.57
TW6	187.92	193.87	206.98	223.61
TW7	175.16	180.08	191.69	208.11
TW8	173.60	177.17	188.52	204.22
TW9	173.58	177.52	189.34	204.73
TW10	116.88	116.52	125.94	136.78
TW11	171.72	175.26	187.40	202.99

Table C.2: Experimental thermocouple readings [°C] for water test cases.

Probe	Water 1	Water 2	Water 3	Water 4
TP1	40.88	46.89	50.62	60.29
TP2	40.33	46.23	49.78	59.38
TP3	41.33	47.51	51.13	60.73
TP4	41.69	47.77	51.47	61.11
TP5	41.74	47.72	51.36	60.88
TP6	41.62	47.66	51.32	61.05
TW1	39.47	44.71	48.92	58.59
TW2	39.95	45.38	49.40	58.81
TW3	39.76	45.14	49.12	58.45
TW4	40.23	45.73	49.75	59.23
TW5	43.68	49.83	54.47	65.06
TW6	47.06	54.29	59.23	70.99
TW7	41.90	47.95	51.47	60.84
TW8	40.42	46.12	50.27	59.53
TW9	41.34	46.77	50.76	60.68
TW10	30.34	32.61	34.23	38.31
TW11	40.45	46.05	50.01	59.43

C.2 Nondimensional Parameters

Table C.3 and Table C.4 provide the full set of nondimensional parameters for each test case.

Table C.3: Nondimensional parameters for salt test cases (global and local).

Parameter	Salt 1	Salt 2	Salt 3	Salt 4
Pr (global)	54.29	54.07	53.89	53.63
Re_L (global)	2715	3044	3778	5491
Gr_L (global)	7.86×10^8	7.96×10^8	8.78×10^8	9.84×10^8
Ri_L (global)	106.7	85.86	61.50	32.64
Ra_L (global)	4.27×10^{10}	4.04×10^{10}	3.66×10^{10}	3.12×10^{10}
Pr_D (local)	50.45	46.48	37.46	27.93
Re_D (local)	74.86	86.0	96.7	110.2
Ra_D (local)	1.25×10^6	1.42×10^6	1.76×10^6	2.23×10^6
C_f (friction)	0.153	0.162	0.114	0.088
$\mathrm{Nu}_{\mathrm{mean}}$	6.76	6.82	7.14	7.53

Table C.4: Nondimensional parameters for water test cases (global and local).

Parameter	Water 1	Water 2	Water 3	Water 4
Pr (global)	4.23	3.78	3.5	2.96
Re_L (global)	27100	36200	46500	68100
Ri_L (global)	6.69	5.09	3.81	2.58
Ra_L (global)	2.07×10^{10}	2.51×10^{10}	2.88×10^{10}	3.54×10^{10}
Re_D (local)	711.8	953.19	1228	1806.52
Ra_D (local)	7.95×10^5	1.09×10^6	1.33×10^6	1.90×10^6
f (friction)	0.016	0.013	0.012	0.010
$\mathrm{Nu}_{\mathrm{mean}}$	8.69	9.17	10.33	11.02

C.3 Boundary Condition Verification Data

The complete verification results for the `rcExternalTemperature` boundary condition are documented in Section 6.1. Key intermediate values for the analytical reference calculation are reproduced here for traceability.

Quantity	Value
Pipe inner radius r_0	10.0 mm
SS-316 outer radius r_1	11.0 mm
Insulation outer radius r_2	61.0 mm
Area ratio r_2/r_0	6.1
Wall resistance R_{ps}	$0.119\,02\,\text{m}^2\,\text{K/W}$
Internal film resistance R_{int}	$0.001\,39\,\text{m}^2\,\text{K/W}$
External resistance R_{ext} (Modes A, C)	$0.029\,39\,\text{m}^2\,\text{K/W}$
Graetz number Gz	3554
Average Nusselt Nu_{avg}	26.67
Internal h_{int}	$719.8\,\text{W}/(\text{m}^2\,\text{K})$
Stefan–Boltzmann constant σ	$5.670 \times 10^{-8}\,\text{W}/(\text{m}^2\,\text{K}^4)$

Table C.5: Numerical Uncertainties for CFD Quantities of Interest (water tests, $k\,\omega$ SSTLM turbulence model, two-grid GCI, medium level)

QoI	Water 1	Water 2	Water 3	Water 4
TP_{mean} [K]	0.004	0.016	0.060	0.076
TW_{mean} [K]	0.033	0.051	0.040	0.046
$\dot{m}_{\text{flow}}$ [%]	0.37	0.50	0.92	0.59
$\dot{m}_{\text{PIV}}$ [%]	0.43	4.90	4.05	6.83
U_{PIV} [%]	0.43	4.90	4.05	6.84
U_{max} [%]	0.34	3.29	3.00	0.76

Table C.6: Numerical Uncertainties for CFD Quantities of Interest (water tests, k ω SSTLM_Gb turbulence model, two-grid GCI, medium level)

QoI	Water 1	Water 2	Water 3	Water 4
TP_{mean} [K]	0.019	0.072	0.066	0.021
TW_{mean} [K]	0.033	0.057	0.053	0.067
$\dot{m}_{flow}$ [%]	0.49	0.31	1.23	0.83
$\dot{m}_{PIV}$ [%]	1.43	14.51	9.71	22.37
U_{PIV} [%]	1.42	14.50	9.71	22.37
U_{max} [%]	0.30	7.84	3.87	4.05

Table C.7: Validation uncertainty for the centerline (TP_{mean}) and wall (TW_{mean}) temperatures. $u_D = 1.1\,K$ (thermocouple) for both variants.

	TP_{mean} [K]			TW_{mean} [K]		
Case	u_{num}	u_{input}	u_{val}	u_{num}	u_{input}	u_{val}
Salt 1 (Laminar)	0.19	3.63	3.80	0.28	3.63	3.81
Salt 2 (Laminar)	0.96	3.68	3.96	0.30	3.68	3.85
Salt 3 (Laminar)	1.34	3.91	4.28	0.31	3.91	4.07
Salt 4 (Laminar)	1.93	4.20	4.75	0.22	4.20	4.35
Water 1 (Laminar)	0.03	1.39	1.77	0.02	1.39	1.77
Water 2 (Laminar)	0.04	1.58	1.93	0.05	1.58	1.93
Water 3 (Laminar)	0.04	1.73	2.05	0.03	1.73	2.05
Water 4 (Laminar)	0.09	2.11	2.38	0.03	2.11	2.38
Water 1 (k ω SSTLM)	0.00	1.39	1.77	0.03	1.39	1.77
Water 2 (k ω SSTLM)	0.02	1.58	1.93	0.05	1.58	1.93
Water 3 (k ω SSTLM)	0.06	1.73	2.05	0.04	1.73	2.05
Water 4 (k ω SSTLM)	0.08	2.11	2.38	0.05	2.11	2.38
Water 1 (k ω SSTLM_Gb)	0.02	1.39	1.77	0.03	1.39	1.77
Water 2 (k ω SSTLM_Gb)	0.07	1.58	1.93	0.06	1.58	1.93
Water 3 (k ω SSTLM_Gb)	0.07	1.73	2.06	0.05	1.73	2.05
Water 4 (k ω SSTLM_Gb)	0.02	2.11	2.38	0.07	2.11	2.38

Table C.8: Validation uncertainty for the mass flow rate: test-section cross-sectional flux ($\dot{m}_{\text{flow}}$) and reconstructed PIV methodology ($\dot{m}_{\text{PIV}}$). u_{input} and u_D are shared; only u_{num} differs between the variants.

Case	$u_{\text{num}}(\dot{m}_{\text{flow}})$ [%]	$u_{\text{num}}(\dot{m}_{\text{PIV}})$ [%]	u_{input} [%]	u_D [%]	$u_{\text{val}}(\dot{m}_{\text{flow}})$ [%]	$u_{\text{val}}(\dot{m}_{\text{PIV}})$ [%]
Salt 1 (Laminar)	2.70	1.47	3.64	3.42	5.67	5.20
Salt 2 (Laminar)	3.55	5.75	3.48	3.31	5.97	7.50
Salt 3 (Laminar)	4.62	8.54	3.47	3.24	6.62	9.77
Salt 4 (Laminar)	5.22	8.42	3.46	3.15	7.01	9.63
Water 1 (Laminar)	0.95	7.04	4.06	2.59	4.91	8.53
Water 2 (Laminar)	1.01	6.56	3.76	2.35	4.54	7.92
Water 3 (Laminar)	0.88	5.79	3.56	2.20	4.28	7.15
Water 4 (Laminar)	0.83	3.57	3.28	2.04	3.96	5.26
Water 1 (k ω SSTLM)	0.37	0.43	4.06	2.59	4.83	4.84
Water 2 (k ω SSTLM)	0.50	4.90	3.76	2.35	4.46	6.61
Water 3 (k ω SSTLM)	0.92	4.05	3.56	2.20	4.29	5.82
Water 4 (k ω SSTLM)	0.59	6.83	3.28	2.04	3.91	7.85
Water 1 (k ω SSTLM_Gb)	0.49	1.43	4.06	2.59	4.85	5.03
Water 2 (k ω SSTLM_Gb)	0.31	14.51	3.76	2.35	4.44	15.17
Water 3 (k ω SSTLM_Gb)	1.23	9.71	3.56	2.20	4.36	10.58
Water 4 (k ω SSTLM_Gb)	0.83	22.37	3.28	2.04	3.96	22.70

Table C.9: Validation uncertainty for the bulk velocity: reconstructed PIV methodology (U_{PIV}) and profile peak ($U_{\max}$, $Y/H = 0.50$). u_{input} and u_D are shared; only u_{num} differs between the variants.

Case	$u_{\mathrm{num}}(U_{\mathrm{PIV}})$ [%]	$u_{\mathrm{num}}(U_{\max})$ [%]	u_{input} [%]	u_D [%]	$u_{\mathrm{val}}(U_{\mathrm{PIV}})$ [%]	$u_{\mathrm{val}}(U_{\max})$ [%]
Salt 1 (Laminar)	1.46	12.72	3.70	2.33	4.61	13.45
Salt 2 (Laminar)	5.79	5.60	3.55	2.17	7.13	6.98
Salt 3 (Laminar)	8.59	6.92	3.54	2.06	9.51	8.04
Salt 4 (Laminar)	8.50	6.57	3.53	1.91	9.40	7.70
Water 1 (Laminar)	7.04	1.28	4.12	2.59	8.56	5.03
Water 2 (Laminar)	6.56	1.59	3.82	2.34	7.95	4.76
Water 3 (Laminar)	5.79	0.89	3.63	2.19	7.18	4.33
Water 4 (Laminar)	3.57	1.89	3.36	2.04	5.31	4.36
Water 1 ($k\ \omega$ SSTLM)	0.43	0.34	4.12	2.59	4.89	4.88
Water 2 ($k\ \omega$ SSTLM)	4.90	3.29	3.82	2.34	6.64	5.56
Water 3 ($k\ \omega$ SSTLM)	4.05	3.00	3.63	2.19	5.86	5.19
Water 4 ($k\ \omega$ SSTLM)	6.84	0.76	3.36	2.04	7.89	4.00
Water 1 ($k\ \omega$ SSTLM_Gb)	1.42	0.30	4.12	2.59	5.07	4.88
Water 2 ($k\ \omega$ SSTLM_Gb)	14.50	7.84	3.82	2.34	15.18	9.03
Water 3 ($k\ \omega$ SSTLM_Gb)	9.71	3.87	3.63	2.19	10.60	5.74
Water 4 ($k\ \omega$ SSTLM_Gb)	22.37	4.05	3.36	2.04	22.71	5.64

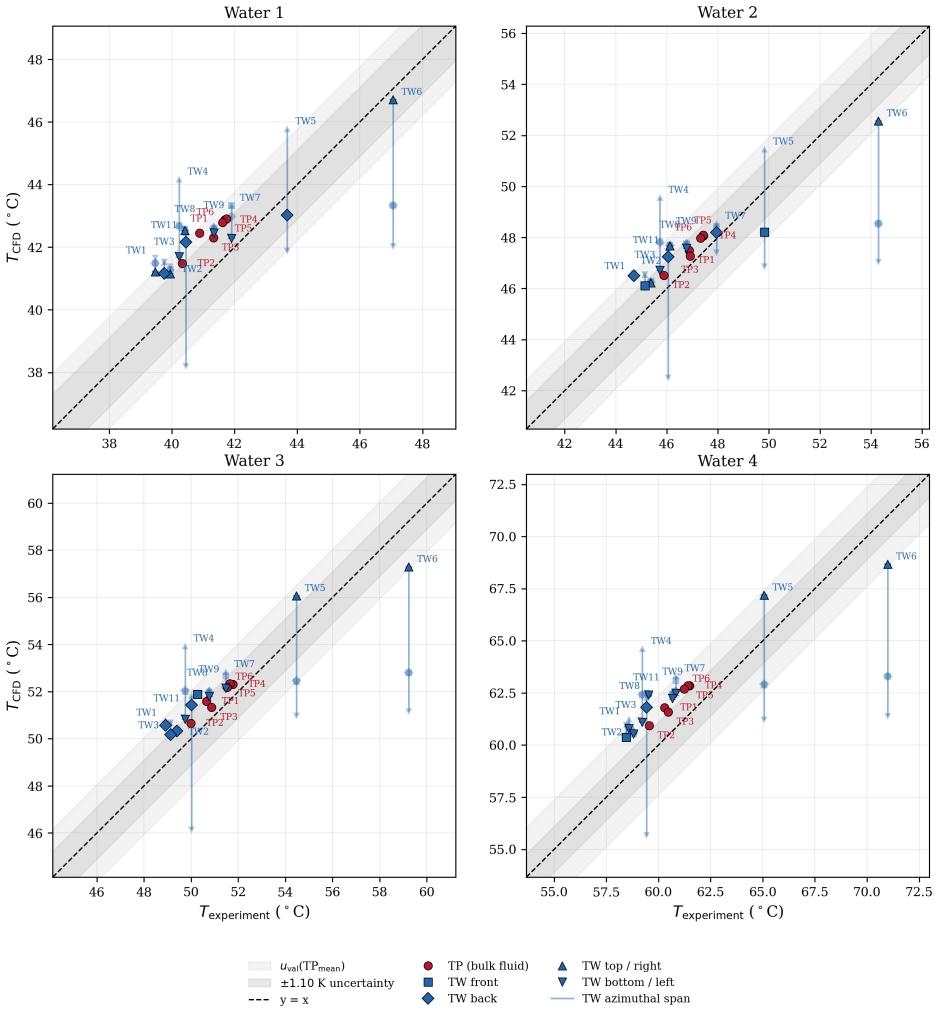

Figure C.1: Temperature Parity for SSTLM Water Test Series

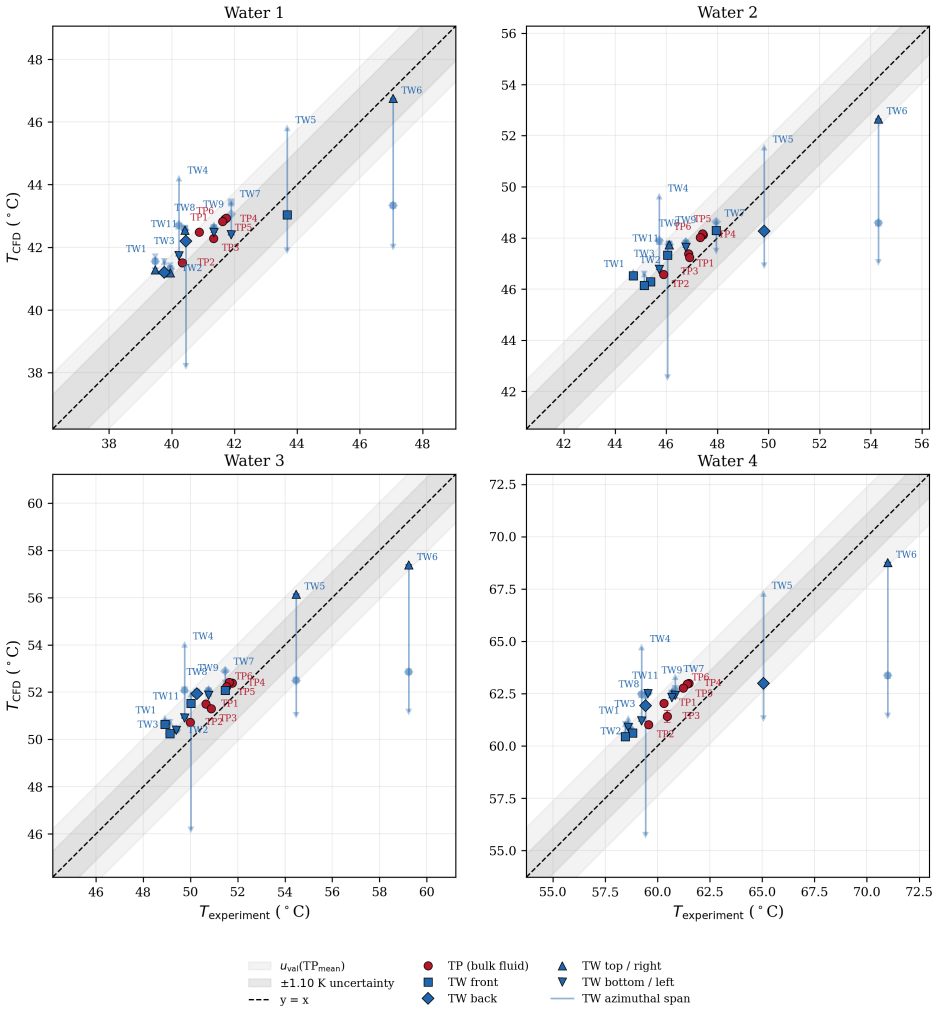

Figure C.2: Temperature Parity for SSTLM Water Test Series

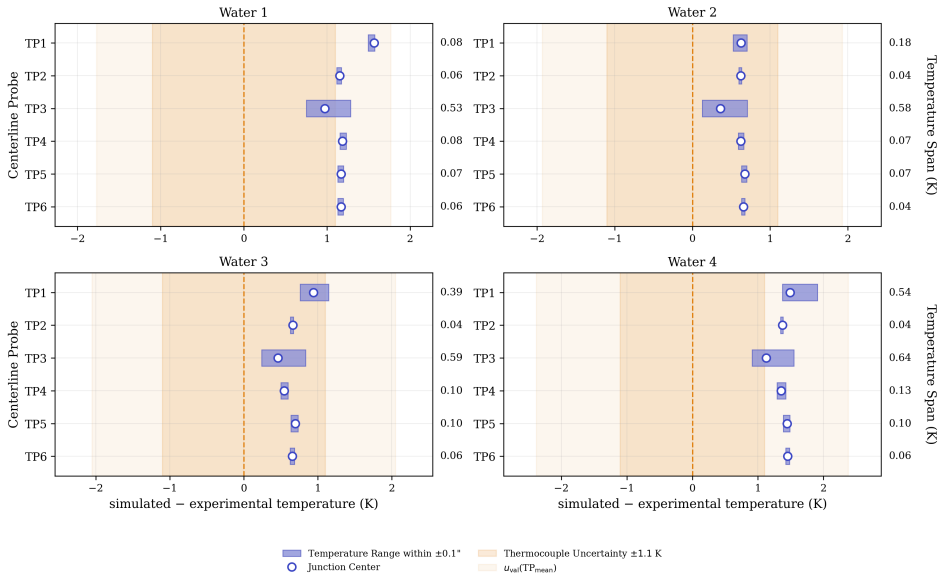

Figure C.3: Temperature Parity for SSTLM Water Test Series

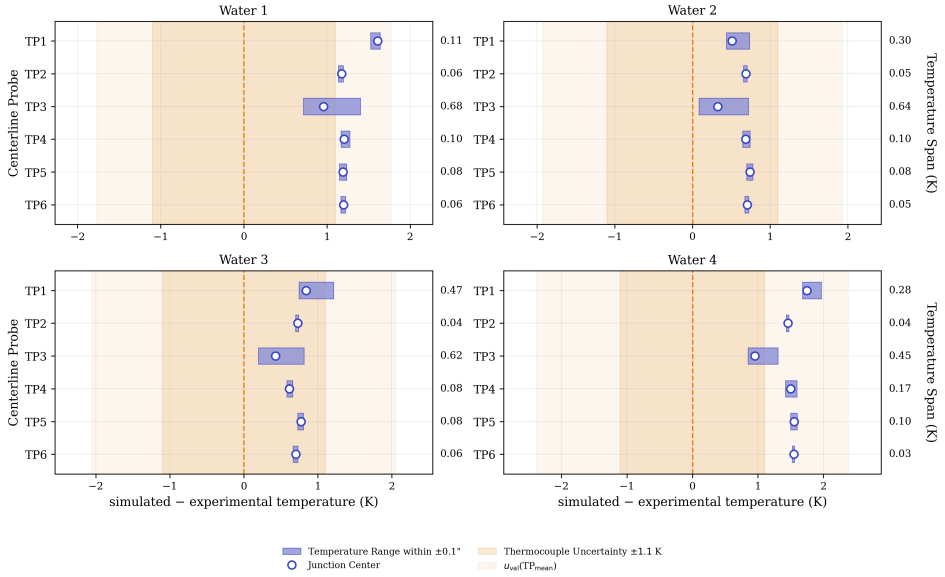

Figure C.4: Temperature Parity for SSTLM Water Test Series

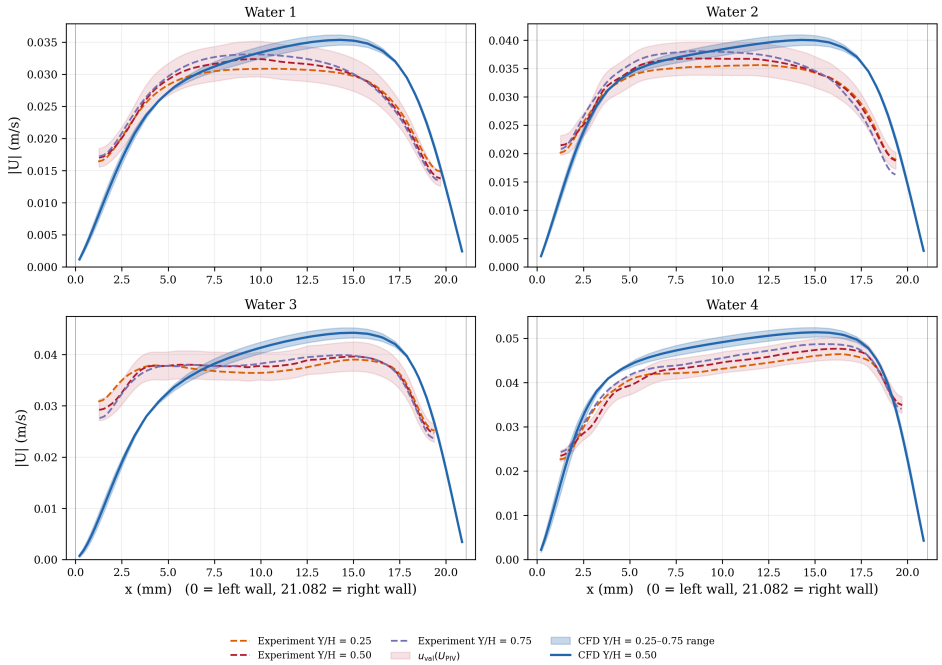

Figure C.5: Temperature Parity for SSTLM Water Test Series

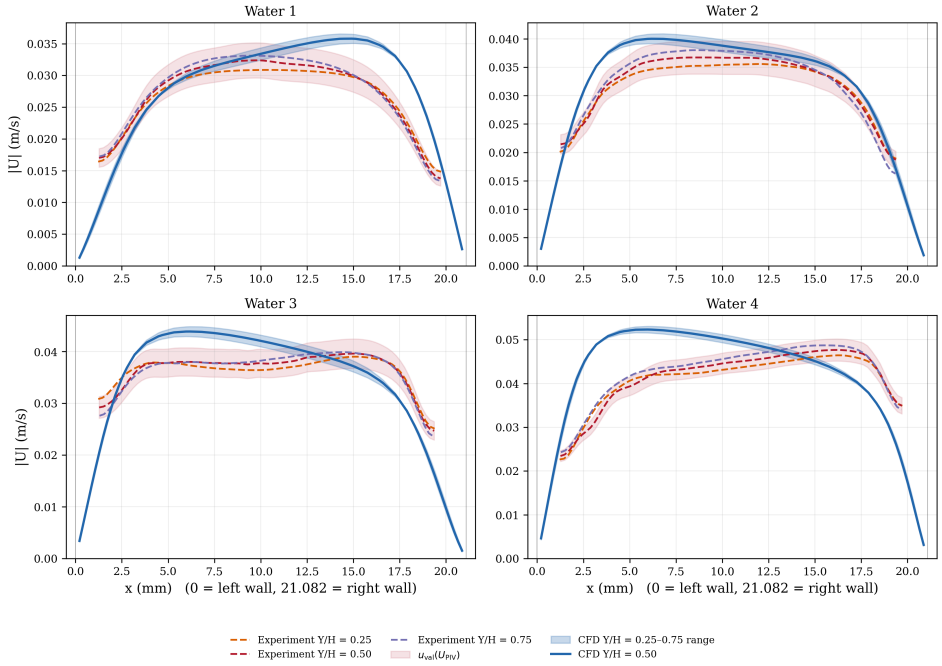

Figure C.6: Temperature Parity for SSTLM Water Test Series